# Changes in cortical organization induced by meditation and its consequences on higher functions

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# Modification de l'organisation corticale induite par la méditation et ses conséquences sur les fonctions supérieures

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# Résumé substantiel

La conscience humaine, ce phare intangible qui éclaire notre expérience subjective du monde, a longtemps été l'objet de fascination, d'interrogation et d'étude dans les domaines de la philosophie, de la psychologie et, plus récemment, des neurosciences. Si nous comprenons la conscience comme l'émergence d'un vaste réseau d'interactions complexes au sein du cerveau, il devient alors possible de postuler que les modulations de ce réseau peuvent entraîner des variations significatives dans notre phénoménologie de l'expérience consciente. L'un des leviers de ces modulations, étonnamment accessible et profondément ancré dans plusieurs traditions anciennes, est la méditation de pleine conscience. Cette pratique, loin d'être une simple relaxation, est une forme d'entraînement de l'esprit qui pousse à un renouvellement de la perception, à une transformation de l'expérience subjective elle-même et donc potentiellement à une restructuration de notre hiérarchie cognitive. Des études récentes en neurosciences ont commencé à cartographier les changements dans l'activité cérébrale chez les méditants expérimentés, révélant des modifications dans les réseaux neuronaux impliqués dans la régulation de l'attention, de l'émotion et du soi. Ces changements suggèrent l'existence d'un lien direct entre les pratiques méditatives et la plasticité neuronale, c'est-à-dire la capacité du cerveau à se modifier structurellement et fonctionnellement en réponse à l'expérience. En modifiant intentionnellement les patterns d'activité de notre cerveau par le biais de la méditation, nous pourrions être en mesure d'influer délibérément sur notre expérience consciente en dehors de la méditation. Ainsi, en entrelaçant la compréhension contemporaine des réseaux neuronaux avec les pratiques ancestrales de méditation, nous nous trouvons à l'aube d'une nouvelle compréhension de la conscience, non seulement en tant que phénomène à observer, mais en tant que domaine d'expérience à transformer activement. L'objectif de ce doctorat est donc de plonger dans l'exploration de cette frontière fascinante où s'entremêlent la science du cerveau, la phénoménologie et les pratiques transformatrices de méditation, afin de déchiffrer comment notre architecture cérébrale la plus intime façonne et est façonnée par notre expérience vécue du monde.

Notre hypothèse était que si la pratique et les états de méditation de pleine de conscience ont une influence sur la hiérarchie cognitive du cerveau dûe aux changements phénoménologiques qu'ils induisent, alors cela devrait se refléter par des changements dans la connectivité fonctionnelle des réseaux cérébraux. Nous avons employé la technique de "diffusion embedding" pour transformer les cartes de connectivité fonctionnelle en un espace de dimensionnalité réduite qui reflète la hiérarchie cognitive du cerveau, facilitant ainsi l'étude des modifications de connectivité entre de larges réseaux cérébraux.

Dans notre première étude, nous nous sommes concentrés sur la manière dont la pratique à long terme de la méditation peut influencer cette hiérarchie cognitive, en cherchant des changements significatifs qui pourraient être considérés comme des traits caractéristiques chez ses pratiquants. Pour cela, nous avons comparé les cartes obtenues par diffusion embedding entre des pratiquants experts (>10.000h de pratique) et novices de la méditation de pleine conscience. Nos résultats ont révélé qu'il était possible d'identifier des modifications, manifestées par une réduction de la dispersion des connexions entre les réseaux, indiquant une augmentation de l'intégration cérébrale. Plus précisément, nous avons observé un tel renforcement des connexions entre le cortex somatomoteur, les réseaux de l'attention, le système limbique et le réseau fronto-pariétal. En outre, ces mêmes modifications de la connectivité cérébrale étaient corrélées avec les scores obtenus sur une échelle évaluant la capacité des individus à prendre du recul par rapport à leurs pensées et émotions, suggérant que les changements induits par la méditation sur la structure de la hiérarchie cognitive pourraient avoir des implications directes sur la régulation émotionnelle et cognitive. Cette corrélation met en lumière le potentiel de la méditation comme outil

thérapeutique, capable d'induire des changements neurocognitifs qui favoriseraient une meilleure gestion des pensées et des émotions.

Suite à notre première étude, nous avons pu déterminer que l'état de méditation centré sur la bienveillance et la compassion (loving-kindness and compassion; LKC) se distinguait de l'état de repos standard, et ce, à travers les deux groupes étudiés. Nous n'avons cependant pas détaillé les spécificités de ces différences. Plus spécifiquement, nous avons observé que la méditation induisait une compression de la hiérarchie cognitive, suggérant une augmentation de la connectivité fonctionnelle entre certains réseaux cérébraux. Dans notre deuxième étude, nous avons approfondi cette question en cherchant à caractériser précisément les différences entre ces états méditatifs et l'état de repos. Nous avons également examiné si ces différences étaient significatives entre les pratiquants novices et experts. L'hypothèse sous-jacente était que, grâce à l'entraînement continu et à l'intégration des principes de la méditation de pleine conscience dans la vie quotidienne, les experts pourraient présenter une modulation moins marquée de la hiérarchie cognitive par rapport aux novices, même en état de repos. Ceci est dû à la possibilité que les experts expriment déjà, en tant que trait stable, les caractéristiques neurocognitives de la méditation de pleine conscience pendant l'état de repos. L'analyse des résultats a révélé des variations significatives entre LKC et l'état de repos. Cependant, ces variations étaient différentes pour les novices et les experts pour le même contraste. Les pratiquants expérimentés montraient une intégration des réseaux cérébraux moins accrue que les novices pendant LKC que pendant RS. Cet effet d'interaction entre états et traits suggère que la méditation de pleine conscience peut conduire à une altération durable de l'architecture fonctionnelle du cerveau suite à une phase d'adaptation et d'apprentissage dans la pratique de la méditation.

Dans notre troisième étude, nous avons cherché à valider les découvertes de notre première étude par une approche longitudinale, examinant si une pratique intensive de la méditation (8 heures par jour) pendant une retraite de 10 jours pouvait induire des modifications de la hiérarchie cognitive similaires à celles observées précédemment. Contrairement aux résultats initiaux, nous avons constaté des différences significatives entre l'état de pleine conscience et l'état de repos, démontrant des effets immédiats de l'état méditatif. Toutefois, contrairement à nos attentes, nous n'avons pas pu mettre en évidence des effets longitudinaux significatifs sur la période étudiée. Les résultats ont révélé que les états méditatifs, en particulier les pratiques dites déconstructives, entraînaient une augmentation de la ségrégation entre les différents réseaux cérébraux, suggérant une moindre intégration fonctionnelle lors de ces états spécifiques. Cela contraste avec les observations de la première étude où la méditation était associée à une plus grande connectivité.

Ces observations mènent à l'hypothèse que le processus méditatif pourrait opérer en deux phases distinctes. Dans un premier temps, les pratiques méditatives déconstructives contribueraient à renforcer la séparation fonctionnelle entre les réseaux, créant peut-être une toile de fond pour des processus cognitifs plus focalisés ou pour une meilleure gestion des informations intra-réseau. À long terme, ces pratiques seraient complétées par des pratiques reconstructives, comme celles liées à LKC, qui favoriseraient une intégration accrue et un meilleur transfert d'informations entre les réseaux cérébraux. Ce mécanisme biphasique pourrait ainsi conduire à des changements durables dans la façon dont les pratiquants expérimentent le monde, influençant de manière significative leur phénoménologie quotidienne. Ces conclusions soulignent la complexité des impacts neurocognitifs de la méditation et l'importance d'une analyse approfondie des différents types de pratiques méditatives afin de comprendre pleinement leurs effets sur le cerveau et la conscience.

**Mots clés :** méditation, pleine conscience, retraite méditative, experts de la méditation, IRMf, connectivité fonctionnelle, connectome, hiérarchie cognitive, défusion cognitive.

# Abstract

Mindfulness meditation, a practice with ancient roots, has been posited to induce significant phenomenological changes in human consciousness through the modulation of complex neural interactions. This research hypothesizes that such modulations, facilitated by the intentional training of the mind, lead to alterations in the structural and functional connectivity within the brain's cognitive hierarchy. Employing advanced neuroimaging techniques, notably diffusion embedding, this thesis aims to transform functional connectivity maps into a lower-dimensional space to better understand and visualize the changes in connectivity among large-scale brain networks. The study explores whether long-term mindfulness practitioners exhibit characteristic changes in their cognitive hierarchy as compared to novices, suggesting a restructuring of neural connectivity that correlates with their enhanced capacity for emotional and cognitive regulation. The investigation extends to differentiating the neural correlates of various meditative states, such as those focused on open monitoring and loving-kindness and compassion, from the standard resting state. It also examines whether these neural signatures differ between novice and expert meditators, potentially indicating that sustained meditation practice could lead to enduring shifts in the brain's functional architecture. Through a multi-faceted approach, this research seeks to elucidate the neurocognitive transformations induced by mindfulness practices and their implications on our understanding of consciousness. The overarching goal is to chart the cerebral landscape shaped by meditation and to uncover how our deepest neurological constructs interact with and are molded by our lived experiences.

# Résumé

La méditation de pleine conscience, une pratique ancestrale, est réputée pour engendrer des modifications profondes dans l'expérience consciente humaine, via la modulation d'interactions neuronales complexes. Cette recherche postule que ces modulations, orchestrées par un entraînement mental délibéré, entraînent des changements dans la connectivité structurelle et fonctionnelle au cœur de la hiérarchie cognitive cérébrale. Recourant à des techniques de neuroimagerie de pointe, et notamment à le “diffusion embedding”, cette thèse s'attache à transposer les cartographies de connectivité fonctionnelle dans un espace de dimension réduite afin de décrypter et de visualiser plus aisément les modifications des connexions entre les grands réseaux cérébraux. L'étude examine si les pratiquants à long-terme de la méditation de pleine conscience manifestent des altérations distinctives dans leur organisation cognitive comparativement aux débutants, ce qui indiquerait une reconfiguration de la connectivité neuronale en lien avec leur aptitude renforcée à réguler leurs émotions et leur cognition. Cette thèse cherche également à distinguer les signatures neuronales propres à divers états méditatifs, tels que la présence ouverte ou la méditation centrée sur la bienveillance et la compassion, par rapport à un état de repos conventionnel. De surcroît, elle s'interroge sur les éventuelles différences entre ces signatures neuronales chez les méditants novices et experts, suggérant que la pratique méditative prolongée pourrait induire des modifications pérennes dans l'architecture fonctionnelle cérébrale. Par le biais d'une démarche globale, cette recherche ambitionne de clarifier les transformations neurocognitives provoquées par la méditation de pleine conscience et d'en évaluer les répercussions sur la compréhension de la conscience. L'objectif ultime est de délimiter le terrain cérébral modelé par la méditation et de révéler la manière dont nos structures neurologiques les plus intimes sont influencées et modelées par le prisme de nos expériences personnelles.

# Laboratory

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# Contributions of the author

**Study 1:** PhDs Jelle Zorn, Antoine Lutz, and Oussama Abdoun designed the study. PhD Jelle Zorn collected the data with help from Eléa Perraud and Liliana Mondragon. I was primarily responsible for the data-analysis, using scripts for the pre-processing furnished by PhD Daniel Margulies and adapted by Loic Daumail. PhD Antoine provided suggestions regarding the data analysis. I wrote the report with inputs from PhD Antoine Lutz and all co-authors provided feedback.

**Study 2:** The same data as Study 1 was used. I analyzed the data and provided the tables and figures. I wrote the first draft of the manuscript with inputs from PhDs Antoine Lutz, Daniel Margulies, and Gaël Chetelat.

**Study 3:** PhDs Arnaud Poublan-Couzardot, Antoine Lutz, Giuseppe Pagnoni and Oussama Abdoun designed the Longimed study with valuable advices from meditation teacher Stephane Offort. PhD Arnaud Poublan-Couzardot and Kacylia Pistoïa organized the recruitment of the 54 participants, the logistics, and planning for three 10-day intensive meditation retreats. I designed the MRI study with help from PhDs Franck Lamberton, Daniel Margulies and Antoine Lutz, and collected the MRI data of the 54 participants over three time points with help from Igor Podolskyi and Eva Morin. I analyzed the data and I wrote the first draft of the manuscript with help from PhD Antoine Lutz.

# Contents

# INTRODUCTION

## Subjective conscious experience and mindfulness meditation

The relationship between subjective conscious experience and physical brain activity remains a profound mystery in cognitive science and neuroscience (Dehaene et al., 2003). However, advances in research on global neuronal workspace theory have begun to shed light on this difficult issue (Dehaene & Changeux, 2011). According to this framework, conscious access arises when incoming information is made globally available to multiple brain systems through long-range connections in a widespread fronto-parieto-temporal network (Dehaene & Naccache, 2001). When neuronal activity reaches a certain threshold, it ignites a self-amplifying avalanche of firing that propagates across the brain, allowing information to be flexibly routed and shared between distant cortical areas (Dehaene et al., 2003; Sergent et al., 2005).

The global neuronal workspace thus provides a first approximation of the neural correlates of consciousness, delineating the boundary between nonconscious and conscious processing (Dehaene & Changeux, 2011). However, while these findings tell us which regions and dynamics are associated with conscious awareness, they do not fully explain how and why subjective experience itself emerges from physical computations in neuronal circuits. Hard problems remain about how to link first-person phenomenology to third-person measurements of brain activity (Chalmers, 1995). This is where contemplative practices like mindfulness meditation may provide a valuable tool.

As an attentional training technique grounded in systematic introspective exploration, mindfulness meditation offers a way to help close the explanatory gap between subjective experience and neurobiology (Lutz et al., 2015). Through refining abilities for monitoring internal states with meta-awareness, meditation enables practitioners to provide clearer first-person reports that can be correlated with neural processes underlying consciousness (Hasenkamp & Barsalou, 2012; Lutz et al., 2008). Thus, integrating phenomenological and neuroscientific methods - an approach known as neurophenomenology - could leverage meditation to generate new hypotheses about the neural correlate of consciousness (Josipovic, 2019). By taking lived experience seriously as a source of data, we may gain crucial insights into how large-scale brain dynamics give rise to qualities of conscious awareness, advancing scientific theories of the mind-brain relationship.

Research indicates that mindfulness meditation can profoundly alter aspects of conscious experience and phenomenology. According to Lutz and colleagues (2015), mindfulness practices refine abilities for monitoring internal states with meta-awareness. This enables practitioners to provide clearer first-person reports that can be correlated with underlying neural processes. Mindfulness is described as enhancing three primary dimensions of phenomenology: meta-awareness, object orientation, and dereification.

Meta-awareness refers to the ability to take experience itself as an object of awareness and monitor it with clarity. Mindfulness is thought to increase meta-cognitive introspective accuracy and metacognitive awareness (Lutz et al., 2015). This allows meditators to better discriminate subtle phenomenological fluctuations.

Object orientation means sustaining attention on the object of meditation, such as the breath. Mindfulness stabilizes attention, leading to less mind wandering and distractibility. This develops concentration capacity and introspective stability.

Finally, dereification is the tendency to conceptualize or reify transitory experiences. Mindfulness reduces dereification by promoting acceptance of the impermanent nature of thoughts and emotions. This modifies cognitive and affective patterns through decreased reliance on conceptual processing.

Together, changes in these three dimensions are hypothesized to underlie a fundamental shift in the phenomenology of mindfulness practitioners compared to ordinary conscious states. Advanced meditators display distinct first-person experiential profiles that can be characterized using neurophenomenological methods and correlated with brain function. Importantly, expert meditation practitioners are able to induce a minimal phenomenal state of consciousness where the intentional structure involving the duality between object and subject is attenuated, as captured by the notion of non-duality (Dahl et al., 2015; Dunne, 2011; Lutz et al., 2007).

As such, if conscious experience depends on distributed neuronal computations, then practices enhancing meta-awareness of mental processes could conceivably reshape large-scale neurocognitive networks. As described, mindfulness meditation develops meta-cognitive introspective skills and changes phenomenology along the dimensions of enhanced meta-awareness, object orientation, and dereification (Lutz et al., 2015). These cognitive shifts likely involve functional and structural alterations in transmodal regions important for awareness, attention, memory, and emotion regulation (K. C. R. Fox et al., 2014; Schoenberg et al., 2018). Meditation may also strengthen fronto-parietal connectivity supporting conscious access to contents across modalities (Brewer et al., 2011).

Given that transmodal pathways enable global broadcasting, modifying their structure and activity could transform overall cognitive architecture. As meta-awareness grows through mindfulness practice, access to embodied emotional states may be enhanced via limbic areas like the insula (Farb et al., 2007; Lutz et al., 2008). Broadened introspective awareness could also increase top-down modulation of early sensory areas, consistent with predictive processing accounts (Lutz et al., 2019). In this manner, by targeting processes tied to consciousness, meditation may reconfigure neurocognitive networks on a systems level. Using mindfulness meditation training to link first-person experiential data with measurements of large-scale brain function can shed further light on these architectural changes.

## Problematic

According to Mesulam's influential model, the cognitive architecture of the human brain comprises transmodal areas that integrate information across sensory modalities, unimodal regions that process specific modalities, and limbic/paralimbic areas involved in emotion and memory (Mesulam, 1998). This distributed, interconnected network underlies higher cognitive functions like attention, awareness, and executive control. As discussed, mindfulness meditation may reshape this neurocognitive matrix by enhancing meta-awareness, introspection, and self-regulation (Hölzel et al., 2011; Lutz et al., 2015). However, directly assessing changes in whole-brain architecture poses methodological challenges.

Recent advances in network neuroscience provide new tools to map meditative modifications of cognitive architecture. Specifically, diffusion embedding (Coifman et al., 2005) applied to resting-state fMRI data enables a data-driven decomposition of functional connectivity patterns into components reflecting large-scale network organization (Margulies et al., 2016). This reveals the latent structural topology underlying inter-regional

organization. By applying diffusion embedding to fMRI data acquired during different meditation techniques, we can map state-dependent alterations in latent connectivity gradients and network modularity linked to cognitive architecture (Valk et al., 2023).
Longitudinally, diffusion embedding can also track lasting trait changes in functional gradients and network topology. These emergent functional pathways could represent the neural signature of a durable architectural change linked to enhanced introspection.

# Thesis layout

The goal of this PhD is to use diffusion embedding of resting-state fMRI data as a powerful approach to characterize both transient state changes and more durable trait alterations linked to mindfulness meditation practice. By applying this technique to longitudinal and cross-sectional fMRI data acquired during different styles of meditation, we can map state-dependent modifications of connectivity gradients and modularity associated with the cognitive architecture underlying attention, self-monitoring, and conscious access. In addition, diffusion embedding can track lasting reshaping of functional gradients and topology as meta-awareness becomes ingrained through extensive practice. This framework promises to elucidate how both transient and enduring effects of meditation reshape cognitive architecture.

## Study 1

In Study 1, the aim was to examine the neural correlates of meditation expertise in a group of long-term Buddhist meditators. The focus was on observing changes in the connectivity patterns of intrinsic networks in the brain. To familiarize participants with meditation techniques, individuals with no prior meditation experience (referred to as novices) underwent a weekend meditation training program before any data collection took place. The concept of meditation expertise was defined using both objective and intersubjective criteria. Firstly, practitioners needed to have learned and practiced the same meditation styles as those employed in traditional three-year meditation retreats within the Kagyu or Nyingma schools of Tibetan Buddhism, where daily meditation periods range from 8 to 12 hours. Additionally, they were required to have accumulated a minimum of 10,000 hours of formal practice and maintain a regular daily practice, indicating a certain level of expertise in the specific meditation practices under investigation. Secondly, a research assistant (C.B.), who possessed extensive experience with these practices, confirmed that the practitioners were regarded as sufficiently skilled and experienced by their Buddhist communities.

The hypothesis put forth was that enduring changes associated with expertise would manifest as specific alterations, relative to novices, in the functioning of large-scale brain networks during both meditation and non-meditative states. In particular, based on previous research on psychedelics such as psilocybin and N,N-Dimethyltryptamine (DMT) inducing ego dissolution and a collapse of the cognitive hierarchy as indicated by the first gradient (Girn et al., 2020; Timmermann, Roseman, et al., 2023), it was predicted that non-dual meditation would also involve a compression of the cognitive hierarchy, albeit to a lesser extent than psychedelics. Both psychedelics and non-dual meditation are thought to reduce self-related processes, potentially resulting in similar effects on the cognitive hierarchy.

## Study 2

In Study 2, we aimed to further characterize the connectivity patterns specifically associated with the loving-kindness compassion (LKC) meditative state. In Study 1, we were able to show a state effect where eccentricity was reduced during LKC compared to rest. However, it remained to be determined whether there were additional specific changes related to the LKC state, including potential interaction effects. We hypothesized that the LKC state would be associated with reduced eccentricity in regions associated with socio-affective processing, such as default-mode, fronto-parietal, and salience networks. Specifically, we predicted LKC would modulate gradient connectivity patterns compared to rest, especially in meditation novices versus experts. The key hypotheses were that LKC would compress the cognitive hierarchy gradient and enhance integration of default-mode, fronto-parietal, and salience networks. Overall, the goal was to elucidate differences in intrinsic functional connectivity between the LKC meditative state and resting state using a data-driven gradient connectivity approach.

## Study 3

The goal of this Study 3 was to investigate the lasting impact of intensive mindfulness meditation training on intrinsic functional brain organization. While prior neuroimaging research shows that standard 8-week mindfulness interventions can alter resting-state functional connectivity, it remains unclear whether the amplified dose of practice during meditation retreats induces further connectivity changes that persist over weeks or months. We hypothesized that the intense, prolonged mindfulness practice during a 10-day retreat would increase integration between large-scale brain networks involved in executive control, attention regulation, and self-referential processing. Specifically, we predicted retreat training would longitudinally increase connectivity within fronto-parietal and salience networks compared to pre-retreat baselines. We also expected retreat-induced connectivity increases between default mode regions and fronto-parietal areas, reflecting improved monitoring and regulation of self-related thoughts cultivated during intensive practice.

To test these hypotheses, we utilized an innovative experimental design combining a 10-day group-format mindfulness retreat with longitudinal fMRI collected at three timepoints: pre-retreat, post-retreat, and critically, 2-3 weeks following retreat completion. This enabled assessment of both acute state effects during the retreat, as well as longer-term trait alterations persisting weeks later. Fifty experienced meditators were recruited and randomly assigned to either undergo the 10-day retreat or remain on a waitlist. At each scan session, participants completed 10-minute resting state, open monitoring meditation, and focused attention meditation scans, presented in randomized order. Importantly, resting state instructions differed from typical - rather than relax, participants were requested to actively engage in mind wandering by focusing on objects of the past and future. This comparison aimed to maximize divergence from meditative states. Whole-cortex functional organization was characterized using gradient connectivity analysis, a novel technique that extracts principal gradients spanning unimodal to transmodal cortex. Comparing gradient expression before versus after retreat enabled broad evaluation of network integration and segregation induced by intensive training. This innovative experimental design will provide novel insights into the lasting impact of extreme mindfulness training on intrinsic functional brain organization. The inclusion of distinct meditative states and controlled mind wandering will elucidate state versus trait effects and clarify inconsistencies in prior research. Findings stand to inform translation of intensive training into clinical applications targeting mental health and wellbeing.

# History of meditation

The first part of this thesis will present the necessary theoretical background to understand and critically apprehend my experimental work. I will begin by providing an historical account of the emergence and development of meditation practices. This will trace meditation's origins in ancient spiritual and philosophical traditions and describe how these contemplative practices have been transformed in the modern world, particularly through the popularization of mindfulness meditation. I will then review the growing body of scientific research on meditation over the past decades. This will encompass findings regarding the effects of meditation on attentional and emotional regulation, as well as challenges that arise when scientifically studying contemplative practices. Establishing this theoretical foundation will be crucial for contextualizing and evaluating the experimental studies on mindfulness meditation and perception that will be presented in subsequent sections.

Meditation refers to a diverse set of practices that pursue a broad range of aims from enhancing relaxation, strengthening focused attention to  cultivating a long-lasting sense of well-being and of human flourishing. Often associated with Eastern contemplative traditions, meditation has a long and rich history spanning over more than two millennia. While precursors existed, Buddhism is considered a major source of meditation practices and of theories about meditation (Gethin, 1998; Wynne, 2007). It appears in India around 500 BCE and gradually spread over Asia and more recently in the West. Beginning in the 1950s, a fledgling scientific literature on meditation emerged, studying psychological and physiological correlates (Benson et al., 1990; Davidson et al., 2003; Kasamatsu & Hirai, 1966; Wallace & Benson, 1972). Over the last three decades, an explosion of research spearheaded by clinical research and cognitive neuroscience has started shedding light on the neural mechanisms and clinical efficacy of different meditation techniques. This chapter aims at clarifying the explanandum of the present research and the historical background of the meditation practices investigated in this research. More specifically, it will provide a selective overview of the history of meditation traditions, their transmission and adaptation, and the growing body of scientific research, focusing on key developments and insights relevant to cognitive science.

## Early Precursors of Meditation practices

The earliest evidence of possible meditation can be traced back to the seals and figurines of the Indus Valley Civilization (c. 3300 – 1300 BCE) in ancient India and Pakistan. Several seals depict figures seated cross-legged in postures reminiscent of yogic asanas (postures) used for meditation, suggesting the existence of proto-yogic practices (Samuel, 2008). While open to interpretation, these artifacts point to contemplative disciplines in the Indian subcontinent predating the historical Buddha (Figure 1.1)

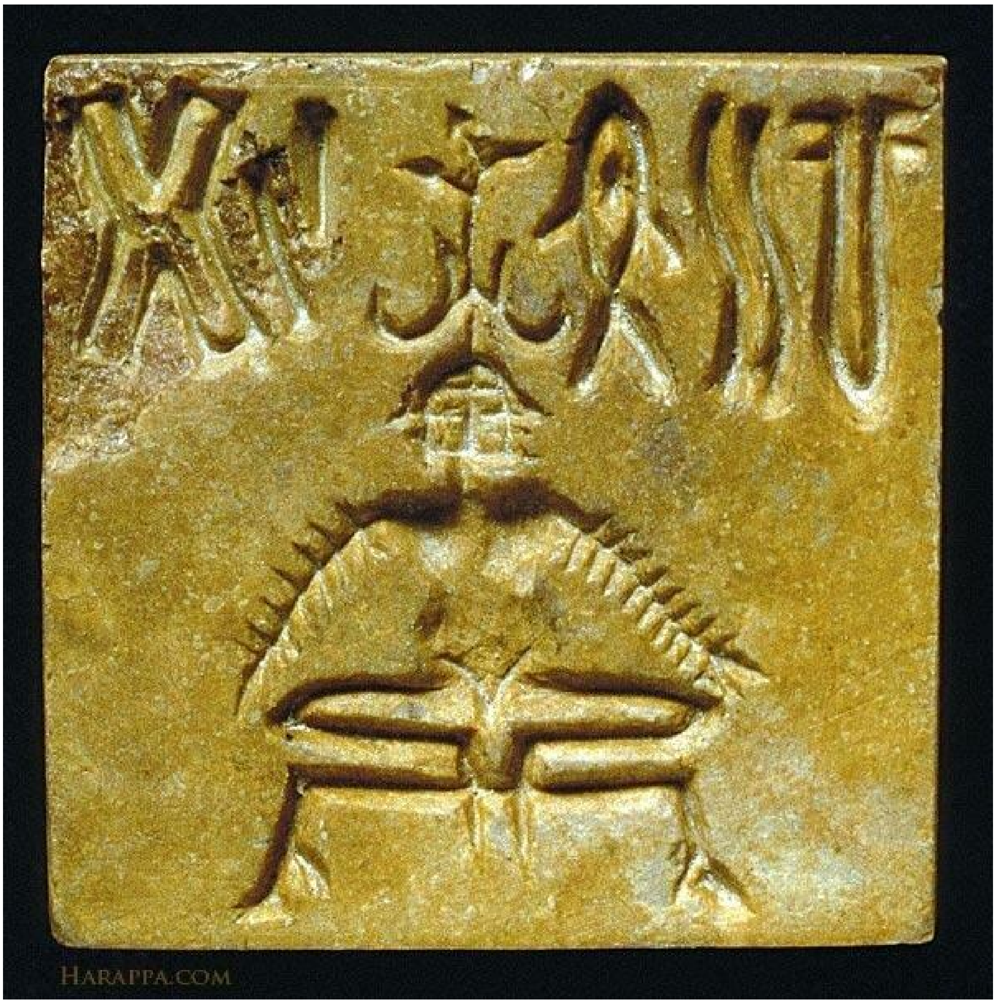


**Figure 1.1:** The seal discovered at Mohenjo-Daro depicts a divine figure seated in a meditative pose commonly associated with meditation or yoga. Source: J.M. Kenoyer/Harappa.com, Courtesy, Dept. of Archeology and Museums, Govt. of Pakistan.

The Vedic period (c. 1500 – 500 BCE) saw the composition of the Upanishads, speculative philosophical texts that formed the foundation for later Indian religions. The Upanishads describe concentration exercises that foreshadow meditation, including breath control (pranayama) and sensory inhibition as means to realize the eternal Self (Atman) (Olivelle, 1998). The Upanishads introduced key concepts like karma, dharma, and reincarnation which shaped Indian meditative traditions. Alongside the Vedic literature emerged the shramana movement, renunciant ascetics who pursued spiritual liberation through austerities and meditative disciplines (Bronkhorst, 1993; Olivelle, 1998).

## Buddhist Beginnings

While precursors existed, Buddhism is considered the original source of Buddhist meditation practices. Siddhartha Gautama, the historical Buddha, was born in Lumbini (present-day Nepal) around 563 BCE. Raised in luxury as a prince of the Shakya clan, he renounced palace life at 29 after encountering the “four sights” that led him to recognize the pervasiveness of suffering. After 6 years pursuing ascetic practices, he abandoned this path as fruitless and instead sat in meditation under the Bodhi tree, attaining enlightenment at 35. In his first sermon at Sarnath, Gautama presented the Four Noble Truths concerning the nature of suffering and the Eightfold Path leading to nibbana (nirvana), which includes meditation (Pali: samadhi) as the sixth factor. Two major forms of meditation practice formulated were samatha bhavana, calm abiding concentration using an object like the breath, and vipassana bhavana, insight meditation observing the impermanent nature of experience (Gethin, 1998).

The Buddha’s teachings were preserved orally by his monastic community until codified into the Pali canon around 100 BCE. The Pali suttas prescribe practices for tranquility (samatha) and insight (vipassana) which ground later Theravada meditation (Cousins, 1996; Gombrich, 1995). Key frameworks like the four foundations of mindfulness and four absorptions (jhanas) were delineated.

After the Buddha’s passing, adherents gathered at the First Council to recite his discourses from memory. Disagreements over monastic discipline at the Second Council around 350 BCE led to the first schism between Sthaviravada and Mahāsāṃghika, the progenitors of the Theravada and Mahayana branches. Further divisions occurred at the Third Council in 250 BCE convened by emperor Ashoka, leading to the Sarvastivada school. The Abhidhamma, an influential systematization of doctrine, emerged. Missionaries sent abroad established Theravada in Sri Lanka and mainland Southeast Asia (Gethin, 1998; Karunadasa, 1996).

## The Emergence of Mahayana

The origins of Mahayana, which became the dominant branch in East Asia, are obscure. Some hold it originated as a reform movement among liberal Sthaviravada monks. Early Mahayana sutras like the Perfection of Wisdom (100 BCE – 100 CE) developed innovative concepts like sunyata (emptiness), upaya (skillful means), and the bodhisattva ideal of compassionately forgoing nirvana to aid others (Williams, 2008).

Key Mahayana philosophical schools included Madhyamaka, founded by Nagarjuna (c. 150 – 250 CE), which used dialectic to reveal the emptiness of all phenomena; and Yogacara, developed by brothers Asanga and Vasubandhu (4th century CE), which posited a ground consciousness (alaya-vijnana) that seeds karmic habitual patterns. Both deeply influenced subsequent meditation theory and practice (Lusthaus, 2014).
The flowering of Tantra between the 7th-11th centuries infused Buddhism with esoteric ritual practices involving mantras, mudras, mandalas, and deity visualizations. Vajrayana Buddhism adopted Tantric methods which were considered more expedient but also dangerous, requiring initiation by a guru (Samuel, 2008). Important meditation manuals were Tsongkhapa's Lam Rim Chen Mo (14th century) in Tibet and Zhiyi's Mohe Zhiguan (6th century) in China.

## East Asian Buddhist Traditions

In China, Buddhism was established in the 1st century CE and initially deemed a foreign cult. But literati drawn to its philosophies helped integrate it with native Daoism and Confucianism. Key figures like Zhiyi (6th century) founded the Tiantai school which systematized meditation, while rival Huayan school employed paradoxical contemplation to reveal interdependence (Zürcher, 2007).

Chán Buddhism arose in the 7th century, reputedly when Bodhidharma passed Zen meditation to disciples like Huike at the Shaolin temple (Broughton, 2023). It rejected scholasticism and doctrinal study, using unconventional techniques like gong'an (koan) contemplation and focusing just on mindful sitting (zazen). Linji Yixuan (d. 866) founded the influential Linji lineage that was later brought to Japan.

In 1168, the monk Eisai introduced Zen to Japan where it was patronized by the warrior class. Dogen (1200 – 1253) advocated shikantaza ("just sitting") meditation. The Rinzai and Soto schools trace their lineage to Linji and Caodong respectively. Nichiren Buddhism also emerged, known for chanting the title of the Lotus Sutra (Dumoulin, 2005; Zürcher, 2007).

## Tibetan Buddhism

In Tibet, Buddhism was introduced in the 7th century during the reign of king Songtsen Gampo. Over the following centuries, lineages from India's monastic universities were translated. Atisha (982 – 1054 CE) visited from India and taught meditation techniques synthesized as lojong (mind training) (Powers, 2007). The Kadampa order was founded by his disciple Dromton. After decline followed the first diffusion of Buddhism, the later diffusion occurred from the 11th-12th century. Marpa (1012-1097) traveled to India to receive transmissions which his disciple Milarepa (1052-1135) propagated. New orders like the Kagyu and Sakya grew powerful. Later, Je Tsongkhapa (1357-1419) founded the Gelug school known for rigorous scholarship. Key texts like the stages of the path (lamrim) genre codified contemplative instructions. The rise of the Gelug and Mongol support led to conflict with the Kagyu. In the 17th century, the 5th Dalai Lama consolidated Gelug hegemony over the region with Mongol backing. Monasteries like Drepung, Sera, and Ganden became huge institutions with tens of thousands of monks (Kapstein, 2014).

## Other Indian Traditions

Beyond Buddhism, other Indic traditions developed sophisticated contemplative disciplines. Jain meditation was systematized in texts like Haribhadra's Yogadṛṣṭisamuccaya (8th century CE). Yogic meditation reaches back to the Yoga Sutras (400 BCE – 200 CE), traditionally attributed to Patanjali, which define yoga as the cessation of mental fluctuations(Bronkhorst, 1993). Tantric practices in Hinduism generated esoteric meditation techniques utilizing mantra, yantra, and identification with deities. Sufism in Islam cultivated spiritual practices of dhikr remembrance, muraqaba visualization, or whirling dancing to induce altered states (Brooks, 1990).

## Transmission to the West

Contacts between India and Greece after Alexander the Great (356 – 323 BCE) may have transmitted early Buddhist ideas westward. Greek philosophers like Pyrrho (c. 360 – 270 BCE) who accompanied Alexander to India developed concepts of ataraxia (imperturbability) reminiscent of Buddhist equanimity. The resonance between Buddhism and Pyrrhonist skepticism influenced later comparisons between meditation and Western philosophy (Beckwith, 2017).

While mentions of Indian meditation entered Greek and Roman literature, direct influence was limited until Asian missionary efforts in the 19th century. A major figure was Vivekananda (1863-1902), who introduced yoga and Vedanta philosophy at the 1893 World Parliament of Religions, helping kindle Western interest (Jackson, 1994). Theosophy, embraced by intellectuals like Schopenhauer and the Transcendentalists, also popularized Indian religions.

In the 1950s-60s, Eastern spiritual teachers like Shunryu Suzuki and DT Suzuki stimulated New Age and counter-cultural enthusiasm for meditation. The “beat generation” poets experimented with Zen, psychedelics, and peyote to attain altered states of consciousness. Their popularization primed mainstream American culture for adopting meditation. (Fields & Bogin, 2022)

The 1970s witnessed an explosion of interest, termed the “meditation boom.” Numerous Indian gurus and Tibetan lamas established teaching centers and ashrams in North America and Europe. Movements like Transcendental Meditation gained celebrity followers like The Beatles, introducing mantra meditation to the masses. Experimentation combined Eastern practices with Western psychology, as in Jon Kabat-Zinn’s Mindfulness-Based Stress Reduction (MBSR) program, one of the first clinical applications of meditation. From this collision of cultures, meditation has been adapted into new forms like Mindfulness-Based Cognitive Therapy (MBCT) tailored for Western secular society.

Mindfulness-based interventions (MBIs) have expanded in scope since their inception in the late 1970s with Kabat-Zinn's development of Mindfulness-Based Stress Reduction (MBSR) (Kabat-Zinn, 1982). While early programs like MBSR focused on alleviating conditions like chronic pain, depression and anxiety with promising results (Kabat-Zinn et al., 1986; Kabat-Zinn et al., 1992; Miller et al., 1995), this approach has been adapted for treating a wider range of mental and physical health issues. For example, Mindfulness-Based Cognitive Therapy (MBCT) was designed to help prevent relapse of major depressive disorder (Teasdale et al., 2000). Research has demonstrated the efficacy of MBIs for diverse issues including addiction, eating disorders, insomnia, and more (Shapero et al., 2018).

In addition to clinical applications, mindfulness training has been incorporated into education, business, justice and other domains. The flexibility of these interventions allows tailoring to diverse contexts and populations. While mindfulness originated in Buddhist practice, it has been successfully secularized for modern Western contexts (Davidson et al., 2003).

However, some scientists have argued more research is needed on the "active ingredients" and optimal "dosage" and delivery formats of MBIs (Crane et al., 2017). Overall, the adaptability of mindfulness-based approaches has supported their dissemination, but further research is required to refine their implementation. While clinical applications of meditation have focused on attentional forms like MBSR, interest is growing in examining the mechanisms and effects of constructive and deconstructive methods. Broadening the scope of contemplative practices studied will provide a fuller understanding of this diverse set of techniques. (see paragraph below for details).

The continued global popularity of practices like mindfulness and yoga testify to meditation's deep appeal, despite the profound differences between traditional Asian and modern Western cultures.

## Scientific Research on Meditation

Scientific investigation of meditation commenced in the 1950s-60s, enabled by new technology like electroencephalography (EEG). Early studies focused on measurement of physiological parameters like skin resistance, heart rate, and brain waves. Kasamatsu and Hirai (1966) recorded alpha wave synchronization during Zen meditation, which was replicated by other researchers (Anand et al., 1961; Wallace, 1970). Case studies of advanced yogis able to voluntarily control autonomic functions also created interest (Wenger & Bagchi, 1961).

Clinical research starting in the 1970s assessed therapeutic applications. Meta-analyses have found meditation effective for treating disorders like anxiety, depression, and chronic pain (Goyal et al., 2014). Jon Kabat-Zinn's work demonstrated Mindfulness-Based Stress Reduction (MBSR) reduced symptoms in patients with chronic pain and anxiety (Kabat-Zinn, 1982). Herbert Benson's Relaxation Response (Benson et al., 1974), which elicited a hypometabolic state dubbed the "relaxation response," also showed clinical promise.
Positron emission tomography (PET) and functional magnetic resonance imaging (fMRI) neuroimaging technologies developed in the 1990s enabled refined mapping of neural correlates. Richard Davidson's EEG work found long-term meditators had increased left-sided anterior activation indicative of positive affect (Davidson et al., 2003). Antoine Lutz found mindfulness meditators could volitionally regulate gamma synchrony associated with focused attention (Lutz et al., 2004). Imaging studies found experienced meditators had

thicker prefrontal cortex (PFC) associated with attention and emotion regulation (Lazar et al., 2005).

As the array of practices studied expanded, attempts were made to map varieties onto cognitive models and neural networks. Focused attention, open monitoring, compassion, and insight meditations were distinguished (Lutz et al., 2008). Works has investigated unique neural signatures for techniques including mantra, breath focus, yoga, mindfulness, visualization, and more.

First-person experiential reports are also drawn upon more frequently to complement third-person data (Lutz & Thompson, 2003). Phenomenological analysis reveals crucial aspects of practices missed by neuroimaging alone and can guide more nuanced research questions.

## Challenges and Opportunities

The burgeoning field faces challenges including methodological limitations, theoretical fragmentation, lack of rigor, and risks of misappropriation. Proper study design is crucial - many early studies lacked control groups, randomization, and replication (Van Dam et al., 2018). Better conceptualization of different practices and their short and long-term trajectories is needed. Commercialization and “McMindfulness” threaten to undermine the ethical foundations of meditation more oriented to selfless service (Purser, 2019).

Yet the opportunities seem boundless for this research renaissance to advance fundamental knowledge of human flourishing, cross-cultural philosophy, consciousness, and the mind-body relationship. Shedding light on the transformative potential of meditation could perhaps offer skillful means for reducing suffering and cultivating prosocial qualities highly relevant to our conflict-ridden world. What seems required is an open and rigorous scientific spirit complemented by respect for the contemplative wisdom traditions from which meditation originates.

In this selective overview, we traced the origins of meditation in ancient Indian asceticism, its systematization in Buddhism, diversification in Indian and East Asian contemplative traditions, transmission through colonial and intercultural encounters to the West, and recent scientific investigation. The history of meditation maps a long, intricate journey - an inner voyage of discovery seeking to liberate human potential which modern science is now beginning to explore, opening up promising frontiers for a greater rapprochement between meditative and scientific understandings of the mind. In the next section, I will review key proposals for classifying meditation techniques based on phenomenological features.

## Scientific classification of meditative practices from a cognitive science perspective

Meditation encompasses a diverse array of practices aimed at training the mind. In recent decades, there has been an explosion of scientific research on meditation and its effects on the brain, mind and behavior. However, one challenge facing this nascent field is the wide variability in contemplative practices coming from different philosophical and religious traditions. As Davidson and Kaszniak (Davidson & Kaszniak, 2015) discuss, operationalizing mindfulness and identifying its key components has been a challenge facing this burgeoning field of contemplative science. Nonetheless, first-person phenomenological reports from experienced mindfulness meditators, along with third-person empirical research, have begun to reveal the ways in which mindfulness practice may lead to cognitive and neuronal changes. Carefully mapping the phenomenology of mindfulness meditation will be an important step in elucidating these contemplative processes of transformation. This chapter will provide an overview of Lutz and colleagues' (2015) heuristic model that decomposes mindfulness into different dimensions that can be used to map multiple practice styles and levels of expertise. The model is grounded in both clinical applications and classical Buddhist scholarship on the phenomenology of mindfulness practice. It identifies core cognitive processes or "dimensions" of mindfulness that maximally distinguish between meditation styles and are deemed crucial for mindfulness-based emotion regulation. Critically, these dimensions are considered dynamic and manipulable, in that they can be directly or indirectly affected by factors like instruction set and expertise level.

The model proposes three primary orthogonal dimensions: object orientation, dereification, and meta-awareness. Orthogonality implies the dimensions can vary independently. Different phenomenological states are hypothesized to emerge from interactions between the dimensions, occupying positions in a multidimensional space. The model also allows for individual differences in typical location within this space, consistent with the notion of mindfulness as a trait. The three core dimensions and their relevance are:

- Object orientation reflects how strongly attention is focused on an object. This dimension helps distinguish focused attention vs open monitoring styles of meditation.
- Dereification refers to viewing internal experiences as transient mental events rather than as inherent aspects of self or reality. This supports decentering, a key mechanism in mindfulness-based therapies.
- Meta-awareness involves monitoring the quality of awareness itself. This facilitates detection of mind wandering and other shifts in phenomenological states.

Overall, the model provides a theoretical framework to bridge first-person phenomenology and third-person measurement in mindfulness research. Mapping meditation practices and expertise levels to phenomenological dimensions also enables comparative analysis and integration of findings across studies.

## Object Orientation

Object orientation refers to the degree to which attention is directed towards a specific object or class of objects. This dimension varies across different styles of meditation practice. In focused attention (FA) meditation, one continuously sustains selective focus on a single chosen object, like the breath. This involves a strong orientation towards that object. In contrast, open monitoring (OM) meditation involves deliberately reducing any intentional focus on objects. Attention is instead monitored in an open, objectless manner.

These differences in object orientation distinguish FA and OM practices. In FA meditation, attention is highly selective and object-oriented whereas in OM meditation, object orientation is minimized (Figure 1.2). This aligns with phenomenological descriptions - FA practitioners report a sense of their attention being strongly directed towards an object, while OM practitioners describe a weak or absent orientation.

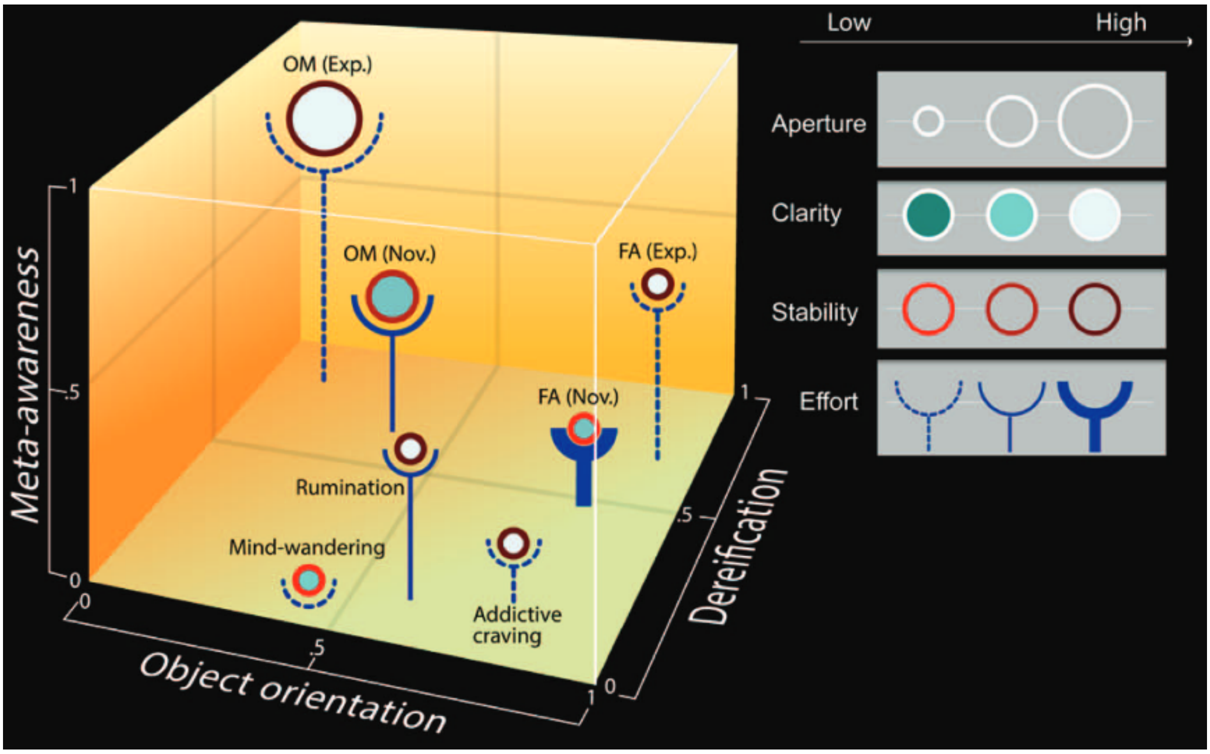


**Figure 1.2:** This figure describes a mapping of two common mindfulness practices, focus attention meditation (FA) and open monitoring meditation (OM), along with three relevant mental states (rumination, mind-wandering, and addictive craving), on a multidimensional phenomenological space. The space consists of three primary dimensions: object orientation, dereification, and meta-awareness, which are represented in a Euclidean space. Additionally, there are four secondary dimensions: aperture, clarity, stability, and effort, which are depicted respectively by the diameter, fill color, perimeter color, and width of a supporting stalk of a circle. Mind-wandering is illustrated as an effortless state with low meta-awareness and low dereification, where the content of experience is perceived as accurate depictions of reality. Addictive craving is portrayed as a state strongly and repeatedly focused on the object of addiction with high object orientation. Rumination is depicted as a state where the person is aware of stable intrusive thoughts with some meta-awareness, though still experienced as "real." From Lutz et al., 2015.

Though not a focus of this thesis, object orientation provides a useful framework for contextualizing different meditation practices. FA practices fall on the high end of this dimension due to their sustained focus on a chosen object. OM practices fall on the low end given their open, objectless monitoring of conscious experience. Tracking where specific practices fall along this spectrum aids in interpreting research findings and comparing results across studies.

## Meta Awareness

The monitoring of experience is a key dimension of mindfulness meditation. Meta-awareness, defined as the mental state that arises when attention is explicitly directed towards the present contents of experience (Smallwood & Schooler, 2015), is cultivated through mindfulness practices. Meta-awareness involves a heightened awareness and monitoring of the embodied processes of consciousness, including the dynamics of thinking, feeling, and perceiving (Dahl et al., 2015).

When meta-awareness is low or absent, we can become experientially 'fused' or absorbed in the contents of consciousness, without awareness of the processes themselves (Dahl et al., 2015). In states of experiential fusion, one remains immersed and entangled in thoughts, emotions, and sensations, unaware of the mental processes generating them. Importantly, psychological disorders like depression, anxiety and chronic pain are characterized by high degrees of fusion with negative thoughts and difficult emotions (Hoge et al., 2013; Lo et al., 2008; McCracken et al., 2004).

The cultivation of meta-awareness through mindfulness enables practitioners to step back and gain perspective, shifting from immersion in experience to being able to observe experience itself . This corresponds to a higher position in the cube in Figure 1.2. With sufficient training, meta-awareness can be sustained for long periods without requiring explicit attentional focus (Lutz et al., 2013). This ability to disentangle from being absorbed in thoughts and emotions by directing awareness towards the experiential process is considered an important precursor to mental health in clinical contexts (Hayes, 2004; Kabat-Zinn, 2013; Segal et al., 2018).

Thus, meta-awareness cultivated in mindfulness can be understood as an embodied and intentional process of sustained, graded monitoring of the dynamics of consciousness, which counteracts states of experiential fusion (Dunne, 2011; Lutz et al., 2013). Deliberate cultivation of meta-awareness is a key aim of mindfulness meditation.

## Dereification

The degree to which thoughts, feelings, and perceptions are interpreted as accurate depictions of reality versus fleeting mental events is a key dimension of mindfulness meditation (Lutz et al., 2015). This capacity for dereification refers to experiencing thoughts and feelings as transient mental processes rather than concrete representations of reality.

Without dereification, thoughts can be reified, or perceived as objective truths. For example, the catastrophizing thought, “It’s terrible and it’s never going to get better,” in response to pain lacks dereification (Sullivan et al., 1995). When reified, such thoughts can induce anxiety and amplify pain. In contrast, with dereification, thoughts lose their sense of representational accuracy, and are simply experienced as passing mental events (Lutz et al., 2015).

Like meta-awareness, dereification can spontaneously arise, such as in moments of realizing one was daydreaming. However, mindfulness meditation aims to systematically cultivate dereification as a stable capacity. At initial stages, dereification may rely on explicit cognitive reappraisals like “This is just a thought” when distracted (Lutz et al., 2015). But at advanced stages, dereification occurs more automatically through the meta-awareness cultivated in mindfulness.

Meta-awareness enables the perspective shift from immersion in thoughts and feelings to experiential observation. From this metacognitive stance, thoughts are naturally experienced as transient mental events within the broader field of sensations, emotions and perceptions. This highlights the close linkage between meta-awareness and dereification. By enhancing meta-awareness, mindfulness meditation fosters the ability to experience thoughts and feelings as passing phenomena, rather than as inherently real.

Developing meta-awareness and dereification are considered two primary mechanisms through which mindfulness enables adaptive self-regulation of emotions like pain (Lutz et al., 2015). The novelty of the phenomenological model is providing testable predictions about how different mindfulness practices and levels of expertise affect these metacognitive capacities.

In previous works (Zorn et al., 2020, 2021), they adopted this model to inform and test predictions about how mindfulness training impacts pain regulation. As I am using the same sample, I will also frame my research questions around cultivation of meta-awareness and dereification. Since well-validated measures of these specific constructs are still lacking, we studied them under the umbrella term of “cognitive defusion,” which combines aspects of both meta-awareness and dereification. In summary, deliberately cultivating meta-awareness and dereification are two key, interrelated mechanisms through which mindfulness meditation enables experiencing thoughts and feelings as transient mental events rather than as inherent reality. This metacognitive shift is hypothesized to be a core pathway through which mindfulness promotes adaptive self-regulation of challenging emotions. Thus, it may correlate with changes regarding the cognitive architecture of experts practitioners.

## Critiques and Limitations

Some limitations temper these classification schemes. Critics argue they impose arbitrary categories on intricately interconnected practices embedded in broader ethical and soteriological frameworks (Sharf, 2015). Typologies risk decontextualizing meditation from the religious milieu that traditionally scaffold and give meaning to the practices.

Others contend that the multiplicity of meditation eludes any attempt at neat classification. Practices often blend techniques, defying rigid boundaries. The same technique might have different meanings and aims in various settings. Scientific schemas inevitably simplify the diversity and fluidity of tradition-based practices (Grossman & Van Dam, 2013).

Additionally, these maps privilege Buddhist models and may not adequately represent non-Buddhist contemplative practices. The focus on techniques risks reducing meditation to mere method, rather than appreciating the orientation towards compassionate living common across traditions.

Finally, some scientists argue classifications should be grounded in empirical evidence about underlying cognitive and neural processes, rather than first-person phenomenology which cannot be objectively verified and operationalized (Josipovic, 2019). However, defenders counter that phenomenological data can complement and guide third-person experiments in an integrated paradigm (Lutz & Thompson, 2003).

## Ongoing Refinement

While current schemas have limitations, they provide an invaluable starting framework. Classifications will be continually refined as research on different practices deepens. Expanding beyond dividing practices into distinct categories, future cartographies could map meditations along multidimensional spectra, for example in terms of:

- Object vs meta-cognitive orientation
- Constructive vs deconstructive in relation to mental contents
- Focused vs unrestricted awareness
- Automatic vs effortful regulation

- Trajectory from beginner to expert stages

Such multidimensional modeling grounded in cognitive and neuroscientific evidence may better capture the subtleties of each practice while retaining their interconnectedness as parts of broader contemplative paths. Integrating first- and third-person methodologies in a respectful dialogue between scientific investigation and lived meditation experience could lead to more illuminating understandings of this uniquely rich dimension of the human mind (Lutz & Thompson, 2003).

Classification supports scientific study by organizing the extensive array of meditation techniques into a coherent framework. Pioneering efforts have delineated categories based on cognitive mechanisms and neurophenomenological signatures. This approach combines first-person subjective reports of experience with third-person measurements of brain activity to study consciousness and cognition. The term was coined by neuroscientists Francisco Varela and Antoine Lutz to emphasize the value of integrating phenomenological accounts of experience with neuroscientific methods (Lutz & Thompson, 2003).

The neurophenomenological approach aims to bridge the explanatory gap between neural processes and the qualitative aspects of experience. It utilizes introspective techniques from disciplines like phenomenology to gather rich first-person data about the contents and structure of conscious states during tasks like meditation. These subjective reports are correlated with quantitative data on brain dynamics collected simultaneously using methods like EEG and fMRI. By gathering mutually constraining data from both perspectives, neurophenomenology seeks to elucidate the neural correlates of consciousness and inform theories of how brain function gives rise to subjective experience (Lutz & Thompson, 2003).

Some key principles of neurophenomenology include:

- A focus on the structure of experience, not just the contents
- Gathering first-person data to inform experimental design and analysis
- Developing rigorous methods for eliciting high-quality subjective reports
- Iteratively relating measures of experience and brain activity
- Recognizing the interdependence of 1st and 3rd person perspectives

This approach holds promise for shedding light on the neurobiology of complex conscious states like those induced by meditation, psychedelics, and hypnosis(Timmermann, Bauer, et al., 2023). The integration of subjective phenomenology with neuroscience provides complementary lenses to elucidate the mind-brain relationship. While current models have limitations, their refinement through rigorously mapping connections to mental faculties and brain networks promises to advance an integrative science of meditation. Developing more sophisticated multidimensional models that integrate both subjective contemplative expertise and objective measurement could provide deeper insights into this uniquely revelatory window into the potentials and workings of the human mind.

I will now describe in the next section the mindfulness practices relevant to the current research – Focused Attention, Open Monitoring and Loving-Kindness and Compassion. Elucidating the cognitive and neural processes cultivated in these practices can shed light on how mindfulness transforms the mind. The current studies gathered neuroimaging data to assess the differential impacts of FA, OM, and LKC meditation states on the cognitive architecture. Linking states different in terms of phenomenology to brain dynamics, this research aims to advance understanding of the contemplative mind. In the following section, we will describe FA, OM and LKC practices.

## Description of mindfulness practices

This section overviews the primary meditation techniques relevant to the present work - Focused Attention (FA), Open Monitoring (OM), Loving Kindness and Compassion (LKC), and Open Presence (OP). I will detail how these practices relate to the metacognitive capacities of meta-awareness and dereification.

First, some general principles proposed to underlie diverse meditation practices (Lutz et al., 2007):

- Practices cultivate specific phenomenally reportable mental states.
- Repeated practice induces lasting trait changes in cognition and affect.
- There is a progression from novice to expert practitioner.

FA trains concentration by sustaining attention on a chosen object like the breath. OM involves nonreactively monitoring the continuous stream of sensations, thoughts and feelings without explicit focus. OM can mature into the advanced practice of OP meditation, where awareness turns back onto itself to transcend subject-object duality (Lutz et al., 2006;

Dunne, 2011). LKC practices extend kindness and compassion first to oneself, then loved ones, strangers, and difficult people, aiming to cultivate altruism.

These techniques derive from Buddhist traditions and secular mindfulness interventions. FA strengthens attentional stability, while OM enhances meta-awareness of mental processes and contents. LKC may foster positive qualities like empathy. Elucidating the complementary cognitive mechanisms developed in these practices can reveal how meditation transforms the cognitive architecture relevant to subjective conscious experience.

## Focused Attention (FA)

Focused attention (FA) meditation involves sustaining attention on a chosen object, such as the breath or a mantra. Traditionally, FA practices have been used to train concentration and develop a calm, tranquil state of mind (Lutz et al., 2008). When the mind wanders away into distraction, this is noticed, and attention is gently brought back to the object. With training, periods of sustained attentional focus become longer, and less effort and monitoring are required for re-orienting attention when distracted (Lutz et al., 2008).

FA meditation engages a distributed network of brain regions involved in cognitive control, including lateral prefrontal cortex, dorsal anterior cingulate cortex, insula, and basal ganglia (Malinowski, 2013). Activity in brain areas linked to monitoring, disengaging attention, and re-orienting to distractions tends to decrease as expertise develops, suggesting greater attentional stability (Hasenkamp & Barsalou, 2012; Manna et al., 2010).

The defining feature of FA is the explicit orienting of attention towards a specific object. In the phenomenological model, FA is characterized by high "object orientation" and initially low levels of meta-awareness and dereification (Lutz et al., 2015). While absorption in the object can lead to temporary suppression of discursive thought, FA does not directly cultivate insight into the nature of experience and self.

However, continued FA training may gradually increase meta-awareness, as the practitioner becomes more aware of the attentional process and contents of consciousness. FA develops concentration skills that provide a foundation for open monitoring styles of meditation aimed at cultivating meta-awareness and dereification.

## Open Monitoring (OM)

Open monitoring (OM) meditation involves nonreactively observing the continuous stream of experiences without focus on any explicit object (Lutz et al., 2008). Whereas focused attention meditation trains concentration, OM aims to develop meta-awareness of the contents and processes of experience.

In OM meditation, attention is kept open to anything that arises in the field of awareness, whether sensations, thoughts, or emotions. The key instruction is to neutrally monitor the flux of consciousness without getting involved in mental narratives or judgments (Lutz et al., 2015). When the mind wanders, this is simply noticed and attention gently returns to an open presence.

Neuroimaging studies reveal OM meditation activates brain regions linked to introspection, body awareness, and emotional processing, including the insula, temporoparietal junction, and anterior cingulate cortex (Lee et al., 2018). Activity in the narrative and conceptual self-processing regions of the default mode network decreases (Brewer et al., 2011).

OM is characterized by relatively low levels of object orientation or focus, along with high meta-awareness and dereification as mental events are observed as transient phenomena (Lutz et al., 2015). meta-awareness and dereification are explicitly cultivated, which may support adaptive self-regulation of challenging emotional states like pain, anxiety, or craving.

Advanced OM practice can lead to a dissolution of subject-object duality and a state of “open presence” where awareness becomes reflexively conscious of itself (Dunne, 2011). While beginning with strengthening meta-awareness, open monitoring meditation can reveal the fundamental nondual nature of consciousness itself.

## Open Presence (OP)

Open presence (OP) meditation is considered an advanced style of open monitoring (OM) practice aimed at recognizing the nature of awareness itself (Dunne, 2011). Whereas OM entails observing the contents of experience like sensations, thoughts, and emotions, OP meditation turns the practitioner's awareness back onto itself, the source, with no duality between subject and object (Lutz et al., 2015).

OP meditation derives from Dzogchen and Mahamudra traditions in Tibetan Buddhism and involves recognizing and sustainingrig pa, one’s “fundamental awareness” (Lutz et al., 2007). The practitioner rests in reflexive awareness devoid of intentional effort or object orientation (Lutz et al., 2015). This state of “open presence” is characterized by a lucid cognition empty of conceptual and dualistic elaboration (Dunne, 2011).

Previous research has characterized OP meditation as inducing a phenomenal state where the subject-object duality of consciousness is attenuated, reflecting the nondual realization sought in OP practice (Josipovic et al., 2019). Compared to focused attention, OP meditation shows lower activation in regions linked to cognitive control and salience detection like dorsal anterior cingulate cortex and anterior insula. OP also decreases default mode network activity associated with narrative self-reference (Josipovic et al., 2019).

The long-term cultivation of OP meditation is hypothesized to produce a trait shift wherein the baseline functional brain organization of experts comes to overlap with the state profile of OP practice itself (Garrison et al., 2015). More research on the neurocognitive impacts of expertise in nondual OP meditation can elucidate mechanisms by which this contemplative practice transforms perception of self and reality.

## Loving-Kindness and Compassion (LKC)

Loving-kindness and compassion (LKC) meditation involves cultivating unconditional kindness and care towards oneself and others (Hofmann et al., 2011). Rather than observing experiences like physical sensations or thoughts, the practitioner actively generates positive emotional states of love, warmth, and compassion.

Typically, one begins by extending good-will towards oneself, repeating phrases like "May I be happy and free from suffering." Attention is then extended outwards towards loved ones, acquaintances, strangers, and finally difficult people (Hutcherson et al., 2008). The ultimate aim is to cultivate these altruistic emotions towards all beings without discrimination.

Functional neuroimaging reveals LKC meditation activates brain networks linked to empathy, affiliation, and theory of mind including the insula, cingulate cortex, and temporoparietal junction (Engen & Singer, 2015). Compared to novices, expert LKC practitioners show greater empathic responses to social emotion stimuli along with increased brain activation in areas involved in understanding others' mental states (Mascaro et al., 2013; Singer & Klimecki, 2014).

Thus, regular LKC practice may enhance prosocial traits like compassion. The generation of unconditioned, positive affect may also support adaptive regulation of difficult emotions and reduce distress (Garland et al., 2015; Zeng et al., 2015). Further research on the long-term impacts of LKC on social cognition and emotion could better elucidate the mechanisms by which compassion transforms the mind. Integrating first-person phenomenology with neuroimaging can link the cultivation of specific qualities like empathy and care to related patterns of brain plasticity.

Having defined these contemplative practices from the cognitive science perspective, I will now review the growing body of scientific research on meditation over the past decades. This will encompass findings regarding the effects of meditation on attentional and emotional regulation, as well as challenges that arise when scientifically studying contemplative practices. Establishing this theoretical foundation will be crucial for contextualizing and evaluating the experimental studies on mindfulness meditation that will be presented in subsequent sections. I aim to provide a comprehensive overview of the current state of meditation research, highlighting key findings, debates, and limitations. This will situate my own experimental work within the broader scientific discourse on mindfulness meditation and subjective conscious experience.

## Behavioral studies of meditation

Scientific investigation of meditation has accelerated in recent decades, enabled by greater research rigor and advances in psychology and neuroscience. Behavioral measures have been central for assessing the impact of meditation on perceptual, cognitive, emotional and social processes. As Britton et al. (Van Dam et al., 2018)note, behavioral methods remain essential for relating neurophenomenological first-person reports to third-person measurements of brain function. This chapter reviews key behavioral findings on how different practices shape various domains of human behavior and psychology.

## Attentional Effects

Research on focused attention (FA) meditation has largely examined how it impacts attentional abilities and perceptual processes. Studies have found that experienced meditation practitioners show enhancements in sustained attention span, the ability to maintain attentional focus on a specific task, and the speed of attentional processing and shifting (Lutz et al., 2008). For instance, when engaging in concentration tasks, long-term practitioners demonstrate better maintenance of vigilance over time compared to non-meditating controls (MacLean et al., 2010). This aligns with the traditional description of FA practices cultivating an unwavering attentional focus.

Experts in FA meditation also exhibit increased perceptual sensitivity. They demonstrate lowered thresholds for detecting faint tactile stimuli, indicating a heightened clarity in moment-to-moment sensory perception (K. C. R. Fox et al., 2012; Kerr et al., 2011). Improvements are also found for auditory discrimination abilities (Delgado-Pastor et al., 2013)and visual perceptual discrimination (Hodgins & Adair, 2010). This suggests that the focused attentional engagement cultivated during FA meditation may transfer to enhancements in sensory clarity.

In addition to improvements in sustaining attention and perceptual sensitivity, FA meditation seems to strengthen voluntary attentional control and stability. Across various neurocognitive measures, practitioners demonstrate enhanced top-down regulation of attention compared to non-meditators (Lutz et al., 2008; Slagter et al., 2007; Tang et al., 2015). This provides empirical support for traditional claims that meditative concentration develops an uninterrupted attentional focus.

Long-term practitioners also exhibit increased meta-awareness and quicker detection of mind-wandering episodes or distractions (Hasenkamp et al., 2012; Mrazek et al., 2013). This aligns with descriptions of mindfulness meditation practices cultivating greater monitoring and awareness of the attentional state.

However, findings are mixed regarding whether these cognitive and perceptual improvements generalize beyond actual meditation sessions (Cahn & Polich, 2013; Chiesa et al., 2011). The degree to which enhancements transfer to non-meditative states and daily life contexts remains uncertain and requires further research. For instance, Cahn and Polich (2013) noted that while meditators exhibit increased frontal theta power during FA meditation, this difference was not observed during non-meditative resting states. Additionally, Chiesa et al.'s (2011) review concluded that there was limited evidence for attentional improvements in meditators outside of meditation. They suggested potential moderating variables like length of training and types of meditation techniques.

The degree to which cognitive gains generalize beyond the meditation session remains an open empirical question. As I have discussed (Gard et al., 2014; Lutz et al., 2015), there are ongoing debates around the mechanisms supporting transfer of learning from meditation states. Some argue that repeated activation of specific brain networks during practice induces long-term plasticity changes that persist in non-meditative contexts. Others propose that transfer requires intentional effort to integrate meditative qualities into daily life. Further research is still needed to clarify the conditions and degree to which contemplative practices generate lasting trait changes versus temporary state effects. Resolving this issue has important theoretical and practical implications for our understanding of meditation. Thus showing that meditation training has long-lasting effects on the cognitive architecture of said practitioners, the evidence could be in favor of the first hypothesis that mindfulness meditation induces lasting trait-like changes that persist in non-meditative contexts.

## Emotion Regulation Effects

Overall, studies have largely found that experienced practitioners display less emotional reactivity and report greater psychological well-being compared to non-meditator controls.

Expert meditators tend to self-report lower intensity and frequency of negative emotions like fear, anger and sadness in daily life (Goleman, 1984; Nielsen & Kaszniak, 2006). They also behaviorally exhibit dampened emotional reactions when exposed to unpleasant stimuli in experimental settings (Sze et al., 2010; Taylor et al., 2013). Neuroimaging evidence further shows lower amygdala activation in response to fearful faces in long-term Vipassana meditators (Desbordes et al., 2012)and Zen practitioners (Taylor et al., 2011). This implicates enhanced emotion regulation capacities from meditation.

However, differences appear based on specific meditation techniques. While focused attention (FA) practices are linked to reduced emotional arousal and reactivity, open monitoring (OM) practices are associated with better detection but non-reactivity to emotions (Teper & Inzlicht, 2013). This suggests OM meditation may uniquely improve meta-awareness and monitoring of emotional states compared to FA concentration techniques.

Regular meditators also describe greater sense of purpose in life and self-acceptance on well-being measures (Greeson et al., 2011). Both mindfulness and compassion-based meditations positively correlate with increased subjective well-being across various scales (Sedlmeier et al., 2012). These self-report findings support traditional claims that meditation helps foster eudaimonic well-being and healthy self-attitudes (Garland et al., 2015).

Research also indicates meditation may have prosocial effects on dispositions and behaviors toward others. After short-term loving-kindness meditation training, participants show heightened social connectivity (Hutcherson et al., 2008) and compassion (Leiberg et al., 2011; Weng et al., 2013). Neurological evidence also demonstrates compassion meditation enhances empathic responses (Mascaro et al., 2013; Lutz et al., 2008).

Experienced meditators additionally display greater emotional intelligence and ability to infer others' mental states (Chu, 2010; Mascaro et al., 2013). These social cognition benefits likely arise from honing interoceptive awareness through mindfulness-based practices. However, not all results are consistent in this domain. For instance, one study found no differences in prosocial behavior toward strangers between meditators and controls (Hafenbrack et al., 2014). Positive social findings may be limited to in-group connections (Condon et al., 2013). More research clarifying links to altruism is still needed.

## Cognitive Effects

While the majority of research has focused on attention and emotion, some studies have also examined meditation's impact on higher-order cognitive functions like working memory and cognitive flexibility.

Initial findings suggest that mindfulness meditation may provide benefits to working memory, which is important for temporarily maintaining and manipulating information. In a few studies, experienced mindfulness meditators performed better on visuospatial processing tasks that engage working memory compared to non-meditators (Jha et al., 2007; Mrazek et al., 2013; Zeidan et al., 2010). Meditation practice also seems to help counteract declines in working memory that can occur due to stress ((Kral et al., 2018).

Practitioners additionally demonstrate enhanced cognitive flexibility, as measured by faster performance switching between tasks or mental sets on neuropsychological tests (Moore & Malinowski, 2009; Teper & Inzlicht, 2013). One proposal is that meditation may improve adaptive cognitive control processes that allow flexibly directing attention (Colzato et al., 2012).

These tentative indications that meditation could improve executive functions like working memory and cognitive flexibility align with traditional descriptions of mindfulness and open monitoring practices cultivating metacognitive awareness and detection of mind states. However, more research is still needed to clarify the specific cognitive mechanisms underlying these potential higher-order benefits.

Overall, while findings are preliminary, the initial evidence suggests meditation may have broader enhancing effects on cognitive capacities beyond just attention. This accords with first-person reports of meditation fostering a clear, responsive mind. Understanding the ways contemplative practices shape higher cognitive faculties remains an interesting direction for ongoing research.

## Limitations

While behavioral research largely supports traditional claims of meditative mental training, some inconsistent findings illustrate the complexity of this field. Apparent expert performance advantages may sometimes reflect pre-existing individual differences rather than causal effects of meditation itself (K. C. R. Fox et al., 2014). Self-selection factors continue to pose challenges (Davidson & Kaszniak, 2015). Doubts remain about the degree findings generalize from laboratory settings involving short-term immediate effects to long-term changes in real-world contexts (Davidson & Kaszniak, 2015; Van Dam et al., 2018).

In conclusion, behavioral research provides initial evidence for meditation facilitating various domains such as attention, emotion regulation, prosocial dispositions and executive function. However, limitations remain concerning underlying mechanisms, replicability, generalizability and the interaction of individual differences with practice effects. As methodology continues improving, behavioral measures will be essential for relating subjective phenomenology to objective neural and cognitive outcomes in order to elucidate the potentials of mental training through systematic meditative practice. Integrating first- and third-person methods can lead to richer understanding of contemplative practice effects (Lutz and Thompson, 2003; Lutz et al., 2015). Overall, behavioral research supports traditional claims of transformative benefits from mental development through meditation.

I have discussed so far the first-person/phenomenological dimensions of meditation practices and have described their associations and consequences on behaviors. Before developing on the neurophysiological measures of interest for my PhD research, which is functional MRI and more broadly reviewing the contribution of neuroimaging techniques on our understanding of meditation , I will now present the methodology used during my research.

# METHODS

My PhD research relies on sophisticated methods which I will present now. In brief, I used neuroimaging (fMRI) to investigate the impacts of meditation practices on the organization of resting state intrinsic connectivity networks in the brain. In this section, I will first remind the reader of the nature of the brain signal which we measured, the so-called BOLD signal. Then I will explain how we can study the brain functions by observing the spontaneous activity of the BOLD signal at rest. Understanding this question will require to discuss the cognitive functions associated to the activity of several large-scale brain networks. In the last part of the neuroimaging section, I will introduce a recent theoretical proposal which organizes these various networks along a gradient representing the brain hierarchical organization. I will present a recent method to measure this gradient, which I used during my PhD research. After having presented these neuroimaging methods, I will describe the general methodology of a cross-sectional and one longitudinal study, which we conducted during my PhD project.

## Neuroimaging

### Bold signal

The blood-oxygen-level dependent (BOLD) signal is a type of magnetic resonance imaging (MRI) contrast that is sensitive to changes in blood flow and oxygenation in the brain. When a brain region is active, there is an increased demand for oxygen which results in a local increase in blood flow to that region. This results in a relative decrease in deoxygenated hemoglobin in that area, which causes a slight increase in signal intensity on what is called T2*-weighted MRI scans. The BOLD signal was first discovered by (Ogawa et al., 1990) and has since become the most widely used method for functional MRI studies of brain activity.

T2*-weighted MRI is a type of scan sequence that is sensitive to the BOLD signal. T2* refers to the transverse relaxation time of protons in the magnetic field, which is shortened by magnetic field inhomogeneities caused by deoxyhemoglobin (Ogawa et al., 1990. ). Deoxyhemoglobin acts like a local contrast agent that distorts the magnetic field, leading to faster T2* decay and signal loss (Thulborn et al., 1982). When neural activity increases blood flow, the resulting reduction in deoxyhemoglobin leads to less magnetic distortions, slower T2* decay, and higher signal on T2*-weighted scans (Boxerman et al., 1995). Therefore, T2*-weighting provides the contrast needed to detect small changes in deoxyhemoglobin concentration that reflect functional brain activation through the BOLD mechanism (Bandettini et al., 1992). The T2* decay time constant typically ranges from 50-100ms in the human brain at standard MRI field strengths (Logothetis et al., 2001), making it well suited to detect BOLD signal changes occurring on a similar timescale (Logothetis et al. 2001).

The physiology underlying the BOLD signal is complex, but can be broadly described by neurovascular coupling and the balloon model. When a population of neurons becomes active, this creates an increased energy demand which triggers localized vasodilation of blood vessels in that brain region (A. J. Smith et al., 2002). The balloon model conceptualizes this as oxygenated blood inflating a balloon, displacing deoxygenated blood which decreases the local deoxyhemoglobin concentration (Buxton et al., 1998). Deoxyhemoglobin is paramagnetic and acts as an intrinsic contrast agent, so a reduction in its concentration leads to less magnetic field distortions and higher T2* signal (Thulborn et al., 1982). The amplitude of the BOLD response reflects the degree of neurophysiological activity, with higher neural firing causing greater oxygen consumption and blood flow (Heeger & Ress, 2002).

However, the BOLD signal is an indirect measure of neural activity and can be influenced by non-neuronal factors. The neurovascular coupling that links neural activity to changes in blood flow and oxygenation involves complex signaling pathways between neurons, astrocytes, and vascular cells (Attwell et al., 2010). This hemodynamic response function varies across brain regions and individuals (Handwerker et al., 2004). Differences in baseline blood flow, blood volume, and metabolic rate can impact the BOLD response (Ances et al., 2009). There are also intrinsic physiological noises such as cardiac and respiratory fluctuations that contribute non-neural variance (Glover et al., 2000). These factors underscore the importance of properly designed experiments and cautious interpretations when studying BOLD fMRI.

The dynamics of the BOLD response are sluggish relative to the underlying neural signals. A rapid burst of neural firing leads to a BOLD response that peaks 4-6 seconds later and lasts up to 30 seconds due to the slow hemodynamics (K. J. Friston et al., 1994). This means the BOLD signal has poor temporal resolution on the order of seconds. However, it provides good spatial resolution on the order of millimeters due to the precise localization of MRI scans, allowing activity patterns to be mapped within cortical layers and subcortical nuclei (Yacoub et al., 2001). The spatial specificity of BOLD can be enhanced using high-field scanners ≥7T and specialized acquisition techniques (Uğurbil et al., 2003).

BOLD fMRI experiments compare the MRI signal during different conditions to infer the brain regions involved in different cognitive processes or states. The prototypical design is a block paradigm that alternates between periods of stimulation and rest (Bandettini et al., 1992). The signal timecourse is extracted from regions of interest and compared between conditions (Worsley et al., 1996). More sophisticated event-related designs can present brief stimuli in a randomized order to characterize BOLD responses to specific events (Rosen et al., 1998). Functional connectivity analyses examine correlations in BOLD activity between regions at rest or during tasks to map brain networks (Biswal et al., 1995). Multi-variate pattern analysis techniques can even decode mental representations based on fine-grained activity patterns (Norman et al., 2006).

A major focus of BOLD fMRI studies has been on mapping the functional organization and connectivity of the human cerebral cortex. P. T. Fox et al., (1988) first demonstrated localized BOLD increases in visual cortex during photic stimulation. Since then, thousands of studies have mapped activation patterns across the cortical surface in response to diverse sensory, motor, cognitive, social and emotional tasks (S. M. Smith et al., 2009). Forming a complex functional mosaic, these studies have revealed the distributed, segregated and integrative nature of cortical information processing (K. Friston, 2002).

Subcortical structures have also been extensively studied with BOLD fMRI. Robust activation of the amygdala, hippocampus, thalamus, basal ganglia and other structures has been demonstrated during learning, memory, motivation and affective processing tasks (LaBar et al., 1998). Analyses of functional connectivity have further characterized the interactions of cortical and subcortical regions within broader circuits (Haber & Calzavara, 2009). This has provided insights into both healthy brain function and dysfunctions in neurological and psychiatric disorders (Greicius et al., 2004).

An ongoing challenge is linking the BOLD signal more directly to underlying neural activity. Advances in imaging physics, neurophysiological recordings and computational modeling are helping address this issue. High-resolution 7T fMRI combined with layer-specific intracranial electrophysiology has shown promising results in localizing BOLD signals to specific cortical layers and cell types (Logothetis, 2008; Polimeni et al., 2010). Biophysical models can simulate microscopic hemodynamics and oxygen metabolism to better understand messenger dynamics (Stephan et al., 2007). Analyses of periodic fMRI signals also show promise in estimating neural properties like excitability and synaptic strengths (Wang et al., 2013).

Future directions for BOLD fMRI research include pushing to higher spatial resolutions, improving temporal resolution using multiband acceleration techniques (Feinberg et al., 2010), combining fMRI with other imaging modalities like EEG/MEG (Huster et al., 2012), and relating signals to emerging theories of population coding (Kriegeskorte et al., 2008), attractor networks (Rolls & Deco, 2010) and predictive processing (Friston, 2010). This continued development and integration of complementary techniques for interrogating brain function hold great promise for furthering our understanding of the neural foundations of human cognition.

In summary, the blood-oxygen-level dependent (BOLD) signal measured by functional magnetic resonance imaging (fMRI) has become an indispensable tool in cognitive neuroscience research. It provides an indirect window into human brain function by utilizing localized changes in blood flow and oxygenation that reflect underlying neural activity. Since its initial discovery over 30 years ago, BOLD fMRI has rapidly advanced to map functional activation patterns, connectivity networks and information representations across the whole brain with good spatial resolution. Ongoing work to improve the interpretation of fMRI signals and relate them to specific neuronal computations will further elucidate the neurobiological basis of cognition in health and disease. Despite limitations in directly quantifying neural activity, the BOLD signal remains one of the dominant method for non-invasively investigating human brain function.

## Resting state

Resting state functional magnetic resonance imaging (rs-fMRI) examines spontaneous fluctuations in the BOLD signal when a subject is not performing an explicit task. First discovered by (Biswal et al., 1995), coherent low-frequency BOLD (<0.1Hz) signal fluctuations were found to occur in functionally related brain regions even at rest. This spurred a new field of resting state functional connectivity research to map intrinsic networks and study baseline brain organization.

A typical rs-fMRI scan lasts 5-10 minutes as the subject lies quietly in the scanner, fixating on a crosshair without any imposed stimuli or behaviors. After standard preprocessing, voxelwise timecourses are extracted and analyzed to identify correlated slow (<0.1Hz) signal fluctuations across distributed brain areas. This reveals resting state networks such as the default mode, dorsal attention, salience and motor networks that recapitulate known functional systems (Smith et al., 2009). I will describe these functional systems below.

Rs-fMRI provides a powerful non-invasive tool to study whole-brain functional organization and communication. It complements task-based fMRI by measuring intrinsic connectivity networks that form the backbone for cognitive processing. Rs-fMRI connectivity patterns have been linked to behavioral traits and clinical symptoms, showing the functional relevance of intrinsic networks (Finn et al., 2015). Longitudinal rs-fMRI can track network development and breakdown in neurodevelopmental and neurodegenerative disorders (Zhang et al., 2010). Spontaneous activity may also play an active role in brain maintenance and consolidation processes (Miall & Robertson, 2006).

Several analytic methods have been developed to characterize rs-fMRI connectivity patterns. Seed-based correlation mapping examines temporal correlations between a seed region and every other brain voxel (Biswal et al. 1995). This reveals functional networks linked to the seed, but is biased by the choice of seed region. Independent component analysis (ICA) decomposes the data into maximally independent spatial maps and timecourses in a data-driven manner (Beckmann et al., 2005). Graph theory quantifies network topology by representing brain regions as nodes and functional connections as edges (Bullmore & Sporns, 2009). Machine learning approaches like support vector machines can even discriminate between patient groups based on whole-brain connectivity patterns (Craddock et al., 2009).

Multiple confounding factors must be addressed when analyzing and interpreting rs-fMRI data. Head motion can introduce substantial connectivity artifacts that require stringent correction methods (Power et al., 2014). Cardiac and respiratory oscillations can also contaminate connectivity measurements and must be removed (Glover et al., 2000). Group comparisons must account for differences in vascular reactivity that could masquerade as neural effects (D'Esposito et al., 2003). Proper study design, modeling of non-neural signals, and cautious interpretations are essential for valid rs-fMRI connectivity findings. A study by Kawagoe et al. (2017) directly compared two common resting-state instructions - "think of nothing" vs "mind wandering" - in healthy subjects. Self-report data confirmed subjects spent more time in the appropriate mental state based on the instructions. Independent component analysis of fMRI data revealed stronger functional connectivity in the MW condition compared to the TN condition between the default mode network and visual, salience, and frontal networks. Furthermore, the strength of connectivity between some networks was correlated with self-reported time spent in the TN state. These findings indicate pre-scan instructions meaningfully impact resting-state functional connectivity, likely by inducing distinct psychological states. Carefully considering the precise instructions given to subjects prior to resting-state fMRI acquisition may therefore be important for consistent and interpretable results. My PhD project involved three studies examining the impact of mindfulness meditation on cognitive and neural function. Study 1 and Study 2 utilized "mind wandering" instructions prior to resting state fMRI scans, while Study 3 used "active mind wandering" instructions. We initially hypothesized that by simply instructing participants to let their minds wander freely as in Studies 1 and 2, they may inadvertently engage in a focused attention meditation state where their thoughts become the object of focus. However, as Study 3 involved meditation training, this posed a potential replication issue compared to other studies utilizing naïve participants who would be unable to meditate during a resting scan. To address this, in Study 3 I specifically requested participants to engage in mind wandering while refraining from meditation. The "active mind wandering" instructions aimed to elicit a distinct mental state from meditation states. By clarifying the precise nature of mind wandering for participants in Study 3, I hoped to avoid inadvertent meditation during resting scans and improve consistency with prior literature. Critically, both Study 2 and Study 3 included comparisons of resting state to open monitoring meditation in novice meditators. By treating open monitoring as a baseline, I can evaluate whether the different pre-scan instructions in Studies 2 and 3 elicited meaningful differences in the cognitive hierarchy suggestive of changes in the functional connectivity. Differences of contrast between resting state and open monitoring, despite similar novice groups, would indicate the instructions strongly shaped network interactions and should be carefully considered for any mindfulness meditative study.

In summary, resting state functional connectivity MRI examines spontaneous low-frequency fluctuations in the BOLD signal that reflect intrinsic brain organization. Since its initial discovery over 20 years ago, rs-fMRI has rapidly evolved into a mainstream brain mapping methodology. It has revealed canonical functional networks that recapitulate task activations and covary with behavior and clinical symptoms. Ongoing methodological improvements, multimodal integration and standardization efforts are improving the reliability of rs-fMRI and establishing stronger links between imaging readouts, electrophysiology and underlying neurobiology. As a non-invasive imaging biomarker that can be widely applied, rs-fMRI holds continued promise for understanding brain network dynamics in health and disease. Careful applications paired with an appreciation of its limitations will allow rs-fMRI to shed further light on the functional foundations of human cognition.

## Functional networks

The analysis of resting state fMRI data has revealed a number of robust and consistent large-scale brain networks that spontaneously synchronize in the absence of any overt task or stimulation. Seminal work by (Smith et al., 2009) and others characterized several canonical resting state networks including the default mode network, dorsal attention network, salience network, and motor networks that recapitulate known functional systems. The connectivity patterns and spatial topography of these intrinsic networks closely correspond to co-activation patterns observed across various task fMRI studies. This suggests the brain is intrinsically organized into distinct functional networks that coordinate ongoing processing and provide a substrate for executing goal-directed behaviors. Characterizing the functional networks during rest has provided key insights into the backbone of normal and pathological brain functions. Mapping the nodes, connections, and interactions between the resting state networks has become a major focus of functional connectivity research across neuroscience and psychiatry. Examining the modulation and disruption of intrinsic functional networks holds promise for understanding not only the normal brain development but also neurodegeneration.

In the case of my PhD, I was interested in characterizing the modulation of these intrinsic functional networks by meditation expertise and meditative states. Below I will review the functional role commonly attributed to  five major large-scale brain networks: the dorsal network, the ventral network, the fronto-parietal network, the limbic network and the DMN. This parcellation of the brain into these five systems makes it possible to study meditation in relation to the specific large-scale networks.

### Dorsal Attention Network

The dorsal attention network is one of the resting state functional connectivity networks that has been consistently identified using fMRI. First described by (Fox et al., 2005), this network includes the frontal eye fields (FEF), intraparietal sulcus (IPS), and superior parietal lobule (SPL), and shows coordinated activity during goal-directed tasks requiring top-down attention.

The dorsal attention network is involved in voluntary, top-down control of attention, in contrast to the stimulus-driven ventral attention network anchored in the temporoparietal junction and ventral frontal cortex (Corbetta et al., 2008). The dorsal network directs attention in a goal-directed manner, selecting relevant sensory inputs and filtering out distractors. Activations are elicited during spatial orienting, saccades, and working memory tasks that require sustained focus (Gazzaley et al., 2007).

Functional connectivity in the dorsal attention network predicts performance on tasks requiring visual selection and tracking (Rosenberg et al., 2015). Disruption of network nodes impairs attention orienting, demonstrating its causal role in controlling the focus of attention (Blankenburg et al., 2010). EEG signatures further reveal selective synchronization within this network during attention deployment (Bressler et al., 2008).

Dorsal network activity likely reflects the selective processing of sensory information to guide behavior (Miller & Buschman, 2013). Information about behaviorally relevant locations is encoded via spatially selective neurons in the FEF, IPS and SPL. Recurrent connectivity between these regions may maintain attention on target locations and filter irrelevant distraction. The network integrates sensory evidence and expectations to update internal attentional priorities mapped across the network (Chong et al., 2017).

This coordinated brain activity stems from structural connectivity of the dorsal attention network revealed by diffusion MRI and tractography. The SLF1 and SLF2 white matter tracts link parietal and frontal regions specialized for spatial cognition and eye movements (Thiebaut de Schotten et al., 2011). Interestingly, the right dorsal network shows greater structural and functional connectivity than the left (O'Reilly et al., 2013). This may reflect left lateralized language networks competing with right-dominant dorsal attention networks.
The dorsal attention network interacts with the ventral attention network to orient attention based on internal goals versus salience cues (Vossel et al., 2014). It is anticorrelated with the default mode network, deactivating when attention focuses inwards (Fox et al., 2005). Fronto-parietal connections allow flexible switching between central-executive and default mode networks to modulate external versus internal attention (Spreng et al., 2010).

Dysfunction of the dorsal attention network is implicated in attentional disorders after traumatic brain injury (TBI) (Bonnelle et al., 2011) and in attention deficit and hyperactivity disorders (ADHD), where reduced connectivity correlates with inattention (Sripada et al., 2014). Over-connectivity of this network in autism may cause sensory over-focus (Keehn et al., 2013). Schizophrenia also involves altered prefrontal-parietal coordination during attention tasks (Gold et al., 2007). Dorsal network integrity is thus essential for proper attentional functioning.

Ongoing research aims to further detail the computational mechanisms by which this distributed cortical network controls attention and selects actions. Representational similarity analyses can map how attention modulates information encoded in dorsal network regions (Chadick et al., 2014). Multivariate pattern classification has also shown promise in decoding attentional targets from IPS activity patterns (Bettencourt & Xu, 2016). Combining neuroimaging with neurophysiological studies in animal models will provide further insight into the neural dynamics of flexible attentional control (Womelsdorf & Everling, 2015).

In summary, the dorsal attention system is crucial for top-down executive control of attention necessary for many cognitive tasks. Its distributed cortical network encodes attention priority maps and selects goal-relevant information through recurrent interactions between frontal and parietal regions. Dysfunction of this intrinsically connected system occurs in many neurological and psychiatric conditions. Further research integrating structure, function and computational modeling will continue to elaborate the network mechanisms underlying flexible attentional control in the human brain.

## Ventral Attention Network

The ventral attention network is another intrinsic connectivity network revealed by resting state fMRI analyses (Fox et al., 2006). Anchored in the temporoparietal junction (TPJ) and ventral frontal cortex (VFC), this network shows increased activity during detection of salient stimuli, especially when attention is re-oriented (Corbetta et al., 2008).
While the dorsal attention network controls voluntary top-down attention, the ventral network is specialized for bottom-up attention capture by unexpected or novel stimuli (Kincade et al., 2005). The right TPJ and VFC act as a stimulus-driven attention system that interrupts ongoing cognition and orients attention to salient events (Hutchinson et al., 2009). TPJ lesions lead to attentional neglect, indicating the ventral network's critical role (Karnath & Rorden, 2012).

The ventral network responds strongly to behaviorally relevant stimuli like one's own name versus others' names (Kampe et al., 2003). Unique predictive coding properties in TPJ neurons may allow detection of important deviations from expectations (Krämer et al., 2011). Activity here encodes the level of subjective salience, which may emerge from comparing expectations against inputs (Geng & Mangun, 2011).

When salient events are detected, the ventral network triggers reorienting signals that engage the dorsal network while disengaging default mode regions (Shulman et al., 2009). Reciprocal inhibition between dorsal and ventral networks leads to suppression of the dorsal system by TPJ during attention re-orienting (Wen et al., 2012). The ventral network can thus interrupt and reset ongoing attentional focus.

Ventral network perturbations are implicated in psychiatric disorders with attention abnormalities. Over-reactivity may contribute to sensory disturbances in schizophrenia (Wynn et al., 2005). Hypoactivity may increase distractibility in ADHD (Mazaheri et al., 2010). Autism has also been linked to ventral network dysfunction, potentially underlying sensory sensitivity and problems re-orienting attention (Keehn et al., 2013).

However, the right-lateralized ventral network shows heterogeneity, with some evidence for dissociable sub-networks (Gillebert et al., 2012). Anterior regions may support response selection, while posterior TPJ regulates attention. Diffusion tractography also indicates separable connectivity of TPJ sub-regions (Mars et al., 2012). Ventral network parcellation remains an active research area.

Ongoing work seeks to integrate evidence from lesion patients, neuroimaging, and electrophysiology to elaborate component processes within the ventral attention network (Geng & Vossel, 2013). For example, phase-locking of right TPJ theta oscillations may enable communication with prefrontal regions (Voloh et al., 2015). TMS disruption demonstrates critical roles for both TPJ and VFC in detecting targets at unattended locations (Chica et al., 2011).

Multivariate pattern analyses can further specify representations encoded across ventral network regions during stimulus-driven reorienting (Blankenburg et al., 2010). Linking fMRI connectivity to measurements of neurotransmitters like norepinephrine may also elucidate neuromodulatory mechanisms that amplify responses to motivationally salient events (Hermans et al., 2011).

In summary, the ventral attention network plays a crucial role in detecting behaviorally relevant stimuli that deviate from expectations, triggering reorienting signals that shift attention and update action plans. Interactions of the ventral network with dorsal attention and default mode systems allow dynamic switching between internally and externally focused cognition. Further research on heterogeneous functions within TPJ and VFC and neurotransmitter systems involved in salience processing will deepen understanding of the network dynamics supporting flexible human attention.

### Fronto-parietal network

The fronto-parietal network, also referred to as the central executive network or multiple-demand system, is a distributed set of brain regions involved in cognitive control and flexible task processing (Duncan, 2010). Core nodes include the lateral prefrontal cortex, inferior parietal lobule, anterior insula, and anterior cingulate cortex (Niendam et al., 2012).

First described in task-based fMRI studies, fronto-parietal activations were found across diverse cognitive demands, from working memory and inhibitory control to reasoning and problem solving (Duncan & Owen, 2000). This suggested a domain-general network supporting goal-directed cognition (Fedorenko et al., 2013). Subsequent resting state fMRI analyses confirmed intrinsic fronto-parietal connectivity, which correlates with general intelligence (Cole et al., 2012).

The lateral PFC contributes executive control processes like representation and maintenance of task rules and goals (Miller & Cohen, 2001). The parietal cortex associates sensory inputs with motor plans, a key computation for goal-oriented behavior (Gottlieb, 2007). Coordinated activity between PFC and parietal nodes likely underpins context-specific sensorimotor mappings (Brass & von Cramon, 2004).

The fronto-parietal network interacts with dorsal attention regions including the frontal eye fields and intraparietal sulcus to guide selective attention and working memory (Scolari et al., 2015). It exchanges signals with default mode and salience networks to regulate the balance between external attention and internal cognition (Spreng et al., 2010).

Fronto-parietal communication relies on the superior longitudinal fasciculus white matter tract allowing information integration for complex cognition (Jung & Haier, 2007). Structural and functional connectivity within this tract correlate with fluid intelligence (Li et al., 2009). Developmental increases in long-range fronto-parietal connectivity parallel improved cognitive control abilities (Fair et al., 2007).

Disruption of the fronto-parietal network is found in many psychiatric and neurological disorders. Reduced connectivity occurs in schizophrenia along with cognitive control deficits (Cole et al., 2011). The network shows hyperconnectivity in autism, which may impair flexible switching between tasks (Solomon et al., 2009). Fronto-parietal circuits degenerate in Alzheimer’s disease leading to executive dysfunction (Seeley et al., 2007).

Current research combines neuroimaging with computational modeling to understand information processing mechanisms within the fronto-parietal network. For example, PFC may contextualize representations in posterior cortex by maintaining task sets (Mante et al., 2013). Dynamic coding models propose distributed maintenance of context across the network through synaptic activity traces (Chatham & Badre, 2015).

Oscillatory synchronization may also coordinate fronto-parietal computation. EEG shows phase coupling between frontal and parietal regions during working memory (Palva et al., 2010). Disruption of alphaband synchronization impairs top-down control (Phillips et al., 2014). Computational models demonstrate how oscillatory interactions could implement cognitive control (Helfrich & Knight, 2016).

In summary, the fronto-parietal network comprises a set of higher-order brain areas specialized for flexible goal-directed cognition, cognitive control and adaptive task-processing. Further research integrating fMRI with electrophysiology, connectivity modeling and multimodal imaging will continue elaborating the network’s complex computational mechanisms supporting executive functions.

### Limbic network

The limbic network refers to a set of interconnected brain structures involved in emotion, motivation, learning and memory. Key regions include the amygdala, hippocampus, hypothalamus, thalamus, cingulate cortex and orbitofrontal cortex (Catani et al., 2013).
First recognized in early 20th century anatomical studies, the concept of the limbic system unified these subcortical areas and medial cortical regions (Papez, 1937). While an oversimplification anatomically, limbic network connectivity revealed by modern neuroimaging confirms functional integration of these regions for emotional-cognitive processes (Pessoa, 2008).

The amygdala is considered the emotional core of this network, rapidly detecting emotional salience and tagging memories for storage (Dolan & Vuilleumier, 2003). Reciprocal connections with association cortex, thalamus and hippocampus allow emotional coloring of perceptions and memories (Phelps & LeDoux, 2005).
The hippocampus consolidates explicit memories for facts and events (Squire et al., 2004). It also encodes spatial representations via place cells, supporting navigation (O'Keefe & Dostrovsky, 1971). Hippocampal theta oscillations coordinate memory formation and spatial mapping (Hasselmo, 2005).

Interactions between the amygdala, hippocampus and prefrontal cortex mediate context-dependent fear learning and expression (Maren & Quirk, 2004). The ventromedial PFC integrates affective value for goal-directed behavior (Roy et al., 2012), while the anterior cingulate monitors performance and detects conflict (Shackman et al., 2011).
The hypothalamus and interconnected nuclei regulate homeostatic drives related to feeding, drinking, energy expenditure, and social bonding via neuropeptides (Saper & Lowell, 2014). Orexin/hypocretin neurons promote arousal and motivated behaviors (Thannickal et al., 2007).

Neuroimaging reveals coordinated activity within this distributed limbic network during emotional tasks. Amygdala-prefrontal co-activation occurs when regulating negative affect

(Banks et al., 2007). Viewing negative stimuli synchronizes amygdala-hippocampal theta oscillations measured with MEG (Cornwell et al., 2012).

Functional connectivity MRI also identifies an intrinsic limbic network at rest, with amygdala as a hub interlinking other regions (Roy et al., 2009). Connectivity between the amygdala and mpFC predicts emotion regulation capacity (Kim et al., 2011). White matter pathways including the cingulum bundle and fornix provide anatomical infrastructure for limbic network communication (Von Der Heide et al., 2013).

Dysfunction in limbic regions is implicated in mood and anxiety disorders. Reduced mpFC regulation of the amygdala may contribute to excessive fear in PTSD (Shin et al., 2006). Altered amygdala-prefrontal coupling occurs in depression (Matthews & Harrison, 2012). Hyperconnectivity between the amygdala and anterior insula, a hub of the salience network, is found in generalized anxiety disorder (Andreescu et al., 2014).

Ongoing research aims to relate limbic network disturbances to specific psychiatric symptoms. For example, amygdala hyperactivity to negative stimuli may link to anxious arousal (Blackford & Pine, 2012), while blunted ventral striatal activation could mediate anhedonia in depression (Forbes et al., 2009). Translating connectivity abnormalities to improved diagnosis and targeted treatment remains a challenge.

In summary, the limbic network encompasses interconnected subcortical and medial cortical regions critically involved in emotion, motivation, learning and memory. Advances in human neuroimaging have further elucidated the network dynamics of this system. Dysfunctions in limbic network connectivity likely contribute to mood and anxiety disorders. Further research on linking neuroimaging markers to specific symptoms will enhance understanding of affective disorders.

## Default mode network

The default mode network (DMN) refers to a set of brain regions that routinely deactivate during goal-directed tasks and activate during rest (Raichle et al., 2001). First recognized in PET studies, this pattern was initially dismissed as uninteresting baseline activity. However, fMRI revealed the DMN encompasses interacting hubs including the medial prefrontal cortex (mPFC), posterior cingulate cortex (PCC), inferior parietal lobule (IPL), lateral temporal cortex (LTC), and hippocampal formation (Buckner et al., 2008).

Rather than representing a passive baseline, high metabolic activity of the DMN during rest indicates it supports core functions like emotion regulation, self-referential processing, remembering the past, and envisioning the future (Andrews-Hanna et al., 2014). The DMN shows coherent low-frequency BOLD signal fluctuations during rest, confirming its intrinsic functional connectivity independent of external tasks (Fox et al., 2005).

As an intrinsic connectivity network, the DMN exhibits anti-correlated fluctuations with dorsal attention networks. Attentional engagement suppresses DMN activity, so internal mentation does not interfere with external goal-directed cognition (Kelly et al., 2008). The salience network dynamically controls this switching between DMN-dominated and task-positive states (Goulden et al., 2014).

DMN hubs make distinct computational contributions. The mPFC represents social/emotional aspects of one's identity (Murray et al., 2012). The PCC integrates self-relevant sensations (Leech & Sharp, 2014). The temporoparietal junction encodes mentalizing and theory of mind (Saxe & Kanwisher, 2003). The hippocampus supports autobiographical memory (Spreng & Grady, 2010).

Structural and functional connectivity allows information integration between these specialized hubs. The cingulum bundle directly links the PCC, mPFC and hippocampus (Jones et al., 2013). Resting state functional connectivity predicts offline memory consolidation, highlighting the DMN's role in memory stabilization (Tambini et al., 2010).
Aberrant DMN function is found in psychiatric disorders. Hypoconnectivity occurs in autism and schizophrenia, which may relate to mentalizing deficits (Whitfield-Gabrieli et al., 2009). Hyperconnectivity in depression (Sheline et al., 2010) and PTSD (Bluhm et al., 2009) could reflect excessive rumination on negative memories. The DMN shows disrupted development in ADHD (Sripada et al., 2014).

Current research combines new neuroimaging methods with computational modeling to reveal mechanisms underlying DMN function. For example, multilayer network modeling demonstrates metastable dynamics allowing flexible DMN subsystem interactions (Vidaurre et al., 2018). Dynamic causal modeling can infer directed effective connectivity between hubs during social tasks (Li et al., 2012).

Spatial Independent Component Analysis parcellates the DMN into distinct subsystems (Doucet et al., 2011). High-resolution fMRI mapped seven distinct functional zones within the PCC linked to different processing modes (Leech et al., 2011). Ongoing work on mapping

DMN functional heterogeneity will further elaborate its complex dynamics. As such, the default mode network refers to a set of interacting brain hubs supporting self-referential cognition, social processing and memory consolidation. DMN dysfunction has been linked to numerous psychiatric conditions. Advances in neuroimaging, analytic methods and computational modeling continue to reveal new insights into network mechanisms enabling the DMN's broad contribution to human cognition.

A recent perspective paper by Smallwood et al. (2021) provides additional insight into the DMN by considering its spatial topography and location relative to unimodal sensory and motor cortices. The authors note that DMN regions are functionally interconnected yet maximally distant from primary sensory regions based on gradients of intrinsic functional connectivity (Margulies et al., 2016).
This physical and functional separation from sensorimotor cortex helps explain why the DMN often operates independently of immediate perceptual input (Smallwood et al., 2021). The DMN's transmodal position at the ends of processing streams originating in unimodal cortex also accounts for its diverse roles in higher cognition that integrate distributed neural signals (Mesulam, 1998).

As Smallwood et al. summarize: "In this way, the topography of the DMN explains why it is involved in cognitive states that combine highly abstract features of cognition and that are often only loosely related to the events in the here and now" (Smallwood et al., 2021).

Several examples from recent fMRI studies demonstrate that DMN activity tracks use of prior knowledge rather than immediate perceptual inputs (Smallwood et al., 2021):

- During delayed match to sample tasks, DMN regions activate when decisions depend on memory from the prior trial rather than current sensory cues (Murphy et al., 2018; Konishi et al., 2015).
- Contextual priming of semantic decisions engages the DMN more when multiple cues are provided versus a single cue, even though the same judgement is made (Lanzoni et al., 2020).
- The DMN increases activity once temporary rules are learned in schema-based decision tasks (Vatansever et al., 2017).

As Smallwood et al. explain, “Viewed collectively, these studies show that alterations in DMN activity can reflect how cognition changes during a task” (Smallwood et al., 2021). The DMN is engaged when decisions are informed by prior experience rather than immediate inputs.

Considering the location of the DMN along principal gradients of intrinsic functional connectivity provides a unified framework to understand its contributions to cognition (Smallwood et al., 2021):

> The distributed yet interconnected nature of the DMN allows it to integrate signals from across the cortex and represent global brain states.
>
> Its transmodal position facilitates involvement in relatively abstract mental processes that do not directly map to immediate sensory inputs.
>
> Physical and functional distance from unimodal regions enables modes of cognition decoupled from the external world, such as memory, social cognition, and imaginative thought.

In summary, the topographic perspective on the DMN highlighted by Smallwood et al. accounts for its varied roles in cognition in an appealingly parsimonious manner based on the principles of cortical organization. The DMN's location at the farthest extremes from sensorimotor systems frees its operations from perceptual constraints and provides a mechanism for diverse contributions to internally-focused, memory-guided abstract thought.

This theoretical account about these principles of brain hierarchical organization emerged only over the last decade in the literature in part to the development of novel methodological tool of functional connectivity like the diffusion embedding approach, which I used during my thesis. In the next paragraph, I will review the computational principles of this novel methodological framework (REF).

# Brain hierarchical organization

### Mesulam modal

The perspective that the spatial topography of the default mode network relates to its functional roles provides a foundation for considering broader principles of cortical organization. As Smallwood et al. discuss, the location of the default mode network at the farthest extremes from sensorimotor systems is consistent with Mesulam's influential model regarding how the topology of the cerebral cortex balances segregated and integrative processing (Mesulam, 1998). According to this model, unimodal regions in the sensory and motor cortices support concrete representations tethered to the external world. In contrast, transmodal association areas located farther from the periphery allow more abstract representations unconstrained by immediate perceptual input (Mesulam, 1998). This framework based on the convergence of connectivity from unimodal to transmodal cortex offers an account of the default mode network's distal position and its contributions to internal mentation decoupled from the external environment. Next, I will explain Mesulam's topological model of cortical organization in more detail to shed light on the functional implications of the default mode network's location along principal gradients.

Mesulam's influential model proposes that the cerebral cortex is topologically organized to balance segregated processing in modular networks with integration of information from distributed systems (Mesulam, 1998). This framework helps explain the functional architecture of transmodal regions such as the default mode network located distant from unimodal sensory and motor areas.

According to this model, the topology of cortical connections shapes the nature of neural representations along a gradient from unimodal to transmodal cortex (Mesulam, 1998). Unimodal regions in primary sensory and motor areas support domain-specific processing and concrete representations directly linked to sensory input and motor output (Felleman & Van Essen, 1991). For example, primary visual cortex encodes basic features of retinal stimuli like orientation, spatial frequency, and contrast (Hubel & Wiesel, 1962).
Beyond these areas, association cortices enable more complex and multidimensional representations by converging inputs from lower order regions (Mesulam, 1998). For instance, the ventral visual stream integrates visual features into increasingly complex object representations to support objects recognition (Ungerleider & Haxby, 1994). Association areas also link representations across modalities, such as visuospatial transformations for grasping (Andersen et al., 1997).

At the apex of this processing hierarchy, transmodal areas receive highly processed inputs from multiple modality-specific streams (Mesulam, 1998). By combining diverse multimodal signals, transmodal regions can encode highly abstract, domain-general representations less tied to any single sensory modality (Binder et al., 2009). For example, anterior regions of the temporal lobe integrate object properties and meaning (Patterson et al., 2007).
Transmodal areas are also maximally distant from the unimodal periphery in their connectivity patterns and physical location on the cortical surface (Margulies et al., 2016).

This remoteness from the sensorimotor constraints imposed by the external environment allows additional flexibility and complexity of representations (Buckner & Krienen, 2013).

Transmodal areas can perform elaborative cognitive operations such as semantic integration, memory encoding/retrieval, and mental simulation (Binder et al., 2009).
In particular, the default mode network located at the extreme transmodal end of cortical gradients is a prime example of how distal topology enables complex cognition (Margulies et al., 2016). The default mode network's physical and functional distance from unimodal areas allows representation of mental contents decoupled from the immediate environment, such as self-generated thought, social cognition, and memory (Buckner et al., 2008).

Thus, Mesulam's model implies that the location of a cortical area along the unimodal-transmodal spectrum has major implications for its functional contributions (Mesulam, 1998). Unimodal regions perform modality-specific computations on immediate sensory data. Multimodal association areas enable abstraction by integrating distributed inputs. Maximally transmodal zones facilitate elaboration into even more complex and environmentally detached representations.

This topological arrangement balances local segregation of specialized processing with global integration (Sporns et al., 2004). Local clusters of densely interconnected regions perform distinct computations. For example, visual regions analyze retinal inputs while motor areas plan movements. However, long-range connections also allow collaboration between distant areas to produce multifaceted behaviors (Park & Friston, 2013).

According to Mesulam, the topology of connections linking unimodal to transmodal cortex gives rise to a convergence-divergence hierarchy (Mesulam, 1998). Feedforward connectivity converges from primary regions towards transmodal areas. Top-down feedback divergence propagates in the reverse direction. This anatomy was classically described in the visual system, but acts as a general cortical organizing principle (Felleman & Van Essen, 1991).

Convergent connectivity allows synthesis of information across modalities, providing substrates for abstraction and generalization (Binder et al., 2009). Divergent connections enable contextual modulation of lower level processing based on higher order representations (Gilbert & Li, 2013). Computational models show that convergence and divergence support both generalization and specialized knowledge in deep neural networks (Richards et al., 2019).

Empirical research supports the presence of functional gradients aligned with Mesulam's model. Task fMRI indicates a posterior-to-anterior axis of increasing abstraction in

frontoparietal cortex (Badre & D'Esposito, 2009). Posterior regions respond to concrete sensory stimuli while anterior zones encode abstract rules and relationships.
Resting-state fMRI also reveals gradations in connectivity from unimodal to transmodal areas (Margulies et al., 2016). The principal gradient described by Margulies and colleagues (2016) progresses from sensory-motor to default mode network. This aligns neural similarity with physical proximity.

Intracranial recordings demonstrate increased temporal integration of information along ventral visual pathways (Murray et al., 2014). Later transmodal stages represent longer perceptual timescales suited for abstraction. Anatomical gradients in synaptic plasticity may support this temporal hierarchy (Vogel-Ciernia & Wood, 2014).

In summary, Mesulam's influential framework proposes that cortical topology and connectivity balance localized modular processing with distributed integration (Mesulam, 1998). This organization gives rise to functional gradients from unimodal areas that perform modality-specific computations to transmodal regions that enable more complex, integrative representations (Margulies et al., 2016). The location of the default mode network at the distant transmodal extreme allows great flexibility and abstraction detached from the immediate environment (Buckner & Krienen, 2013). Empirical research provides converging evidence for gradients aligning with Mesulam’s model. Overall, this topographical perspective elucidates the functional contributions of brain regions based on their location within cortex.

Recent advances in neuroimaging data analysis provide new ways to empirically map functional gradients aligned with this model. Specifically, techniques to identify principal axes of variation in resting-state functional connectivity data have revealed continuous gradations in network properties (Margulies et al., 2016).

These functional connectivity gradients extend from unimodal sensory-motor areas to transmodal association zones, recapitulating Mesulam’s topology. The default mode network sits at the abstract transmodal extreme of the principal gradient, consistent with its proposed role in highly integrative cognition (Margulies et al., 2016). Examining the alignment between intrinsic connectivity gradients and functional specialization can shed new light on the topological principles proposed by Mesulam. Next, I will explain how functional connectivity gradients can be calculated from neuroimaging data and interpreted as reflections of the brain's topological organization.

## Diffusion embedding

Functional connectivity gradients are data-driven mappings of continuous variations in resting-state functional connectivity patterns across the cerebral cortex. These gradients align with topological principles of cortical organization, transitioning from unimodal sensory-motor to transmodal association zones (Margulies et al., 2016). Diffusion embedding techniques are used to reduce high-dimensional neuroimaging data into low-dimensional gradient representations.

Diffusion mapping is a nonlinear dimensionality reduction method well-suited for extracting dominant connectivity gradients from functional MRI data (Margulies et al., 2016). This approach models transitions between cortical regions based on similarities of their connection profiles using a random walk process (Margulies et al., 2016). The diffusion mapping algorithm converts regional fMRI time series into a cortex-wide graph, with nodes representing brain regions and edge weights reflecting connectivity strength (Coifman et al., 2005). A Markov matrix describing random walk transitions between nodes is constructed by normalizing graph edge weights. This matrix is decomposed into eigenvectors and eigenvalues describing connectivity gradients (Coifman et al., 2005).

Eigenvectors provide a compact embedding of network topology. The principal eigenvector corresponds to the dominant connectivity gradient. Subsequent eigenvectors capture secondary gradients explaining decrasing variance. Eigenvalue magnitude indicates gradient importance. Diffusion mapping gradients are aligned using Procrustes rotation (Margulies et al., 2016). This technique applied to resting-state fMRI reveals a continuous gradient transitioning from unimodal sensorimotor to transmodal association cortex (Margulies et al., 2016). Primary visual, auditory, somatomotor, and ventral attention networks anchor the unimodal end. The default mode network and frontoparietal control network occupy the transmodal extreme. The principal gradient tracks both physical proximity and functional similarity (Margulies et al., 2016). Unimodal regions cluster are based on functional specialization rather than spatial location. But transmodal areas converge together from distributed loci. Spatial distance and connectivity-based similarity are intrinsically coupled. (Margulies et al., 2016)

Secondary gradients describe additional functional dimensions (Margulies et al., 2016). The second gradient differentiates visual versus somatomotor processes. A posterior-anterior gradient separates representational versus control systems. Later gradients fractionate specific networks.

Diffusion mapping gradients align closely with those from alternative techniques like principal components analysis. Results are reproducible across datasets and robust to methodological variations (Hong et al., 2018). Gradients topography is conserved over development, however the second gradient explains more variance from the connectivity matrix than the first gradient during infancy (Dong et al., 2021).

Empirical studies demonstrate functional relevance of connectivity gradients. The principal gradient tracks graded abstraction during language processing and social cognition (Vogel et al., 2022; Zhang et al., 2022). Functional gradients shift in neurological disorders (Parkes et al., 2022). Neural representations become less specific towards the transmodal end (Planchamp et al., 2022). Computational models show how gradients may emerge from white matter connectivity constraints between areas (Vázquez-Rodríguez et al., 2022). Ongoing work links gradients to cortical dendritic architecture, molecular expression, and genetic patterning (Huntenburg et al., 2018; Richiardi et al., 2022).

In summary, diffusion embedding applied to functional connectivity data reveals continuous gradients reflecting the brain's topological organization (Margulies et al., 2016). The principal gradient extends from unimodal to transmodal cortex, providing an in vivo manifestation of principles described by Mesulam (Mesulam, 1998). Diffusion mapping techniques reduce high-dimensional neuroimaging data into an interpretable low-dimensional embedding while preserving important functional relationships (Coifman et al., 2005).

Bethlehem et al. (2020) and Valk et al., (2023) further expanded this framework in what they call dispersion metrics. Dispersion metrics capture the spread or distribution of nodes within networks in multidimensional gradient space. Specifically, within-network dispersion quantifies the summed squared Euclidean distance between all nodes in a network to that network's centroid (Bethlehem, 2020; Valk 2023). This provides an index of how diverse or dissimilar connectivity profiles are within a given network. Between-network dispersion can also be calculated as the Euclidean distance between network centroids (Bethlehem, 2020; Valk 2023).

These dispersion measures are applied in the gradient space derived from diffusion map embedding. Applying dispersion metrics in gradient space is useful for several reasons. First, gradient space better captures the topology of functional connectivity compared to examining single connections (Bethlehem, 2020). Second, dispersion can be calculated in multidimensional gradient space, providing a more comprehensive characterization of connectivity patterns compared to dispersion in any single gradient. Finally, diffusion embedding gracefully handles noise and sparse connectivity matrices (Margulies et al., 2023), making it well-suited for functional connectivity analyses.

In summary, dispersion measured within gradient space derived from diffusion embedding provides an informative and robust approach to quantify changes in the topology of functional networks. This technique has shed light on age-related changes in network organization (Bethlehem, 2020) and their relevance for cognition (Valk, 2023).

Examining how cognitive processes map onto intrinsic functional gradients can offer new insights into the significance of topological principles for brain function (Vogel et al., 2022). Connectivity gradients provide an invaluable reference frame for exploring transitions between modular specialization and distributed integration. Ongoing work to relate these gradients to other aspects of cortical anatomy and physiology promises continued advances in unraveling the brain's network architecture. As such, I propose to use this methodology to characterize the mindfulness meditation trait and state effect.

Having laid out the methodology utilized in my experimental work, I will now review the growing body of scientific research on neuroimaging and meditation over the past decades. This will encompass key findings from structural imaging studies, which have examined how meditation impacts brain morphology, as well as functional imaging studies using techniques such as fMRI to characterize the effects of meditation on neural activity and connectivity. Establishing this background on the current state of neuroimaging research in meditation will be important for contextualizing the original neuroscientific studies on mindfulness meditation and perception that will be presented in the subsequent sections of this thesis. My aim is to provide a comprehensive synthesis of the structural and functional neural correlates of meditation in order to situate my own experimental research within the broader scientific discourse on unraveling the brain mechanisms underlying contemplative practices.

## Neuroimaging studies of meditation:

Neuroimaging techniques have enabled significant insights into the neural correlates of meditation states or meditation expertise and into the functional and structural brain changes induced by meditation practices.  Structural MRI studies investigate whether the brain's gray and white matter morphology is shaped by the long-term practice of meditation. This line of research aims to map structural changes underlying the transformations in mental faculties and traits cultivated through sustained contemplative practice. Over the long-term, the repeated activation of neural circuits over hours of meditation may induce plastic changes that accrue into stable changes in brain structure. (Fox et al. 2014). This chapter reviews key structural neuroimaging findings regarding cortical thickness, regional volumes, white matter connectivity and their behavioral implications.

## Cross-sectional studies

Several studies have examined the effects of long-term meditation experience on brain morphology and function using structural MRI. Lazar et al. (2005) found that focused attention meditation significantly increased cortical thickness in the right anterior insula and prefrontal cortex (BA 9 and 10). Increases were less significant in the somatosensory, auditory, and visual cortices. Luders et al. (2012)'s cross-sectional study compared experienced meditators (mean ± SD 19.8 ± 11.4 years of experience) to healthy controls and found increased cortical gyrification, maximal in the right anterior dorsal insula, suggesting that mean curvature increased in experienced meditators. Their prediction was based on the hypothesis that group differences and/or correlations would be most pronounced in cortical regions known to increase in volume in meditators, such as the right anterior insula (Lazar et al., 2005; Hölzel et al., 2008). They also found lesser increases in curvatures in the left anterior dorsal insula, left precentral gyrus, right fusiform gyrus, and right cuneus. Alterations are also found in the insula, involved in interoceptive and socio-emotional processing (Fox et al., 2014). This accords with mindfulness meditation honing awareness of internal states. Volume differences in the putamen and caudate linked to attentional control are additional common findings (Hölzel et al., 2011; Murakami et al., 2012). However, the overall pattern across structures remains complex. Overall, findings suggest attentional focus during meditative calm abiding strengthens cortical representation in executive control areas as well as interoceptive brain regions (Tang et al., 2015; Fox et al., 2014). In the most recent meta-analysis, the right anterior ventral insula appears as the most consistent effect across 25 studies on structural effects associated with the practice of mindfulness meditation (Pernet et al 2021). Alongside gray matter, some studies find alterations in white matter connectivity patterns that may support enhanced attentional abilities. For instance, greater integrity of the anterior corona radiata, related to cognitive control, was found in meditators compared to controls (Luders et al. 2011).

Higher fractional anisotropy in the corticospinal tract (Kang et al., 2013), superior longitudinal fasciculus (Luders et al. 2011) and corpus callosum (Luders et al., 2009; Kang et al., 2013) also suggest potential strengthening of sensorimotor and interhemispheric connectivity through long-term practice. However, evidence remains preliminary and the vast majority of this evidence is cross-sectional. Not all findings replicate across studies and methodological constraints persist (Fox et al., 2014). As Van Dam et al. (2018) caution, despite isolated positive reports, no research to date has adequately assessed the relationship between amount of meditation training and structural brain change. Establishing dose-response curves using longitudinal studies remains an important goal.

## Longitudinal studies

In a first step toward address this question, the ReSource Project, which was a 9-month secular mental training program with three different modules (Presence, Affect, and Perspective), investigated the effects on brain cortical morphology. The Presence module focused on cultivating present-moment attention and interoception, similar to mindfulness interventions. This was found to lead to increases in cortical thickness in prefrontal, anterior cingulate, and parietal regions involved in attention, as well as interoceptive regions like the insula. The social intersubjective training modules Affect and Perspective led to dissociable plasticity patterns in frontoinsular regions involved in empathy and inferior frontal and temporal regions involved in perspective-taking. Functional networks related to the specific capacities trained also showed training-related structural changes. However, structural differences are not found universally. A recent study (Kral et. all 2022), the largest and most rigorously controlled study to date, failed to replicate prior findings and found no evidence that MBSR produced neuroplastic changes compared to either control group, either at the whole-brain level or in regions of interest drawn from prior MBSR studies. This provids strong evidence that the short and low dose 8-week meditation intervention such as the standard mindfulness-based interventions are too weak to produce detectable structural changes.

More recently, the Age Well Project (Chetelat et al., 2022) study, which was an 18-month randomized controlled trial, investigated the effects of meditation training on brain structure and function in older adults. 137 cognitively unimpaired, community-dwelling older individuals were randomly assigned to either an 18-month meditation training program, an active control condition of English language classes, or a passive control condition. The primary outcomes were changes in volume and perfusion in the anterior cingulate cortex (ACC) and insula, measured with MRI scans at baseline and 18 months. The results showed no significant differences between the meditation group and either control group in change in ACC or insula volume over 18 months. There were also no significant between-group differences in perfusion changes in these regions. Thus, the primary hypotheses that meditation training would increase ACC and insula volume and perfusion compared to control conditions were not supported.

## Methodological Challenges

A number of outstanding methodological challenges temper structural neuroimaging findings on meditation (Van Dam et al., 2018; Fox et al., 2014). As we saw, most studies are cross-sectional comparisons between meditators and controls at single time-points. Longitudinal designs tracking within-subject structural changes over time would better assess intra-individual plasticity, but they are difficult and often requires large samples to show any effect. The lack of active control groups makes it difficult to separate general motivational factors from meditation-specific cognitive training. Differences in type and life-time hours of practice are often poorly reported or controlled between groups. Cognitive domains linked to observed neural changes also remain underspecified. Finally, the interaction between pre-existing individual differences versus causal effects of meditative training on brain structure needs to be disambiguated.

In summary, structural MRI provides initial evidence that dedicated mental training through meditation practice over years and decades may induce plastic reorganization in gray and white matter morphology. Observed differences in cortical thickness, regional volumes and connectivity align with the cognitive and affective faculties meditation is traditionally described as cultivating. However, findings remain preliminary and establishing clear structural biomarkers of contemplative expertise through rigorously controlled research remains an ongoing challenge. This endeavor to chart how sustained meditative practice sculpts the material foundations of the mind holds exciting potential for unlocking the brain’s capacities for cognitive and emotional self-transformation.

Having reviewed the current state of research on structural brain changes associated with meditation expertise and training, I will now turn to findings from functional neuroimaging studies using techniques such as fMRI to examine the impacts of meditation on functional connectivity. Functional connectivity analyzed via resting-state fMRI provides insight into how interactions within and between large-scale brain networks involved in cognitive and emotional processing are altered as a function of meditation practice. This line of research has revealed differences in network connectivity patterns when comparing long-term meditators and novices, as well as changes in functional connectivity following mindfulness training, and in psychiatric context following MBI. Synthesizing results from the growing body of functional connectivity meditation research will further elucidate the neurophysiological mechanisms underlying the cognitive and affective transformations cultivated through contemplative practices. Establishing this background will situate the original resting-state fMRI studies on functional connectivity changes following mindfulness training that I will present in later sections.

## Meditation trait

Trait mindfulness refers to an individual's innate tendency towards mindful states and behaviors. Recent research has begun to investigate how differences in trait mindfulness manifest in the brain's resting state functional connectivity. The resting state refers to intrinsic brain activity occurring when an individual is at rest and not engaged in any explicit task. During rest, the brain exhibits synchronized low-frequency fluctuations that reveal functional connections between distributed regions. Several resting state networks show reliable patterns of functional connectivity, including the default mode network (DMN), salience network (SN), and frontoparietal network (FPN) (Uddin et al., 2019). Variations in the strength of connectivity within and between these networks has been linked to differences in cognitive processes and behavioral tendencies relevant to mindfulness (Brewer et al., 2011).

A growing number of studies have now examined how individual differences in trait mindfulness correlate with or predict patterns of resting state functional connectivity. This research suggests that higher trait mindfulness is linked to a distinct configuration of the DMN, SN, and FPN. However, findings have sometimes varied across studies, likely reflecting differences in how mindfulness was operationalized and measured. Broadly speaking, two approaches have been taken: 1) correlating questionnaire measures of trait mindfulness with network connectivity in meditation-naïve individuals (Bilevicius et al., 2018; Parkinson et al., 2019), and 2) comparing intrinsic network connectivity between long-term mindfulness meditation practitioners and meditation-naïve controls (Bauer et al., 2019; Froeliger et al., 2012). Each approach provides complementary insights into the neural bases of trait mindfulness.

## Questionnaire Studies in Meditation-Naïve Individuals

A common method has been to correlate trait mindfulness scores from self-report questionnaires with resting state connectivity patterns in individuals without formal meditation experience. For example, Bilevicius et al. (2018) administered the Mindful Attention Awareness Scale (MAAS) and examined correlations with connectivity within and between the DMN, SN, and FPNs using independent components analysis (ICA). The MAAS focuses specifically on attentional aspects of mindfulness. Bilevicius and colleagues found that higher MAAS scores were linked to decreased functional connectivity between the SN and right cuneus, a region of the default mode network involved in visual processing (Beason-Held et al., 1998). Additionally, the DMN showed reduced connectivity with the left middle frontal gyrus (MFG) and left superior temporal gyrus (STG) at higher levels of trait mindfulness. The STG has been implicated in visuotemporal attention (Shapiro et al., 2002), suggesting a decoupling of DMN regions linked to attentional processes. Higher MAAS scores also predicted increased functional connectivity between the SN and left insula, a core node of this salience network. Finally, elevated MAAS scores were associated with greater connectivity between the DMN and right MFG. The right MFG has been posited to act as a gateway that facilitates switching between dorsal fronto-parietal and ventral attention networks, which is critical for orienting attention (Corbetta et al., 2008). Taken together, these results point to a distinct modulation of DMN and salience network connectivity related to the attentional aspect of mindfulness as measured by the MAAS (Bilevicius et al., 2018).

In another questionnaire study, Parkinson et al. (2019) examined associations between the Five Facet Mindfulness Questionnaire (FFMQ) and intrinsic network connectivity. The FFMQ includes five subscales: Observing, Describing, Acting with Awareness, Non-judging of Inner Experience, and Non-reactivity to Inner Experience. Using ICA, Parkinson and colleagues found that overall FFMQ scores correlated positively with connectivity between the left insula and an attentional network component, as well as connectivity between the SN and the cuneus. Specific facets also showed distinct connectivity patterns. The Observing subscale predicted greater connectivity between the insula and attentional network, while the

Describing subscale was linked to reduced connectivity between the superior temporal gyrus (STG) and the DMN, SN, and FPNs. The Non-reactivity subscale was associated with weaker STG-FPN connectivity. Notably, the Non-judging facet correlated with increased connectivity between the mid-cingulate cortex and DMN. The mid-cingulate is considered part of the salience network and is involved in processing affective and cognitive information (Uddin et al., 2017). These facet-specific effects highlight how different dimensions of mindfulness relate to distinct resting state connectivity signatures, even within a single sample.

In summary, questionnaire research suggests that trait mindfulness is linked to a decoupling between nodes of the DMN, especially the cuneus and STG regions, and other networks including the SN and FPNs (Bilevicius et al., 2018; Parkinson et al., 2019). Individuals higher in trait mindfulness also exhibit elevated within-network connectivity in the salience network, particularly between the insula and other nodes (Bilevicius et al., 2018; Parkinson et al., 2019). Evidence regarding the DMN itself is mixed, with both increased and decreased connectivity reported between DMN nodes like the middle frontal gyrus and the cuneus. This may reflect differences in the specific aspect of mindfulness measured by the MAAS versus FFMQ. Nevertheless, these studies converge in pointing to a functionally distinct Triple Network configuration associated with mindful traits.

## Comparisons of Long-Term Meditators and Meditation-Naïve Individuals

Rather than using questionnaires as trait mindfulness proxies, another approach has been to compare intrinsic network connectivity directly between experienced meditators and meditation-naïve controls. The assumption is that long-term practitioners will exhibit heightened trait mindfulness as a consequence of their sustained training. For example, Bauer et al. (2019) contrasted resting state networks between meditators with an average of over 1600 hours of practice in Vipassanā meditation and meditation-naïve individuals. Meditators showed reduced DMN connectivity to the left superior frontal gyrus (SFG), right MFG, inferior parietal lobe (IPL), and STG compared to novices (Bauer et al., 2019). Functional connectivity between the DMN and fronto-parietal network was also decreased in meditators. Additionally, mPFC-MFG connectivity was negatively associated with the amount of meditation experience across groups. This pattern suggests that trait mindfulness honed through extensive practice is linked to reduced within-DMN connectivity and decoupling between the DMN and executive networks.

Similarly, Froeliger et al. (2012) found increased within-network connectivity in the dorsal attention network (DAN), as well as greater connectivity between the DAN and DMN and between the FPN and SN, when comparing experienced meditators to novices. The strength of connectivity between the mPFC and MFG was positively correlated with meditation experience. Thus, in line with the questionnaire research, comparisons of long-term practitioners and meditation-naïve individuals imply that enhanced trait mindfulness is associated with elevated within-network integrity but reduced between-network connectivity involving the DMN (Bauer et al., 2019; Froeliger et al., 2012).

## Proposed Cognitive Implications

The distinct resting state functional connectivity signature tied to trait mindfulness raises questions about the related cognitive processes and mechanisms. Though reverse inferences should be made cautiously, prevailing accounts of the DMN, SN, and FPN allow some tentative hypotheses (Poldrack, 2006). First, the reduced connectivity observed between SN regions and the cuneus, a key node of the DMN, could indicate that individuals with greater trait mindfulness are less prone to self-referential thought and mind wandering (Bilevicius et al., 2018; Parkinson et al., 2019). The cuneus has been linked specifically to internally-directed attention (Benedek et al., 2016). Thus, decoupling from salience network regions critical for switching between internal and external focus (Sridharan et al., 2008) may facilitate present-moment awareness and experiential acceptance, core features of mindfulness (Doll et al., 2015).

Second, increased coupling between posterior cingulate cortex (PCC) of the DMN and dorsolateral PFC (dlPFC) of the FPN could reflect improved attentional control related to trait mindfulness (Creswell et al., 2016; King et al., 2016; Kral et al., 2019). The PCC is involved in autobiographical memory, planning, and self-reflection, while dlPFC regions contribute to goal-oriented cognition like attentional control (Davey et al., 2016; Ptak, 2012). Their increased connectivity may support training the mind away from self-focused rumination and towards present-centered awareness, mediated by strengthening of top-down attentional systems (Kral et al., 2019; van Vugt et al., 2012). This DMN-FPN link could alternatively be explained by the subsystem of the FPN connected to the DMN modulating connectivity via this “bridge” (Dixon et al., 2018).

Third and finally, mindfulness training has been found to increase connectivity between the dorsal anterior cingulate (dACC) and other salience network nodes like the anterior insula (Bilevicius et al., 2018; Parkinson et al., 2019; Su et al., 2016). The dACC is considered important for cognitive control and attention regulation (Bush et al., 2000). Thus, people with intrinsically higher trait mindfulness may exhibit strengthened within-SN connectivity that facilitates attentional control and insulating ongoing cognitive processes from distraction (Benedek et al., 2016).

In summary, the unique Triple Network configuration tied to trait mindfulness seems to reflect enhanced attentional control and present-focused awareness, along with reduced self-referential thought and mind wandering. While reverse inferences based on resting state connectivity must be made cautiously, the observed patterns fit theoretical accounts of mindfulness. Further research is still needed to more systematically test the psychological implications of these network-level neural traits.

Despite emerging consistencies, some discrepancies in the literature remain. For example, while Bilevicius et al. (2018) found that trait mindfulness correlated with decreased DMN connectivity to the right MFG, Bauer et al. (2019) reported reduced connectivity between the mPFC and right MFG in meditators compared to novices. Furthermore, Bilevicius and colleagues observed lower functional connectivity between the DMN and insula at higher mindfulness levels, whereas Parkinson et al. (2019) identified increased connectivity between the insula and SN that correlated with overall trait mindfulness and specific facets like Observing.

Several factors may contribute to these inconsistencies. First, differences in the particular aspect of mindfulness measured likely play a key role. The MAAS emphasizes attentional components, while the FFMQ is intended to capture a broader conceptualization of mindfulness including facets like Non-judging (Bergomi et al., 2013). Second, sample characteristics could moderate effects. Most studies have relied on relatively small sample sizes of less than 50 participants. Larger cohorts that increase representativeness are important for stable estimates of brain-behavior relationships. Third, variations in the resting state analytical approach could contribute to divergent connectivity patterns, an issue considered further below. Finally, an inherent limitation for all individual difference studies is that they cannot determine causality. Associations between trait mindfulness and intrinsic connectivity may reflect consequences as well as antecedents of mindful capacities (Fox et al., 2014).

Interventions manipulating mindfulness through training could complement these correlational findings and help establish causal connectivity changes. However, most mindfulness training studies have focused on task-based fMRI rather than examining the resting state. Just two published studies have evaluated resting state effects of mindfulness interventions without an active control group (Doll et al., 2015; Yang et al., 2016). These uncontrolled pre-post designs found reduced connectivity between the DMN and SN after brief mindfulness treatments. But only rigorously controlled randomized trials can demonstrate intervention-specific connectivity changes. Other studies have included active control groups but were underpowered, with 12 to 20 participants per arm (Creswell et al., 2016; King et al., 2016; Kral et al., 2019; Lifshitz et al., 2019; Taren et al., 2015). Larger randomized controlled trials are critical for clarifying the causal impact of mindfulness training on Triple Network connectivity and the extent to which induced plasticity mirrors intrinsic differences in trait mindfulness.

Beyond these conceptual issues, the research linking individual differences in trait mindfulness to resting state networks has methodological limitations. Small sample sizes in many studies make subtle connectivity effects hard to detect reliably. Emotion regulation(Bush et al., 2000; Stevens, 2011). The rACC is further divided into pgACC and sgACC (Stevens, 2011). Extensive cytoarchitectural, lesion and neuroimaging evidence implicates a functional distinction in the ACC wherein the dorsal component, dACC, is related to regulation of cognitive processes, and the rostral component, rACC, in regulating emotional processes (Bush et al., 2000; Stevens, 2011). The dACC, a key node of the SN, is linked to increased connectivity with the SN in mindful individuals (Bilevicius et al., 2018; Parkinson et al., 2019). dACC-SN connectivity is also increased in chronic-pain afflicted individuals after mindfulness training compared to controls (Su et al., 2016). Dorsal ACC additionally has been implicated in pain processing and the anticipation of pain (Vogt, 2005), which suggests a possible mechanism for mindfulness meditation training on altered functional connectivity in chronic pain (Su et al., 2016) and pain relief by mindfulness.

Apart from the physical and cognitive aspects of pain, the emotional aspect of pain is thought to be processed, in part, in the rACC (Etkin et al., 2011). Relatedly, mindfulness training-mediated modulations of the rACC have shown antidepressant effects (Yang et al., 2016). As aberrant sgACC-DMN hyperconnectivity has been observed in depression (Connolly et al., 2013; Drevets et al., 2008), it has been speculated that decreased connectivity in these regions mediated by mindfulness training may relate to an antidepressant effect (Yang et al., 2016). Additionally, in the pgACC, the modulation of functional connectivity through mindfulness training was related to increased resilience measured through the Resilience Quotient Test (Kwak et al., 2019), a measure of protection against psychiatric conditions. Indeed, regarding mindfulness-mediated rsFC modulations of the affective component of the ACC – the rACC, findings unveil decreased rACC-amygdala connectivity (Taren et al., 2017) and increased rACC-dmPFC connectivity (Kilpatrick et al., 2011; Kwak et al., 2019). As those pathways are evidenced to play major roles in emotion regulation (Zotev et al., 2013), dysfunctions are frequently linked to mood disorders. It is indeed notable that cognitive control of emotions mediated by prefrontal cortices coupling to rACC are aberrant in MDD (Disner et al., 2011), and connectivity dysfunctions of rACC and the amygdala component of limbic circuitry are prevalent in mood disorders (Alexander et al., 2020; Connolly et al., 2013; Hakamata et al., 2020; Kamphausen et al., 2012; Scharnowski et al., 2020). All this illustrates that there could a be modulation of corticolimbic circuitry mediated by mindfulness, resulting in a regulation of emotional processes. However, neuroanatomical specificity in the rACC needs to be further defined as a study highlighted functional decoupling between the dmPFC and the pgACC subregion of the rACC (Yang et al., 2016). Overall, even though further studies should highlight clearer neural signatures of emotion regulation mediated by mindfulness, there seems to be an important role played by mindfulness in neuronal plasticity regarding affect monitoring.

In summary, these studies suggest that mindfulness is related to the modulation of the functional connectivity of several large-scale brain networks implicated in attention control, self-awareness, pain processing and emotion regulation, providing a neural understanding of underlying mechanisms. I hypothesized that the modulation of large-scale brain networks through mindfulness practice may lead to modifications in the cognitive architecture and hierarchy of the brain and that I would find the same pattern in my Study 1. Notably, even though the neuroscience of mindfulness is in its nascency, several neural patterns emerge that may relate to core psychological features of mindfulness training. The self-awareness component of mindfulness may be related to cuneus/DMN-SN decoupling and the attention regulation component of mindfulness may relate to PCC/DMN and dlPFC/FPN coupling. Together, these mindfulness-related effects suggest an overall reconfiguration of DMN connectivity with the SN and FPN. The ACC may play a unique functional connectivity role mediated by mindfulness. Indeed, mindfulness is associated with modulation of dmPFC-rACC-amygdala circuitry with increased rACC-dmPFC connectivity and decreased rACC-amygdala connectivity. These mindfulness-induced changes in the connectivity of the affective region of ACC – the rACC – may relate to improved emotional processing. Finally, cognitive aspects of emotion processing, and pain processing in particular, involve dACC-SN coupling, or in other words, increased within-SN coupling. In this sense, mindfulness seems to increase connectivity within the SN, a network implicated in salient event processing, which may help practitioners re-evaluate pain and associated negative affective processing. Overall, these neural signatures could help inform our understanding of therapeutic effects mediated by mindfulness that I will describe now.

## MBI

As we saw, Mindfulness-based interventions (MBIs) refer to psychological programs that train core mindfulness skills and techniques as an approach for treating clinical conditions or improving wellbeing. The most prominent and standardized mindfulness intervention is Mindfulness-Based Stress Reduction (MBSR), an 8-week group program teaching formal and informal mindfulness practices (Kabat-Zinn, 1990). Other examples include Mindfulness-Based Cognitive Therapy (MBCT), which integrates mindfulness skills with traditional cognitive therapy, and Acceptance and Commitment Therapy (ACT) programs incorporating mindfulness of present experience and values-oriented behavioral change strategies.

Mindfulness-based interventions (MBIs) like mindfulness meditation training and mindfulness-based cognitive therapy have become increasingly popular as complementary approaches for treating clinical conditions and improving wellbeing. A growing body of research has started examining how these interventions targeting mindfulness skills affect intrinsic functional connectivity of large-scale brain networks during resting state. Characterizing the effects of MBIs on resting state connectivity can help elucidate underlying neurobiological mechanisms and advance models of mindfulness-based therapeutic change.

Prevailing theoretical frameworks propose that mindfulness training reshapes the dynamics and functionality of core neurocognitive networks involved in emotion regulation, attentional control, and self-awareness (Hölzel et al., 2011; Vago & Silbersweig, 2012). In particular, the default mode network (DMN), salience network (SN), and frontoparietal network (FPN) have been most consistently linked to mindfulness-related modulation of their resting state functional connectivity.

Overall, current evidence from neuroimaging studies indicates MBIs may strengthen functional integration between attentional control (FPN) and self-referential (DMN) systems while reducing connectivity within emotion appraisal (SN) circuitry. This reconfiguration of intrinsic connectivity dynamics between DMN, FPN, and SN nodes provides a preliminary framework for understanding the improvements in psychiatric symptoms and wellbeing linked to mindfulness training. However, substantial methodological limitations regarding study design, specificity of findings to mindfulness, and inadequate controls temper conclusions about causal effects of MBIs on resting state functional connectivity.

## Modulation of Between-Network Interactions

Among intervention studies examining MBIs in patient populations, one of the most consistent effects has been increased resting state functional connectivity between nodes of the DMN and FPN. For example, King et al. (2016) observed increased connectivity between posterior cingulate cortex (PCC) of the DMN and dorsolateral PFC of the FPN following mindfulness-based exposure therapy versus control therapy in combat veterans with PTSD.

Creswell et al. (2016) similarly found that a brief 3-day mindfulness meditation retreat increased PCC-dlPFC connectivity compared to relaxation training in stressed unemployed adults. In patients with major depressive disorder, Lifshitz et al. (2019) reported that 2 weeks of mindfulness training strengthened resting connectivity between the PCC and a region of ventromedial PFC linked to the DMN.

Connectivity increases between PCC and medial temporal lobe nodes of the DMN have also been observed following MBIs in patients with Parkinson's disease (Moore et al., 2020). The PCC and interconnected cortical midline structures play key roles in internally-directed cognition (Leech et al., 2011), while lateral PFC regions are important for attentional control (Miller & Cohen, 2001).

Thus, increased coupling of these DMN and FPN nodes following MBIs may reflect improved attentional regulation of self-referential mental activity and mind wandering (King et al., 2016; van Vugt et al., 2012). This functional integration of brain systems involved in attentional control and internal cognition represents a potential neural mechanism for therapeutic improvements linked to mindfulness training.

Another relatively consistent finding has been decreased intrinsic connectivity between key nodes of the SN after MBIs. For example, in the aforementioned study of combat veterans with PTSD, King et al. (2016) found that mindfulness-based exposure therapy reduced resting connectivity between the amygdala and subgenual anterior cingulate cortex (sgACC) compared to a control intervention.

Similarly, Taren et al. (2015) reported that a 3-day intensive mindfulness meditation retreat weakened amygdala-sgACC connectivity relative to relaxation training in chronically stressed unemployed adults. The amygdala and interconnected paralimbic cortical regions play central roles in threat, fear, and emotion processing (LeDoux, 2003).

The sgACC in particular has been identified as a regulatory nexus that can modulate amygdala reactivity (Gee et al., 2013). Thus, the observed decoupling between these SN nodes following MBIs may indicate improved regulation of negative emotional responses, which could underlie therapeutic improvements in conditions like PTSD and mood disorders characterized by emotional dysregulation (King et al., 2016).

In addition to modulating between-network interactions, MBIs appear to alter intrinsic connectivity patterns within the DMN itself. For example, Gu et al. (2019) found that MBCT strengthens integration among anterior and posterior DMN subsystems in patients with residual depression. Increased within-DMN connectivity has also been observed following mindfulness training in combat veterans with PTSD (King et al., 2016) and patients with Parkinson's disease (Moore et al., 2020).

This enhanced network integration may facilitate metacognitive forms of self-reflection that support therapeutic benefits linked to mindfulness (Vago & Silbersweig, 2012). However, these within-DMN effects have been more variable across studies compared to increased DMN-FPN and decreased SN connectivity.

## Relation to Therapeutic Effects

Importantly, some studies have linked resting connectivity changes following MBIs to improvements in clinical symptoms or wellbeing, which provides preliminary evidence that network modulations may mediate therapeutic effects. For example, in unemployed adults with high stress, the increases in PCC-dlPFC connectivity following mindfulness training were associated with reduced interleukin-6, a biomarker of inflammatory stress responses (Creswell et al., 2016).

In patients with residual depressive symptoms, increased integration among DMN subsystems following MBCT was correlated with longitudinal reductions in depression (Gu et al., 2019). Furthermore, combat veterans with PTSD who exhibited the greatest increases in DMN-FPN connectivity following mindfulness-based therapy showed the largest decreases in avoidant and hyperarousal symptoms (King et al., 2016).

Finally, in patients with Parkinson's disease, strengthened connectivity between DMN and FPN nodes after mindfulness training was linked to self-reported improvements in overall quality of life (Moore et al., 2020). Although preliminary, these findings suggest network connectivity changes may be functionally important for health-related outcomes.

However, not all studies examining symptom improvement have found correlations with specific connectivity changes. This may be due in part to substantial differences in methodology across studies assessing clinical outcomes. Nevertheless, taken together, current evidence indicates MBIs can alter functional connections within and between brain networks involved in self-referential, attentional, and emotional processes – changes that appear related to mindfulness-based therapeutic effects.

Despite these suggestive findings, considerable limitations temper causal inferences about the effects of MBIs on resting state functional connectivity. Many studies lacked active control groups, had small sample sizes, or focused exclusively on short-term effects post-intervention rather than long-term maintenance of connectivity changes. The field currently lacks large randomized controlled trials methodologically equipped to make robust conclusions about causal effects of MBIs on brain functional connectivity.

Additionally, MBIs are complex, multi-component interventions, and findings have not always dissociated connectivity changes specific to mindfulness training from other elements like cognitive reappraisal skills or social support. Control groups providing equivalent time, attention, and expectation of benefits are needed to isolate mediation effects associated with mindfulness itself. Studies are also often underpowered for assessing individual differences like varying levels of mindfulness practice.

Furthermore, considerable heterogeneity exists in the type and duration of MBIs implemented across studies, ranging from 8-week MBSR programs to 3-day intensive retreats. Future research directly comparing different “doses” and delivery methods will be essential for delineating mindfulness-specific neural mechanisms. Longitudinal assessments can also examine trajectories and maintenance of connectivity changes.

Finally, the majority of studies have involved seed-based correlations between select network nodes rather than assessing whole-brain connectivity patterns. Data-driven methods like independent components analysis can complement seed-based approaches by providing a more comprehensive, unbiased mapping of functional networks. Multimodal neuroimaging integrating resting fMRI with structural, diffusion, and spectroscopy measures can enrich understanding of tissue-level plasticity underlying mindfulness training effects.
In summary, current evidence suggests MBIs may produce clinically relevant shifts in intrinsic functional connections within and between core neurocognitive networks that support attention regulation, emotion regulation, and self-referential processing. However, substantial methodological limitations temper the ability to make definitive causal conclusions about mindfulness-specific effects on functional brain connectivity at this stage.

Looking forward, rigorously designed randomized controlled trials using active comparison groups, larger samples, and multivariate imaging approaches are critically needed to advance mechanistic models of how mindfulness training impacts brain functional dynamics and mental health.

I will now turn to discuss the impact of longitudinal mindfulness meditation training interventions on functional connectivity patterns in the brain. This area of research is highly relevant for Study 3 of my dissertation, in which I investigated changes in resting-state networks following an 8-week mindfulness training program. As we will see, longitudinal training studies provide critical insights into how functional connectivity is shaped over time through repeated practice of mindfulness techniques. Elucidating the brain mechanisms underlying these dynamic network changes will further our understanding of how mindfulness sculpts cognition through the remodeling of large-scale neurocognitive circuits.

## Meditation training

A growing body of research has investigated how undergoing training in mindfulness meditation impacts resting state functional connectivity patterns in the brain. As opposed to trait mindfulness, mindfulness meditation involves techniques and practices aimed at cultivating mindful skills and states. When practiced repeatedly over time, mindfulness meditation is theorized to induce neuroplastic changes that reshape the functionality and connectivity of large-scale brain networks relevant to attention, interoception, meta-awareness, and emotion regulation (Tang et al., 2015).

While early neuroimaging studies focused on task-based changes, there has been increasing interest in characterizing the effects of mindfulness training on intrinsic connectivity during rest. Resting state networks show synchronized low frequency fluctuations between distributed brain regions in the absence of external tasks. Well-characterized networks exhibit reliable connectivity patterns, including the default mode network (DMN), salience network (SN), and frontoparietal network (FPN) (Buckner et al., 2008; Seeley et al., 2007). These networks contribute to cognitive capacities theoretically shaped by mindfulness, such as attentional control, self-monitoring, and emotional processing (Hölzel et al., 2011).

A series of studies have now probed changes in resting state functional connectivity following interventions training novice participants in mindfulness meditation skills and techniques. While results have varied, likely reflecting differences in methodological approaches and specific meditation practices trained, some consistent network-level effects have emerged. Broadly speaking, mindfulness meditation appears to increase connectivity within salient and attentional networks while reducing connectivity within default mode regions and between the DMN and executive control networks. This reconfiguration of the Triple Network system provides preliminary neurobiological models for how mindfulness meditation may enhance attention regulation, meta-awareness, and emotion regulation capacities (Tang et al., 2015).

## Connectivity Changes Within and Between the Triple Networks

Several studies have reported increased connectivity within core hubs of the salience network following mindfulness meditation training. For example, Kilpatrick et al. (2011) examined participants before and after 8 weeks of Mindfulness-Based Stress Reduction (MBSR) training versus a waitlist control. MBSR is a standardized program teaching a variety of mindfulness skills and techniques. Using independent component analysis (ICA), Kilpatrick and colleagues found increased connectivity between the dorsal medial prefrontal cortex (dmPFC) and rostral anterior cingulate cortex (rACC) within a composite salience/auditory network after MBSR training (Kilpatrick et al., 2011).

Similarly, Bilevicius et al. (2018) observed that higher trait mindfulness correlated with elevated connectivity between the left insula and the salience network in individuals without formal meditation experience. Additionally, Su and colleagues (2016) reported increased connectivity between the anterior insula and dorsal ACC nodes of the salience network in patients with chronic pain after undergoing MBSR training compared to healthy controls. As these core network hubs are thought to support attentional control and the processing of salient events, strengthened within-network communication may represent a key neural mechanism for improvements in attentional capacities following mindfulness meditation (Seeley et al., 2007; Uddin, 2015).

Along with augmenting salience network connectivity, mindfulness meditation appears to reconfigure connectivity between the DMN, SN, and FPN. For instance, Doll et al. (2015) found that brief daily mindfulness meditation delivered via smartphone app led to decreased connectivity between the insula node of the SN and the posterior "core" of the DMN after two weeks of training. Taren et al. (2015) also reported reduced connectivity between the amygdala and subgenual ACC, a limbic node of the SN, in participants undergoing a 3-day intensive mindfulness retreat compared to relaxation training.

These findings converge with observed correlations between trait mindfulness and intrinsic network connectivity patterns, including decreased SN-DMN connectivity (Bilevicius et al., 2018). Together they indicate a decoupling between salience network regions and key hubs within the default mode system following mindfulness training. This may reflect strengthened attentional control facilitated by the SN over internally-oriented DMN processes.

In addition, several studies have reported increased connectivity between core nodes of the DMN and FPN after mindfulness meditation training. For example, King et al. (2016) found increased connectivity between posterior cingulate cortex (PCC) of the DMN and dorsolateral PFC (dlPFC) of the FPN in combat veterans undergoing mindfulness-based exposure therapy compared to a control therapy program. Similarly, Creswell et al. (2016) observed increased PCC-dlPFC connectivity in stressed unemployed adults after a 3-day intensive mindfulness retreat relative to relaxation training.

Kral et al. (2019) also demonstrated increased connectivity between PCC and bilateral dlPFC over the course of 8 weeks of MBSR training in comparison to active and passive control groups. Notably, PCC-dlPFC connectivity increases correlated with reduced mind-wandering assessed via experience sampling. As the PCC is linked to self-referential processing (Leech et al., 2011) while the dlPFC contributes to attentional control (Miller and

Cohen, 2001), increased coupling between these DMN and FPN nodes may point to improved attentional regulation of internally-oriented cognition following mindfulness training (van Vugt et al., 2012).

## Regional Effects Within Specific Networks

Beyond network interactions, mindfulness meditation training appears to alter functional connectivity within particular nodes and subsystems of the Triple Network model. For instance, consistent reductions in connectivity have been observed between the SN and posterior cortical midline structures after mindfulness interventions. This includes decreased connectivity between the cuneus, a region of the DMN, and a) the salience network in its entirety (Bilevicius et al., 2018), and b) the dorsal ACC node of the SN specifically (Yang et al., 2016).

The cuneus is involved in basic visual processing and potentially plays a role in internally-directed attention and visual mental imagery (Benedek et al., 2016; Beason-Held et al., 1998). Thus, decoupling between the cuneus and salience network after mindfulness meditation aligns with the theoretical premise that mindfulness reduces self-referential narrative thought and mind-wandering (Doll et al., 2015). This effect parallels findings linking trait mindfulness to reduced cuneus-SN connectivity (Bilevicius et al., 2018).

Within the DMN, mindfulness meditation appears to decrease connectivity of the rACC, specifically its pregenual subdivision (pgACC), to midline posterior cortical structures. For example, Yang et al. (2016) reported reduced connectivity between pgACC and the PCC, precuneus, and medial temporal lobes after a 40-day mindfulness-based mental training program. Similarly, Kwak et al. (2019) observed decreased connectivity between pgACC and PCC after an intensive 4-day mindfulness retreat compared to relaxation training. The pgACC is thought to support emotion regulation and visceromotor control processes (Stevens et al., 2011). Thus, its decoupling from key DMN nodes may reflect improved regulation of self-focused emotional thought following mindfulness training.

Finally, several studies have pointed to training-related increases in connectivity between the dorsal ACC and both salience network and default mode regions (Kwak et al., 2019; Yang et al., 2016). The dACC is strongly implicated in attention control and cognitive regulation (Bush et al., 2000). For example, Yang et al. (2016) found increased dACC-PCC connectivity after mindfulness training, while Kwak et al. (2019) reported strengthened connectivity between the dACC and medial temporal lobes of the DMN after a mindfulness retreat. These DMN effects may indicate improved attentional control of memory-based self-referential processing. Kwak and colleagues also observed increased connectivity between dACC and the anterior insula, suggesting simultaneous strengthening of cognitive control within the SN.

Through repetitive mindfulness practice, enhanced attention control mediated by dACC may become generalized across salient, attentional, and default mode networks. As a result, DMN-based self-focused thought may become more readily regulated, consistent with therapeutic models of mindfulness (Hasenkamp et al., 2012).

## Methodological Limitations

While these studies provide preliminary evidence that mindfulness training reconfigures resting state connectivity within and between the Triple Networks, several limitations should be highlighted. First, the research has been characterized by heterogeneity in study design and analytical methodology. Mindfulness interventions have included multi-week structured programs like MBSR (Kilpatrick et al., 2011; Kral et al., 2019), brief smartphone-based trainings (Doll et al., 2015), and intensive retreats (Creswell et al., 2016; Kwak et al., 2019), along with varying levels of methodological rigor. Analytically, both seed-based and ICA approaches have been employed, each with their own strengths and weaknesses (Lee et al., 2013). This diversity across studies has likely contributed to some inconsistencies in specific connectivity findings.

Second, sample sizes in mindfulness intervention studies have generally been quite modest, often with 10-20 participants per group. These small samples provide low statistical power for detecting real effects, while increasing the probability that significant findings reflect spurious associations (Button et al., 2013). Large, pre-registered randomized controlled trials are imperative for robust tests of mindfulness training effects on intrinsic connectivity.

Third, most longitudinal mindfulness training studies have lacked an active control group. Comparing meditation training only to waitlist or no-training groups provides insufficient experimental control. Active control interventions that are structurally matched but lack the hypothesized active ingredients of mindfulness, such as health education or relaxation training, are necessary to rule out non-specific effects like expectancy biases or social support (MacCoon et al., 2012). Just a few resting state studies have included active control conditions so far (Creswell et al., 2016; Kwak et al., 2019).

Fourth, the field currently lacks consensus on a theoretical framework for selecting seed regions and networks to investigate in relation to mindfulness. This has led to variability in chosen seed regions across studies. For example, the PCC, vmPFC, and pgACC have all been used to probe the DMN, while connectivity of the dlPFC or inferior frontal gyrus has been assessed to characterize the FPN. Standardizing seeds and networks of interest would benefit cumulative knowledge development. Emerging taxonomies provide useful guidance, particularly the six-network parcellation proposed by Uddin and colleagues (Uddin et al., 2019).

Finally, reverse inferences about the psychological implications of observed intrinsic network connectivity changes should be made cautiously (Poldrack, 2006). While prevailing theoretical accounts allow hypotheses about how Triple Network reconfigurations may support attention and emotion regulation processes, these brain-behavior links require direct empirical validation. As a complementary future direction, covariation between changes in both functional connectivity and relevant behavioral performance would provide powerful evidence elucidating neural mechanisms of mindfulness training effects.

## The Way Forward

Despite these limitations, extending neuroimaging research on mindfulness to the resting state promises to provide vital insights into brain network connectivity changes induced through meditation practice. Growing evidence suggests mindfulness strengthens within-network communication in salience regions while reducing connectivity between midline DMN areas and attentional control networks. This aligns with conceptual models that mindfulness enhances present-focused meta-awareness while reducing mind-wandering and self-rumination (Brewer et al., 2011).

Still, current findings remain preliminary. Rigorously designed randomized controlled trials with larger samples, active control groups, and standardized analytical methods are needed to permit stronger causal inferences regarding the effects of mindfulness meditation training on Triple Network functional dynamics (Tang et al., 2015). Relating intrinsic connectivity changes to behavioral and clinical outcomes will further enrich neurobiological models.

There is also a need to map sequence effects, examining network plasticity occurring at different stages over the course of mindfulness training.

As research in this nascent area continues advancing, the impacts of various meditation techniques and programs on resting brain function can be characterized more definitively. Here, we propose to use diffusion embedding, which does not rely on narrow seed-based hypotheses (Margulies et al., 2016), providing a more data-driven approach than selecting relevant seeds and networks compared to conventional methods. This may lead to novel, theory-agnostic discoveries about intrinsic networks modulated by mindfulness meditation.

## Project questions

As we saw previously, research on mindfulness meditation and resting state functional connectivity has produced intriguing yet inconsistent findings, likely due to significant heterogeneity in study designs, analytical approaches, and conceptualizations of mindfulness. I highlighted several major limitations that future research should address. Here, I describe how diffusion embedding models, could help overcome many of these challenges and propel the field forward.

A primary limitation noted is the diversity of operational definitions and constructs related to mindfulness. Researchers have examined trait mindfulness in meditation-naïve individuals using self-report questionnaires, compared long-term meditators to novices, and assessed effects of various mindfulness training programs. Trait mindfulness itself, even when assessed with the same scale, may represent distinct phenomena depending on context (Sezer et al., 2022). This heterogeneity makes it difficult to integrate findings and develop cohesive theoretical models. Diffusion embedding methods applied to fMRI data could help derive more objective, data-driven characterizations of mindfulness-related neural patterns that transcend constructs imposed by researchers (Ras et al., 2021).

Specifically, diffusion embedding involves applying graph algorithms to fMRI data to embed each participant in a coordinate space based on similarity of whole-brain activation patterns (Margulies et al., 2016). This data-driven approach groups individuals based on the distributed neural signatures underlying their fMRI data, rather than relying on predefined categories or assumptions. Researchers could then examine whether embedded coordinates reflect known categories like meditator status or align with cognitive variables assessed outside the scanner. Any clusters emerging could represent more intrinsic phenotypes detectable through diffuse brain patterns. This could help disambiguate mixed findings for catch-all “mindfulness” variables and enable mapping of distinct neural signatures corresponding to specific cognitive capacities enhanced through various forms of mindfulness training.

Critically, the embedded coordinates locate each subject along continuous dimensions, avoiding discrete labels that obscure within-group heterogeneity. This aligns well with the conceptualization of mindfulness as multidimensional phenomena representing skills that can be cultivated to varying extents (Sezer et al., 2022). Diffusion embedding can quantify individual differences along these dimensions in a more fine-grained, brain-based manner than descriptive categories or subjective self-report measures permit. Researchers could relate dimensions uncovered to specific forms and durations of practice. We may find that different styles of meditation tune separable cognitive capacities, reflected in distinct neural signatures and locations in the embedding space. Elucidating these multidimensional neural phenotypes could clarify mechanisms and help optimize training protocols for particular goals.

The current literature also suffers from variability in resting state analytical approaches (Sezer et al., 2022). Seed-based methods rely on a priori node selection, making results difficult to integrate when different studies examine distinct seeds. Findings also depend on accurate seed placement, which poses challenges for regions with functionally heterogeneous subdivisions, like the ACC and PCC. ICA decompositions likewise vary across studies using different algorithms and parameter settings. Diffusion embedding circumvents these issues by capturing whole-brain patterns without requiring seed selection (Ras et al., 2021). Study-specific variations drop out when embedding is applied to aggregated data, providing a common space for meta-analysis. Algorithmic choices also minimally impact coordinates for any given participant, facilitating replication (Ras et al., 2021).

Another key weakness of prior work is the preponderance of cross-sectional comparisons between meditators and controls (Sezer et al., 2022). Group differences may reflect confounds like lifestyle factors rather than meditation itself. Longitudinal designs better isolate causal effects of training, but too few studies have measured functional connectivity before and after an intervention. Diffusion embedding of pre- and post-training scans would help address this gap. We could quantify shifts in individuals' embedded neural phenotypes corresponding to changes in functional connectivity induced by the training regimen.

Comparing the magnitude and direction of these positional shifts could reveal differential impacts of distinct practices. We may find that certain forms of meditation drive movement along particular dimensions, providing insight into mechanisms. Inter-individual variance in positional change could also help explain heterogeneity in cognitive and clinical effects.

Beyond methodological issues, conceptual integration of findings into cohesive models has been limited (Sezer et al., 2022). Reverse inference has often been used to speculate links between connectivity and psychological constructs. Diffusion embedding represents a substantial advance, as coordinates have no intrinsic labels but rather quantitatively situate individuals based on intrinsic whole-brain patterns (Ras et al., 2021). Researchers could conduct secondary analyses to test correspondence with behavioral metrics or clinical outcomes. Any uncovered relationships emerge from the data rather than presumed mappings. We can infer that dimensions reflecting salient individual differences likely track meaningful variability in how mindfulness manifests neurally and impacts cognition and well-being. This datadriven approach allows models to emerge organically.

In summary, diffusion embedding holds great promise for addressing several key challenges and knowledge gaps impeding progress in this exciting field. Deriving brain-based dimensions that chart variability in how mindfulness manifests neurally could help integrate disparate findings and elucidate relationships to types and durations of training. Longitudinal application could isolate effects of interventions while controlling individual differences. Analyses grounded in a data-driven perspective, avoid limitations of constructs imposed by researchers. Embedding offers a common space for meta-analysis and direct integration of findings across analytical approaches. Building more coherent neurocognitive models will require transcending variability in mindfulness conceptualizations and manifestations. Diffusion embedding provides a powerful method for achieving this unification and developing brain-based biomarkers that link mindfulness practices to enhancements in specific cognitive capacities and clinical outcomes. Large-scale initiatives to compile and embed extensive datasets will help the field move forward.

## ERC STUDY: project BRAIN and MINDFULNESS

Recent research exploring the effects of meditation on cognition and brain function has been limited by heterogeneity in meditation experience across study participants (Davidson & Kaszniak, 2015). While useful for initial exploration, samples including mixed meditation traditions and wide variations in practice hours complicate determining the impacts of specific contemplative methods or expertise levels. This variability also poses challenges for replicating findings between studies. The current work aimed to address these limitations by rigorously selecting a qualitative high and uniform group of expert meditators for detailed investigation.

Our general hypothesis was that sustained, high-level meditation practice may refine cognition and alter brain function in specialized ways compared to occasional or novice practice (Lutz et al., 2004). However, to begin teasing apart the impacts of various meditation styles and expertise, a carefully screened sample of highly experienced practitioners was needed. Thus, we recruited from French and European Tibetan Buddhist communities practicing within the Nyingma (Mahamudra) and Karma Kagyu (Dzogchen) lineages, which share contemplative methods (Perlman et al., 2010). These traditions were also represented in an earlier study we sought to conceptually replicate.

To ensure participants met stringent expertise criteria, a multi-step selection process was implemented. Potential experts needed a minimum of 10,000 hours lifetime practice, completion of at least one traditional 3-year meditation retreat, and sustained daily practice of 45 or more minutes per day in the past year. These benchmarks were derived from literature suggesting thresholds for meditation-induced neuroplastic change and emergence of specialized cognitive skills among long-term practitioners (Lippelt, Hommel, & Colzato, 2014). Attaining such extensive practice hours requires long-term, full-time commitment, often spanning decades.

Experts were initially contacted by a member of our research team who was herself an experienced practitioner within the selected traditions. This helped establish rapport and credibility for assessing expertise. During these interviews, detailed practice histories were acquired encompassing lifetime retreat hours and estimates of daily sitting. Only participants clearly meeting all criteria regarding practice type, duration and consistency were selected for our expert sample.

Practice histories were supplemented by a short guided practice interview to ensure familiarity with core methods like Mahamudra or Dzogchen and appropriate motivation for the research aims. For selected participants, average lifetime practice hours from reported retreats and daily sitting was estimated at 39,000 hours (± 18,000 SD), far exceeding thresholds proposed to induce neuroplastic alterations (Lutz et al., 2004).

To conceptually replicate an earlier study investigating meditation and attentional skills (Perlman et al., 2010), our meditator sample focused on the same Nyingma and Karma Kagyu Tibetan lineages. This helped control for potential tradition-specific impacts that could emerge from distinct contemplative methods. For instance, different styles may variably train mindfulness, sustained attention, emotional regulation and other cognitive capacities relevant to brain and behavior (Lippelt et al., 2014). By selecting adepts from within these matched lineages, we aimed to isolate effects specifically related to practice amount and mastery level attained.

Our expert recruitment and selection process improved upon past work including inexperienced meditators or mixed styles by ensuring a uniform population exceeding published thresholds for meditation-induced neuroplastic change. This qualitative high approach facilitated precisely exploring relationships between contemplation mastery, cognition and neural function while controlling for variability in practice type or history. Details of each expert's contemplative background obtained through extensive interviews established their qualifications and fit within the targeted traditions.

Going forward, this rigorous selection procedure can allow future replication attempts to conceptualize our results within the refined context of advanced practitioners achieving specialized meditation states through years of dedicated training. Numerous cognitive neuroscience experiments are now exploring meditation, yet examining its diverse styles and mastery levels requires carefully operationalizing variables like practice type and history of attainment. Our recruitment framework aimed to address such nuanced factors by filtering for a tightly controlled, expert sample of the highest caliber.

The refinement of meditation samples represents an important step towards parsing its diverse impacts. Mixed and incomparable populations conflate tradition, methods, experience and attainment in a field grappling to isolate meditation's mechanisms. By precisely characterizing expertise, we shed light on relationships between cognition, the brain and intensive contemplative training achievable through a lifetime of dedicated practice. Of course, future work can continue optimizing recruitment to capture the full extent of potential meditation-induced changes, from novice to adept. This initial study applied a rigorous approach to explore the specialized impacts of profound attainment.

A core aim of our meditation research program was establishing an appropriately matched control group to allow rigorous comparisons with expert practitioners. Simply recruiting meditation-naive individuals would not adequately control for variability in meditation experience, knowledge and reported phenomenology. Instead, we developed a novel training protocol to systematically introduce novices to key contemplative practices and phenomenological dimensions relevant to our neurophenomenological investigation (Abdoun et al., 2018)

The two-day training weekend was designed to equip formerly naive participants with both a conceptual foundation and experiential familiarity with meditation. This was crucial since our experimental approach relied on novices' ability to reliably report subjective experiences related to dimensions of interest. To achieve such phenomenological sensitivity among novices, the protocol integrated various meditation styles and acquainted trainees with relevant categories through experiential exercises.

Our training drew from Joy of Living, a secular program emphasizing mindfulness and compassion created by renowned Buddhist teacher Yongey Mingyur Rinpoche for Western audiences (Rinpoche & Swanson, 2007; Tergar, 2018). This selection controlled for partial alignment with expert practitioners' contemplative background. An instructor with 13 years personal practice under Mingyur Rinpoche and 8 years teaching experience expertly guided trainees.

Core components included teachings supported by instructional videos, guided meditations cultivating focused attention (FA) and open monitoring (OM), experiential exercises, Q&A sessions, and group discussion aimed at cultivating insight and ensuring retention. Interspersed breaks and reflection periods optimized learning. Trainees were invited to sustain a minimum 20 minutes daily practice after the weekend until study completion, balancing styles to aid phenemenological sensitivity.

Motivated novices interested in regular long-term practice for several months were recruited to ensure adherence. Compliance was tracked using individual practice logs. This refined protocol familiarized novices with targeted contemplative methods, related phenomenology and nuances of subjective experience reporting to more precisely approximate expert practitioners.

## Training Meditation Practice

A central goal was providing novices direct experience with key FA, OM and LKC techniques for in-depth phenomenological examination. Guided FA meditations focused attention on the breath, progressively broadening to include body scanning. OM introduced loose monitoring of all experience without fixation. Brief LKC practices cultivated care, empathy and kindness.

Experiential Exercises

Several activities aimed to acquaint trainees with the subtleties of phenomenology in an engaging manner. For example, noticing thoughts, emotions or sensations during walking helped reveal mind-wandering and perceptual details. Structured discussions unpacked insights into conceptual constructs and sensory qualities underlying reported experience.

Theory and Discussion

Didactic teachings summarized findings on meditation, contemplative traditions and phenomenology. Q&A sessions encouraged clarification and personal reflection. Group discussions after experiential exercises facilitated sharing to build an experientially-grounded conceptual understanding.

In summary, our aim was developing an optimally effective novice meditation training protocol suited to the methodological needs of neurophenomenological research. By systematically exposing trainees to contemplative practices, phenomenological dimensions and reporting nuances through experiential activities grounded in proper instruction, we intended to establish a control group sharing substantive conceptual and experiential familiarity with target investigated domains. Early feedback indicates this multifaceted approach succeeded in sensitizing novices as intended to facilitate rigorous controlled explorations and replications of meditation expertise.

## Psychometric scales

### Drexel Defusion Scale

The Drexel Defusion Scale (DDS) is a recently developed 10-item self-report questionnaire designed to measure psychological defusion, a core process in Acceptance and Commitment Therapy (ACT). Defusion refers to the ability to observe one's thoughts and feelings as passing mind events rather than as true reflections of reality. This allows them to be experienced with less literal belief and impact.

The DDS aims to assess respondents' typical ability to gain distance from a range of internal experiences including negative thoughts, feelings, physical sensations and urges. After explaining the defusion concept, participants rate their perceived capacity to defuse from hypothetical scenarios involving each experience type on a 6-point Likert scale ranging from "Not at all" to "Very much." Higher total scores indicate greater self-reported defusion skills. Importantly, the DDS assesses defusion from a broader range of experiences compared to other measures like the Cognitive Fusion Questionnaire, enhancing its usefulness. Research also suggests the items reflect the key cognitive mechanisms of meta-awareness and dereification (disidentification) from inner content targeted in ACT protocols. Initial validation found strong internal consistency, with alpha coefficients of 0.83. Convergent and divergent validity were also established relative to conceptually related and unrelated constructs. Overall, the DDS appears to be a valid tool for assessing defusion ability across affective, cognitive and sensory domains.

### Five Facet Mindfulness Questionnaire

The Five Facet Mindfulness Questionnaire (FFMQ) is one of the most widely used self-report instruments for evaluating dispositional or trait mindfulness. It measures five differentiated components theorized to represent the multifaceted mindfulness construct: Observing, Describing, Acting with Awareness, Non-Judging, and Non-Reacting.

The 39-item FFMQ asks respondents to rate how frequently each statement applies to them on a 5-point Likert scale, with higher scores on the five subscales reflecting higher levels of the corresponding facet. It was developed through factor analysis to organize items from existing mindfulness scales. Exploratory and confirmatory analyses support the five-factor structure, and subscale internal consistency is good to excellent.

However, the FFMQ has faced substantial methodological criticism. Items are subject to response biases due to positively and negatively worded formats. Differences in endorsement of such items vary systematically with meditation experience, introducing variance irrelevant to the construct. Additionally, most items remain relatively general and don’t precisely operationalize hypothesized cognitive-affective mechanisms. This clouds the measure's true construct validity for indexing mindfulness. Further issues include lack of specificity regarding the multidimensional construct it aims to capture.

Overall, while the FFMQ is popular in clinical research, its validity as a "process-pure" mindfulness assessment faces significant challenges. Caution is warranted in overly conflating FFMQ scores with the complex, context-dependent experience of mindfulness itself according to various contemplative traditions. Continued refinement of self-report tools is still needed.

### Beck Depression Inventory

The Beck Depression Inventory (BDI) is one of the most widely used psychometric tests for measuring the severity of depression. It is a 21-question multiple-choice self-report inventory that asks individuals to rate how they have been feeling in the past week. Scoring is based on a 4-point scale ranging from no symptoms to severe symptoms. It measures characteristic attitudes and symptoms of depression such as hopelessness and irritability, cognitions such as guilt or feelings of being punished, as well as physical symptoms such as fatigue, weight loss, and lack of interest in sex. Higher total scores indicate more severe depressive symptoms.

The BDI demonstrates high reliability, with alpha coefficients of 0.86 and 0.81 for psychiatric and non-psychiatric populations respectively. It also exhibits good concurrent validity with other depression measures as well as diagnostic criteria. While not intended to diagnose depression itself, the BDI provides a standardized measure of depressogenic cognitions and affect useful for tracking changes over time, either spontaneously or in response to treatment. It remains one of the most widely utilized scales for measuring depression severity due to its strong psychometric properties and ease of administration.

In summary, the DDS, FFMQ and BDI are commonly employed self-report tools within meditation and psychological research, despite some limitations. The DDS offers a targeted measure of defusion across inner experiences. The FFMQ assesses mindfulness facets but potential issues confound scores. And the BDI robustly gauges depression. Careful consideration of psychometric strengths and weaknesses guides interpretation.

## Experimental Protocol for Investigating Neural Correlates of Meditative States and Traits

The present study employed functional magnetic resonance imaging (fMRI) to explore neural activity during rest, open monitoring meditation, and compassion meditation. All procedures received institutional ethics approval. A single session was conducted for each expert meditators and matched novices who underwent intensive training ((Abdoun et al., 2018)).

The session began by acquiring a high-resolution structural scan. Functional scans then commenced with a 10-minute resting state baseline. Eyes remained closed as instructions requested letting thoughts wander naturally without falling asleep. This served to capture default mode network activity untrained by meditation.

Following rest, two 10-minute scans collected during engaged meditation practice. Order of meditative conditions — open monitoring and compassion — was counterbalanced across participants using randomization. For open monitoring, careful instructions guided relaxing the body-mind while maintaining present-focused awareness of any experiences arising without judgment or reaction ((Lutz et al., 2004)).

The compassion instruction focused participants on bringing to mind the image of a loved one, then filling the mind with care, goodwill and the wish to relieve suffering for both that person and oneself ((Klimecki et al., 2014)). Imagery and intentions aimed to evoke states of socio-affective connection and empathy.

Between each scan, participants completed trait and state self-reports using Likert scales ranging from 1-10. Clarity of mind, peacefulness, and valence of experience were assessed to verify distinct phenomenology between conditions and compare experts' experience to trained novices.

This design and stringent recruitment procedures should allow us isolating neural activity specific to open monitoring, compassion, and their relative differences from a resting baseline. Strictly controlled meditation instructions aimed to elicit reliable states for experts while remaining accessible to novices. Self-reports further validated each condition’s target qualities were engaged as intended.

## Elucidating the Transient and Enduring Impacts of Intensive Meditation: Examining a 10-Day Mindfulness Retreat

Meditation retreats offer a unique opportunity to study the effects of intensive meditation practice in an ecologically valid setting. Previous work has highlighted residential retreats as a key component of most meditators' long-term development (King et al., 2019) and an important avenue for bridging lab-based research on novices with cross-sectional studies of experts. However, systematically investigating meditation retreats from a scientific perspective also presents several challenges.

King and colleagues (2019) were the first to propose a theoretical framework for conceptualizing retreats as intensive meditation "dosages" and contextual factors that may influence outcomes. They emphasized common structural elements across retreat formats, including rigorous daily practice schedules of 6-11 hours, a combination of formal and informal meditation, and environments designed to minimize distractions. Retreat centers aim to establish optimal conditions for sustained focus through seclusion, noble silence, and abstention from technology, substances and other potential triggers (King et al., 2019).

This specialized context is theorized to facilitate deeper cultivation of meditative skills and mindsets over intensive short periods compared to conventional lab or community-based interventions. The withdrawal from typical daily habits and immersion in supportive practices are posited to foster exponential gains through reinforcement and “spillover” into other domains of experience (King et al., 2019). While retreats inevitably differ in specific teachings, the intensive format aims universally to cultivate meditation as "a way of life".
Despite the potential insights to be gained from residential retreat research, several practical and methodological challenges arise. Scientifically accessing these secluded, time-limited settings requires overcoming logistical barriers. Monitoring intensive, unstructured practice periods also introduces complexity compared to standardized lab paradigms. Additionally, recruiting appropriate control or comparison groups for retreat participants with varying levels of prior experience is difficult.

To date, most meditation research has focused on more accessible intervention formats using randomized controlled trial (RCT) designs, such as mindfulness-based stress reduction (MBSR) courses (REF). While valuable, these standardized 8-week programs involve significantly lower "doses" than retreats and typically recruit psychosocially stressed populations rather than experienced meditators. As such, they may capture different mechanisms and underestimate practice impacts.

To address these gaps, our research group has developed innovative methods for conducting rigorous scientific investigations within the retreat environment. We utilize a observational longitudinal design to naturalistically assess multidimensional outcomes before, during and after residential programs of 10 days duration.

One of the most comprehensive longitudinal studies on meditation to date is the Shamatha Project led by Clifford Saron and colleagues at the University of California (Saron, 2013). The study aimed to investigate the effects of 3-month intensive shamatha (calm abiding) meditation practice taught by Alan Wallace during two residential retreats at the Shambhala Mountain Center in Colorado. A rigorous experimental design was employed. Sixty healthy participants with prior meditation experience were randomly assigned to either an intensive meditation retreat group (n = 30) or a waitlist control group (n = 30). Extensive measures were collected before, during, and after the respective retreat periods using cognitive tasks, questionnaires, physiological recordings, interviews and biological samples. This allowed for between-group comparisons across the first retreat as well as within-group analyses as control participants engaged in their own second retreat.

The retreat schedule involved around 6 hours per day of individual shamatha practice, plus morning and evening sessions with Wallace. Participants experienced a highly supported environment conducive to sustained focus, characterized by seclusion, noble silence, and minimized distractions (King et al., 2019). As such, it provided a unique opportunity to study the effects of intensive meditation in its natural long-form setting. To date, over a dozen publications have emerged from this rich dataset, beginning in 2010. Several key findings have implications for cognitive and neuroscientific models of attention, emotion regulation and downstream impacts. Improvements were observed in sustained (MacLean et al., 2010) and selective attention (Zanesco et al., 2019), response inhibition (Sahdra et al., 2011), and executive control (Zanesco et al., 2013; Shields et al., 2020). Reductions in mind wandering were also seen (Zanesco et al., 2016). Studies of neural changes found decreased central beta power during meditation, suggesting greater neural stability (Saggar et al., 2012). Computational modeling linked these EEG shifts to enhanced oscillatory dynamics thought to support attentional skills (Saggar et al., 2015). Microstate analyses further revealed retreat-related alterations in the patterning and sequencing of large-scale functional brain networks at rest (Zanesco et al., 2021).

Retreat participants also exhibited increased emotional responses suggestive of greater compassion, such as more sadness and less anger/disgust when viewing violent images (Rosenberg et al., 2015). Physiological responses similarly pointed to heightened empathic engagement with human suffering (King, 2019). Improvements in psychological well-being were measured out to 18 months post-retreat (Sahdra et al., 2011). Remarkably, benefits to interference control were partially sustained seven years later, modulated by ongoing practice (Zanesco et al., 2018). Cellular aging markers like telomerase activity also increased after intensive training (Jacobs et al., 2011).

## LONGIMED Project

A primary objective of my PhD was to investigate the effects of intensive mindfulness meditation practice on the cognitive architecture during meditative and resting states using a controlled longitudinal design. We conducted the LONGIMED study to address gaps in understanding meditation's effects during residential retreats. Our general hypothesis was that intensive meditation practice would alter the brain cognitive architecture in a way that would be similar to the contrast between expert and or novice practitioners (Study 1)

54 participants with varied mindfulness experience were recruited via clinical trial protocols approved by an independent ethics board (clinicaltrials.gov registration NCT04449913). Inclusion required formal meditation training, a daily 20+ minute practice for over a year including prior intensive retreat experience. Exclusion criteria avoided potential confounds.
Participants were randomly assigned to "active" or "control" groups matched for demographics. Although, this design is not relevant for my PhD project, I will still describe it. The active group underwent testing before, during and after participating in a 10-day intensive meditation retreat. To control for repetitive testing effects, controls were tested at equivalent intervals but did not attend the retreat until later. Group labels were anonymized during testing to minimize bias.

Testing involved three behavioral/EEG experiments probing auditory and tactile perception, and pain perception using validated paradigms adapted from previous work (Demarchi et al., 2019; Shergill, 2003; Hoskin et al., 2019). The auditory experiment presented ordered and random tone sequences under focused attention, open monitoring or control conditions while recording EEG. The tactile experiment measured capacity for sensory attenuation - the phenomenon where replicating an externally-applied force leads to stronger felt intensity than passively receiving the same force (Bays et al., 2005; Walsh et al., 2011; Kilteni and Ehrsson, 2017). The pain perception task examined modulation of pain unpleasantness and sensitivity (Hoskin et al., 2019). The active group underwent baseline testing before the retreat, during days 6-7, and three weeks post-retreat. Controls were tested at matched intervals without the retreat experience intervening.

Additional self-report measures and analyses including the full sample were also conducted.

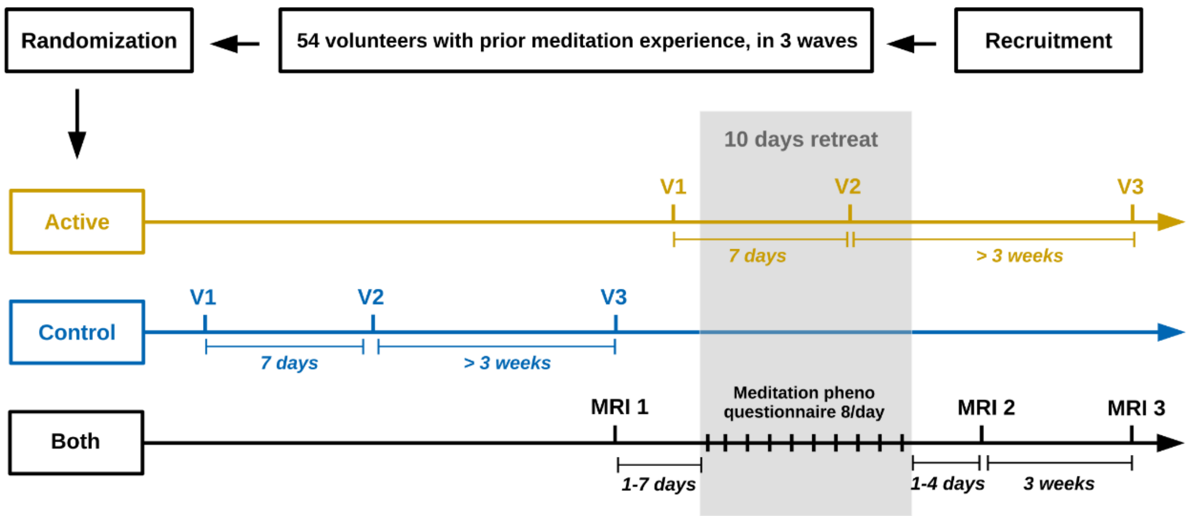


**Figure: Design of the LONGIMED study:** A total of 54 volunteers who had prior meditation experience were recruited for the study and randomly assigned to either an "active" group or a "control" group, with 27 participants in each group. The "active" group underwent behavioral and EEG experiments at three different time points: just before the 10-day intensive mindfulness retreat (referred to as V1, occurring 2 to 1 days prior to the retreat), during the retreat (V2, on day 6 or 7 of the retreat), and at least three weeks after the retreat (V3, occurring 21 to 30 days after the retreat). The "control" group also participated in the experiments, following the same timeline, but before the retreat. Additionally, structural and functional MRI scans were collected from both groups at three different time points: before the retreat (referred to as MRI 1, occurring 7 to 1 days prior to the retreat), immediately after the retreat (MRI 2, occurring 1 to 3 days after the retreat), and at least three weeks after the retreat (MRI 3, occurring 21 to 31 days after the retreat). All participants who attended the retreat completed a set of psychometric questionnaires during their initial visit to the laboratory (V1 for the "control" group and MRI 1 for the "active" group). Furthermore, after each meditation session during the 10-day retreat (which consisted of 8 sessions per day), participants were asked to complete a phenomenological questionnaire. From Poublan-Couzardot, 2022

Our study aimed at exploring the effects of intense meditation retreats on psychological and neural functioning. We employed a battery of quantitative and phenomenological measures. Both control and intervention participants completed assessments at several timepoints before, during, and after participating in a 10-day meditation retreat

At their initial laboratory visit (V1 for controls, MRI 1 for intervention group), all participants completed a series of well-validated psychometric questionnaires measuring personality, affect, cognition, and mindfulness (Plaisant et al., 2010; Caci et al., 2007, Carver and White, 1994, Demianczyk et al., 2014, Morean et al., 2014, Studer et al., 2016; Baer et al., 2006, Heeren et al., 2011). Specifically, we administered measures of the big five traits (Plaisant et al., 2010), behavioral inhibition/activation tendencies (Caci et al., 2007, Carver and White, 1994), defusion skills (Forman et al., 2012), trait mindfulness (Baer et al., 2006, Heeren et al., 2011), handedness (Nicholls et al., 2013), interoceptive awareness (Mehling et al., 2012, Edwige, 2014), depression/anxiety (Kroenke et al., 2010, Norton et al., 2007, Spitzer et al., 1999), pain catastrophizing (Sullivan et al., 1995, French et al., 2005), perseverative thought (Ehring et al., 2011, Devynck et al., 2017), rumination (Treynor et al., 2003), and spiritual well-being (Gomez and Fisher, 2003).

In addition to questionnaires, participants underwent structural and functional magnetic resonance imaging (MRI) at three timepoints - one week prior to the retreat (MRI 1), immediately following (MRI 2), and three weeks post-retreat (MRI 3). This allowed us to investigate changes in function associated with intensive meditation practice, as well as the duration of observed effects.

During the 10-day retreat, all participants (whether in the control or intervention group) engaged in the same meditation activities, consisting of 8 one-hour sessions per day guided by an experienced instructor (c.f Abdoun et al., 2019). To capture participants' first-person experiences, we developed a phenomenological questionnaire based on prior qualitative research (Kok and Singer, 2017, Abdoun et al., 2019, Petitmengin et al., 2017) completed after each daily session.

Notably, this questionnaire was tailored specifically for this study based on feedback from our research team's extensive meditation experience in various traditions, as well as from the instructor himself who had over 15 years and 80,000 hours of personal practice (See Lutz et al., 2015 for discussion of collaboration with contemplative experts in qualitative research design). We first piloted a draft with experienced meditators on a test retreat and revised it accordingly.

The final version consisted of seven domains exploring various facets of participants' session experiences. This included prompts about the general meditation instructions, an estimate of time spent actively vs. passively meditating, levels of fatigue and confidence/doubt regarding the practice (See Petitmengin et al., 2017 for discussion of anchoring phenomenological reports in context). While the rich qualitative retreat data is still under analysis, including phenomenological findings could provide crucial context for interpreting psychological and neural results.

At completion of the retreat, participants again responded to the adapted Meditation Depth Questionnaire (Piron, 2001), Life Change Inventory (Greyson and Ring, 2004), and select pre-retreat scales to assess any lasting personal or attitudinal shifts from the intensive practice period (Ehring et al., 2011, Devynck et al., 2017; Treynor et al., 2003; Mehling et al., 2012, Edwige, 2014; Sullivan et al., 1995, French et al., 2005).

Three weeks following the intensive intervention (MRI 3 visit), subjects filled out these post-measures as well. This design allowed examination of short-term, medium-term, and potential long-lasting effects on factors like rumination, mindfulness, pain perception, and life satisfaction (See Kok and Singer, 2017 for review of time-dependent meditation effects).

By incorporating quantitative neuroimaging and qualitative methodologies at multiple stratified timepoints before, during, and after a standardized 10-day meditation retreat, we aimed to develop a sophisticated, multimodal understanding of contemplative practice's impact on cognition, well-being, and brain function. While analysis of these rich dataset is ongoing, triangulating self-report, neural, and first-person phenomenological data holds promise for generating new insights into meditation's mechanisms of change. Definitive conclusions require continued rigorous exploration, but incorporating qualitative methods can enhance mechanistic explanations emerging from cognitive neuroscience.

Though strictly quantitative findings were not outlined here per the prompt's instructions, the phenomenological methodology was report to provide relevant context for interpreting forthcoming psychological and neurological results. Our ongoing work analyzes not only pre-post changes but also individual trajectories across assessments, allowing more nuanced examination of process over time. With consideration of participant feedback during development, the phenomenological questionnaire may elucidate how meditation experiences unfold dynamically over an extended period of immersive practice (Timmermann, Bauer, et al., 2023).

These qualitative narratives, situated meaningfully with reference to existing theoretical models (Lutz et al., 2015), could help cognitive neuroscience move beyond "what" occurs to address contemplation's "how" and "why". While meticulous phenomenology requires skill to conduct without bias, its integration with 21st-century neuroscientific methodologies may advance more holistic, experience-near accounts of meditation's impacts on mind and brain (See Klein & Petitmengin, 2019). Continued refining of mixed-method approaches coordinated closely with contemplative experts may thus leverage qualitative and quantitative tools synergistically.

In closing, our study aimed to elucidate changes relating to mental health, well-being, and neural functioning resulting from brief intensive meditation using multidisciplinary assessment. Though findings are pending further investigation, strategically incorporating first-person reports represents an innovative means of characterizing practice-induced alterations from multiple complementary viewpoints. Elucidating meditation's mechanisms and developing experientially-informed models remains an ongoing challenge that may be partially addressed through rigorously combined qualitative and quantitative perspectives.

## Experimental protocol for investigating neural correlates of meditative states and longitudinal effects of an intensive meditation retreat

The present study employed functional magnetic resonance imaging (fMRI) to explore neural activity during resting state (RS), open monitoring (OM) and focused attention (FA) meditation. All procedures received full ethics approval. As described, three fMRI sessions were conducted for each participant who underwent intensive training - before, immediately after, and three weeks following the retreat.

Each session began by acquiring a high-resolution anatomical scan. The order of functional scans was randomized, with identical durations and parameters as Studies 1 and 2. However, RS instructions differed - participants were requested to actively engage in mind wandering while refraining from meditation. This aimed to avoid inadvertent meditation and improve consistency with prior literature. For OM, participants received identical instructions to relax and maintain present-focused awareness of any experiences without judgment, as in past work ((Lutz et al., 2004)). The loving-kindness instructions were replaced with FA guidance. Using an original paradigm, participants focused present-moment awareness on one of their arms, with side counterbalanced and kept consistent longitudinally across scans.

After each scan, participants completed trait and state self-reports of stability, object orientation, meta-awareness, fusion, reification, sleepiness, effort, nervousness, and mind wandering on 1-10 Likert scales. We also checked participants focused on their instructed side. This allowed verification of distinct phenomenology between conditions and longitudinal effects.

Overall, this design should isolate and replicate cognitive architectures specific to OM, FA, and their differences from RS baseline. The longitudinal aspect enables demonstration of intensive retreat effects on cognitive architecture and persistence over time.

# EXPERIMENTAL RESULTS

## Study 1: Exploring the embodied mind: functional connectome fingerprinting of meditation expertise

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# Connectome fingerprinting of meditation expertise

**Keywords: meditation, connectome, traits, mindfulness, expertise, DDS**

Abbreviations: open presence (OP), open monitoring meditation (OM), loving-kindness and compassion meditation (LKC), back-to-back (B2B), resting-state (RS), mindfulness-based interventions (MBI), Visual (Vis), dorsal attention (DA), ventral attention (VA), salience network (SN), frontoparietal (FP), default-mode-network (DMN), Drexel defusion scale (DDS), Beck depression inventory (BDI), five facets mindfulness questionnaire (FFMQ), N,N-Dimethyltryptamine (DMT), Supplementary Materials (SM)

# Abstract:

Short mindfulness-based interventions have gained traction in research due to their positive impact on well-being, cognition, and clinical symptoms across various settings. However, these short-term trainings are viewed as preliminary steps within a more extensive transformative path, presumably leading to long-lasting trait changes. Despite this, little is still known about the brain correlates of meditation traits. To address this gap, we investigated the neural correlates of meditation expertise in long-term Buddhist practitioners, comparing the large-scale brain functional connectivity of 28 expert meditators with 47 matched novices. Our hypothesis posited that meditation expertise would be associated with specific and enduring patterns of functional connectivity present during both meditative (open monitoring/open presence and loving-kindness compassion meditations) and non-meditative resting states, as measured by connectivity gradients. Our finding revealed a trend toward the overall contraction in the gradient cognitive hierarchy in experts versus novices during open presence meditation. The signature of expertise was further characterized by an increased integration of large-scale brain networks, including the

somatomotor, dorsal and ventral attention, limbic and frontoparietal networks, which correlated with a higher ability to create psychological distance with thoughts and emotions. Applying a support vector classifier to states not included in training, we successfully decoded expertise as a trait, demonstrating that its non-state-dependent nature. Such heightened integration of bodily maps with affective and attentional networks in meditation experts could point toward a signature of the embodied cognition cultivated in these contemplative practices.

# INTRODUCTION:

Short 8-week mindfulness-based interventions (MBIs) are routinely used in various clinical and educational settings. Meta-analyses indicate that they positively impact well-being and cognition, and decrease clinical symptoms, in particular in mood disorders (1,2). MBI can induce functional changes in the neural processes underlying affect and attention (3,4) which are not always associated with structural changes (5), the latter being reported following longer meditation training (6). According to traditional meditation theories, these short-term training effects are only preliminary steps within a longer and more transformative path leading to long-lasting trait changes in cognition and self-related processes, and which require the combined practice of a variety of meditative techniques (7). Theoretical models of meditation have examined the psychological processes of such meditation practices, grouping them into attentional, constructive, and deconstructive families (7). The attentional family trains attention and meta-awareness and is exemplified by focused attention (FA) meditation, or open monitoring (OM) meditation (8). The constructive family, exemplified by compassion and loving-kindness meditation (LKC), trains perspective taking and cognitive reappraisal capacities and aims at transforming maladaptive self-schema (9). The deconstructive family trains in self-inquiry and aims at recognizing the nature of maladaptive mental schemas (7,10) that cause suffering and prevent a long-lasting form of well-being from emerging; it has thus far received less research interest (11). Deconstructive practice aims in particular at recognizing the constructive and transient nature of basic cognitive structures such as time, space, and subject–object orientation. To explore and gain such insights about the nature of perception and the nature of the self, some Buddhist practitioners are trained in particular into non-dual mindfulness meditations (10), such as Open presence meditation (OP). OP meditation is said to induce a minimal phenomenal state of consciousness where the intentional structure involving the duality between object and subject is attenuated, as captured by the notion of non-duality  (10,12). OP meditation is typically practiced with attentional and constructive practices involving mental imagery and compassion meditations. The alterations of these self-related and affective and attention processes throughout these various practices is said to have long-term impact on cognition as a trait (8) captured here by the notion of "meditation expertise". This developmental trajectory suggests that the baseline brain functional profile of experts will gradually tend to overlap with the dynamical profile of these meditative states themselves. Despite the potential therapeutic and scientific interest in characterizing long-term meditation expertise, little is still known about its neurophysiological mechanisms (13,14). The purpose of the present study is to investigate the neural correlates of meditation expertise in a sample of long-term Buddhist meditators, as measured by changes in the organization of intrinsic connectivity networks in the brain. We hypothesize that any long-lasting, trait-like changes in experts should correlate with specific changes compared to novices in large-scale brain functions detectable both during meditative states as well as at rest during non-meditative states.

To investigate the effect of long-term meditation practice on the intrinsic functional organization of the cerebral cortex (15), we employed a data-driven technique that captures its organization along continuous gradients, also known as connectivity gradients (16,17). This technique can reveal multiple dimensions of cortical organization, with the first dimension describing the cognitive hierarchy (15), starting from sensory cortex and ending with transmodal regions such as the default-mode network (DMN). The second gradient separates visual regions from the other networks (16), and the third gradient spans the multiple-demand network and the networks at opposite end (18). This network is composed of the inferior frontal sulcus, regions of the dorsolateral prefrontal cortex, the anterior insula, the dorsal anterior cingulate, the pre-supplementary motor area, and the intraparietal sulcus. Previous studies have demonstrated that these gradients can be influenced by various factors, including disorders such as depression (19) and autistic spectrum disorder (20), as well as cognitive and psycho-affective training (21). Interestingly, psychedelics, such as psilocybin (22) and N,N-Dimethyltryptamine (DMT) (23) can induce ego dissolution and are correlated to a collapse of the cognitive hierarchy as measured by the first gradient. Given that both psychedelics and non-dual meditation are said to lessen self-related processes, we predicted that non-dual meditation would be associated with a similar compression of the cognitive hierarchy, as Timmerman and colleagues (23) reported, albeit weaker in its effects. Given the paucity of data in the literature on meditation expertise and novelty of the gradient connectivity method, further developing a specific functional hypothesis is challenging. However, one can identify brain networks candidates which may be impacted within the connectome. Meditation has been associated with brain structural and functional changes mainly in frontal and limbic networks (6,24) with the insula and anterior cingulate cortex, part of the salience network (SN), being the regions most sensitive to meditation training according to a meta-analysis (25). Studies on meditation traits and functional connectivity (26,27) or differences between long-term meditation practitioners and novices (13,14,28) have also shown that individuals with meditation experience exhibit reduced connectivity between the DMN and fontoparietal network/SN, while connectivity between the SN and FPN increases. Yet, these effects remain inconsistent across studies, as meditation training has also been linked to increased connectivity between the DMN and FPN/SN (29,30). A recent study reported effects of various forms of meditation training on the connectome consisting of increased functional segregation of regions, including parietal and posterior insular regions, following training in attentional family meditation, indicating that these networks are functionally different from the rest of the cortex. Conversely, perspective training (i.e. constructive family) resulted in increased functional integration of these regions with other brain networks (21).

Investigating expert meditators could thus help develop hypotheses or theoretical understanding about the long-term developmental trajectory of meditation and minimal phenomenal states of consciousness.

# METHODS:

## Participants

Participants were recruited for the Brain and Mindfulness ERC-funded project, which includes a cross-sectional observational neuroscientific study on the effect of mindfulness meditation on experiential, cognitive and affective processes conducted in Lyon, France, from 2015 to 2018. Participants included novice and long-term meditation practitioners (referred to as ‘experts’), who were recruited through multiple screening stages (for details, see the Brain & Mindfulness Project Manual (31)). 75 cognitively normal participants aged 35-66 (SD 7.7) including 28 expert meditators and 47 control participants (referred to as ‘novices’) matched on age and gender ($p>0.5$, see Table 1) were included in this study.  Novices attended a one-weekend meditation training program prior to any measurement to get familiarized with the meditation techniques. Inclusion and exclusion criteria have been previously reported (31) (see (SM)). Finally, subjects had to be affiliated to a social security system. All participants received information on the experimental procedures during a screening session, and provided informed written consent. The study was approved by the regional ethics committee on Human Research (CPP Sud-Est IV, 2015-A01472-47). After excluding participants who exhibited more than 0.3mm/degree movement to control for potential motion effects (2 experts and 3 novices), the analysis was conducted with a reduced sample size of 70 participants (32).

| | Novices | Experts | p |
|---|---|---|---|
| n | 47 | 28 | |
| Sex | 24M/23F | 16M/12F | 0.91 |
| Age | 51.9 (7.5) | 51.8 (8.0) | 0.95 |
| Education | 3.6 (2.0) | 3.2 (2.4) | 0.52 |
| DDS | 29.0 (6.7) | 39.0 (6.6) | <0.001 |
| FFMQ | 128.8 (20.4) | 158.0 (18.2) | <0.001 |
| BDI | 5.4 (4.6) | 3.4 (4.3) | 0.074 |
| Hours | 26 (17) | 39k (18k) | <0.001 |

**Table 1:** Demographics comparison between experts and novices. Continuous variables are presented as mean (standard deviation) and their p-values were calculated using t-test. For sex, the p-value was calculated using a chi-squared test. DDS: Drexel Defusion Scale. FFMQ: Five Facets Mindfulness Questionnaire. BDI: Beck Depression Inventory.

## Paradigm

All participants attended a single fMRI session in which we first acquired their structural image. We then acquired functional scans, starting with a resting state (RS). We also acquired meditative states of LKC meditation. In addition, for novices, we acquired states of OM meditation, and for experts we acquired states of OP meditation (see SM for a description of the meditation practices). All states lasted 10 minutes. The order of acquisition of the two meditative states was random. For the present study, we used three psychometric scales, which are Drexel defusion scale (DDS) (33), Five facets mindfulness questionnaire (FFMQ) (34), and Beck depression inventory (BDI) (35) (for details, see SM).

## Data acquisition and preprocessing

Data was collected on a 3T Siemens Prisma scanner. Functional data was acquired with EPI (TR=2100ms, TE=30ms, 39 slices, voxel size 2.8x2.8x3.1mm3). Structural scans were T1w (1mm iso), T2w (1mm iso) and T2*w (1mm iso). Preprocessing used fMRIprep v1.2.6 (36). This included motion correction, co-registration, normalization to MNI space, CompCor for physiological noise removal, ICA-AROMA denoising, and FreeSurfer surface reconstruction (see SM for details).

## Connectome gradient construction

The construction of the functional connectome gradient followed the procedures detailed in Hong et al. (2019) (20) and in the SM.

## 3D gradient metrics

To investigate multidimensional differences in cortical organization, we focused on the first three components, which explained over 50% of total variance. We combined these gradients by forming a 3D space (21,37), where each gradient constitutes an axis of this space described in Figure 1A.

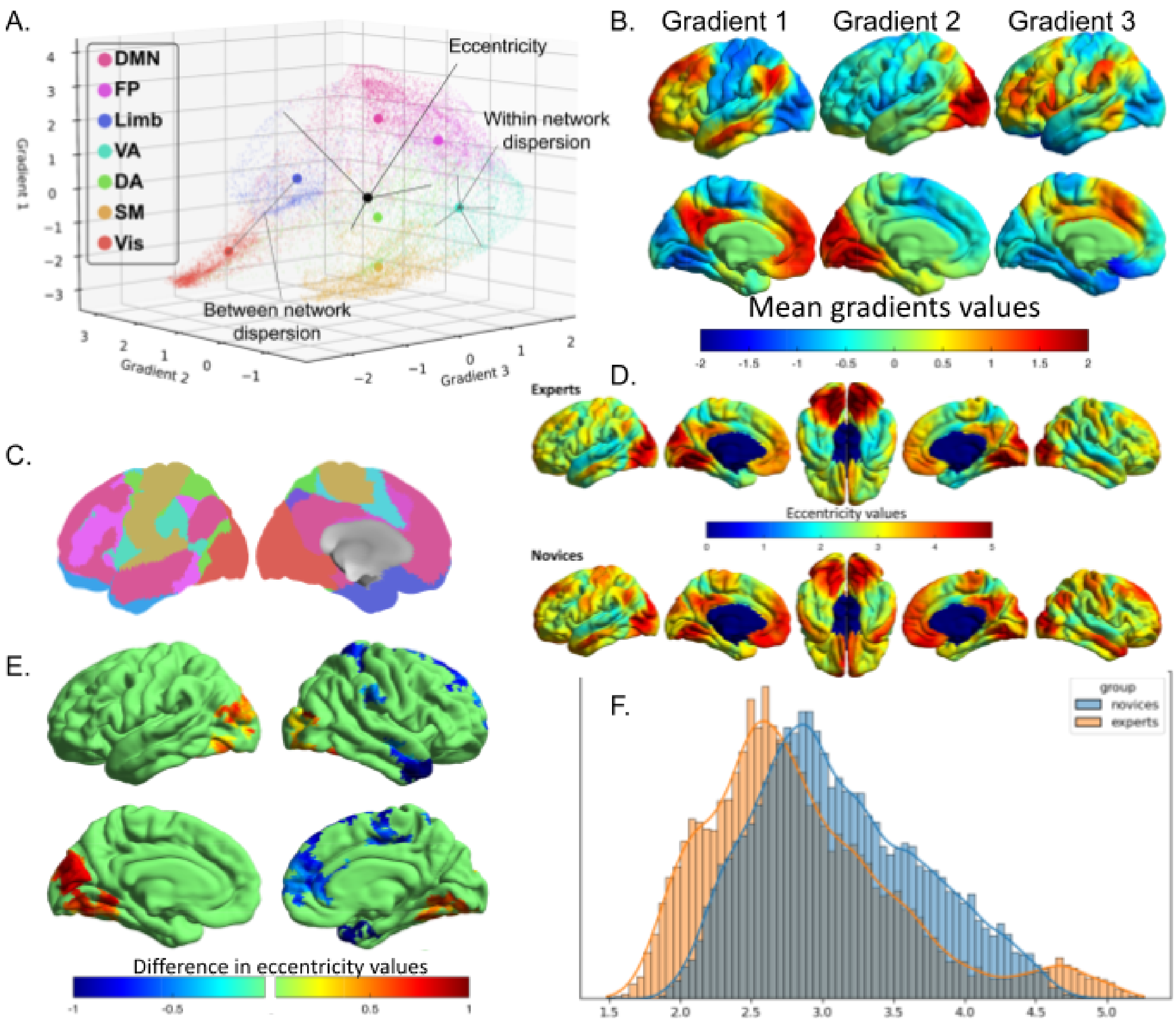


**Figure 1 Effect of long-term meditation practice on eccentricity maps:** (A) Visualization of dispersion metrics within the three-gradient space. The eccentricity value of a vertex corresponds to the Euclidean distance from the barycenter (depicted by a black dot) of the 3D space. For each individual 3D map, we computed an eccentricity map defined by the Euclidean distance from each vertex to the individual barycenter of the 3D space (21,37). These maps reflect the integration (low eccentricity) and segregation (high eccentricity) within the connectome for each voxel of each participant. We then quantified the dispersion metrics, which represents the segregation of large-scale networks (21). The within-network dispersion is calculated as the sum squared Euclidean distance of network vertices to the network barycenter. Between-networks dispersion is quantified as the Euclidean distance between network barycenters (21). (B) The first gradient (left) denotes the cognitive hierarchy of the brain, ranging from the unimodal cortex to the DMN. The second gradient (middle) differentiates the visual cortex from other networks, while the third gradient (right) segregates the limbic network. (C) Visualization of the cortical parcellation (63) used in (A). The color code corresponds to the one in Figure 1A (D) The eccentricity map describes a continuous coordinate system, where a lower value signifies that the vertex is closer to the barycenter of the 3D space. Sensory regions, including the visual and somatomotor cortex, and the DMN tend to be the least integrated regions. Although the average maps of experts and novices are similar overall, it is evident that eccentricity values are visually lower for experts than novices, except within the visual cortex. (E) Surface-wide statistical comparisons between novices and experts are presented, with increases/decreases in experts shown in red/blue. Findings were obtained using surface-based linear models implemented in SurfStat during RS. They show statistically significant differences in eccentricity values between experts and novices particularly in the bilateral visual cortex and within

the right hemisphere of the brain. Specifically, experts had decreased eccentricity in the right temporal pole, dorsal posterior cingulate cortex, supramarginal gyrus and visuomotor cortex (Table 2). On the other hand, only one bilateral cluster within the visual association cortex showed increased eccentricity for experts. This finding indicates that the visual cortex's cluster is more segregated in experts than in novices, whereas all the other clusters were more integrated. (F) A global histogram analysis confirms that, overall, eccentricity values are qualitatively lower for experts than for novices, except for vertices belonging to the visual cortex.

| Cluster region | *p*FDR | *kE* | *Tmax* | *x* | *y* | *z* |
|---|---|---|---|---|---|---|
| L visual association | <0.001 | 967 | 3.4 | -28 | -75 | -15 |
| R visual association | <0.001 | 576 | 3.6 | 39 | -85 | 13 |
| R temporal pole | <0.001 | 206 | -3.7 | 48 | 13 | -37 |
| R dorsal PCC | <0.003 | 127 | -3.4 | 13 | -30 | 44 |
| R supramarginal | 0.005 | 106 | -2.8 | 55 | -32 | 30 |
| R visuomotor | 0.014 | 114 | -3.8 | 12 | -53 | 71 |

**Table 2:** Surface-based analysis of the RS's eccentricity. All eccentricity cluster were lower for experts than for novices, except for the bilateral visual cortex. PCC: Posterior Cingulate Cortex.

## Statistical analyses

We compared gradient component scores between experts and controls using surface-based linear models in SurfStat (http://www. math.mcgill.ca/keith/surfstat/). Surface findings were corrected for family-wise errors using random field theory (pFWE < 0.05). To show group differences in dispersion metrics, we used multivariate non-parametric two-sample testing (38). Post-hoc tests were computed using Studentized bootstrap-t tests with 10.000 repetitions (39). We trained classifiers on dispersion metrics to predict expertise using scikit-learn (40) with modified-huber loss and 5-fold cross-validation repeated 5000 times.

We used area under the curve rather than accuracy to avoid bias from unbalanced samples. To disentangle collinear demographic factors, we used a back-to-back regression (B2B) (41), finding DDS score had a significant contribution to dispersion measures (details in SM).

# RESULTS:

## RESTING STATE ANALYSIS

We hypothesized that the large-scale fMRI connectomics measures would be modulated by trait-like effects of expertise not only during meditative states but also at rest during non-meditative states. To test this hypothesis, we first studied the RS, which is often viewed as a baseline, and its study is a standard approach to characterize expertise or traits (4,13,14) (see Figure 2A). The density of eccentricity map values for both groups (Figure 1F) illustrates how, globally, experts' vertices were embedded closer to the connectome centroid, resulting in more integrated vertices for experts. As a similar pattern was reported in the psychedelic literature, we explored whether the average eccentricity of experts was different from that of novices, but the result was not significant ($t(68)=0.35$; $p=0.16$; (95% CI: -0.13,0.81)). These findings indicate that expertise affects the functional connectivity of the brain, resulting in changes in eccentricity that likely signify differences in information processing and integration.

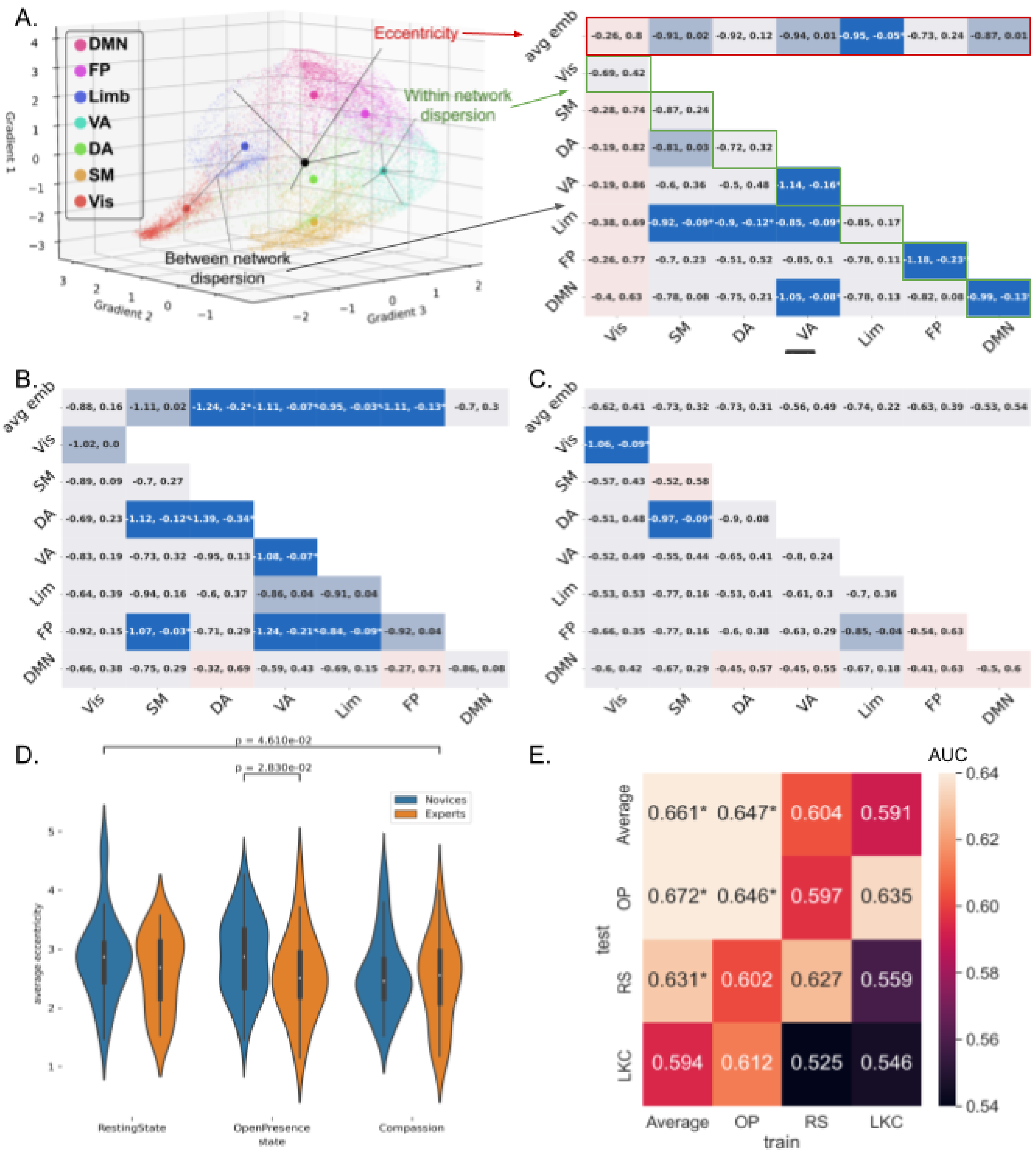

**Figure 2 Comparison of networks' dispersion between experts and novices during several mental states:** (A) Visual explanation of the dispersion metrics using the RS as an example. The first row of each matrix shows the average vertex-wise eccentricity of each network, referred to as the average embedding (21). The remaining dispersion metrics display the within-network dispersion (diagonal in green) and the between-network dispersion (the remaining squares). Significance was tested using bootstrap-t tests (39), and 95% confidence intervals are plotted in each box. The color blue (respectively red) indicates that the average eccentricity value was lower (respectively higher) for experts than novices. Bright-colored boxes indicate significant tests ($p<0.05$), medium-colored boxes indicate a trend ($0.1>p>0.05$), while light-colored boxes indicate non-significant tests ($p>0.1$). The tests are not corrected for multiple comparisons. The results indicate that the several dispersion metrics decreases in experts compared to novices during the RS, which was expected based on previous analyses (Figure 1), except for the visual cortex, which is not significant here. Here, a higher between-network dispersion reflects a weaker connectivity between two networks and a higher within

network dispersion, a lower connectivity between the voxels of a given network. We performed a brain parcellation in large-scale networks (63), commonly utilized in conjunction with diffusion map embedding (16,20). Using Studentized bootstrap tests (see details in SM), we found that, overall, the dispersion of the visual network did not increase for experts within the 3D space during the RS, as we would have expected from the surface-based analysis. However, the dispersion of the limbic network decreased (t=0.51; p=0.045), and marginally decreased for the somatomotor (t=0.45; p=0.069), ventral attention (t=0.46; p=0.064), and DMN networks (t=0.44; p=0.073). Furthermore, the dispersion within the ventral attention (t=0.63; p=0.015), frontoparietal (t=0.68; p=0.007), and default-mode (t=0.55; p=0.022) networks also decreased. We observed a decrease of dispersion between the limbic network on one side and the somatomotor (t=0.51; p=0.031), dorsal attention (t=0.52; p=0.025), and ventral attention (t=0.49; p=0.035) on the other side, and between the ventral attention and the DMN (t=0.55; p=0.03). (B) Dispersion metrics during open presence. Similarly to the resting state, all significant results show a decreased dispersion for experts. However, the results are not identical between the two states, which suggests a modulation of the states by expertise. Specifically, the average eccentricity of the dorsal attention (t=0.69; p=0.007), ventral attention (t=0.57; p=0.027), limbic (t=0.5; p=0.048), and frontoparietal (t=0.61; p=0.018) networks decreased within the 3D space, and marginally for the somatomotor (t=0.52; p=0.056) cortex for the experts compared to the novices. The eccentricity also decreased within the dorsal (t=0.82; p=0.002) and ventral attention (t=0.55; p=0.033) networks. We also observed a decrease of dispersion between the frontoparietal network on one side and the somatomotor (t=0.52; p=0.049), ventral attention (t=0.71; p=0.007) and limbic network (t=0.49; p=0.032) on the other side and between the somatomor cortex and the dorsal attention network (t=0.59; p=0.018) for experts. (C) Dispersion metrics during loving-kindness compassion. This state appears to be more similar for both groups than the other states. (D) Mixed-ANOVA analysis between group and state of the average eccentricity. Only the state effect was significant. Post-hoc tests reported on the figure are not corrected for multiple comparisons. (E) A stochastic gradient descent classifier was trained on the dispersion metrics (A-C) to decode expertise. The training was performed on a subsample of the three states or the average of the three states. The classifier was tested on the same metrics of the remaining sample, either from the same state or from a different state, using AUC. Results showed that the RS state cannot be used to significantly decode the effect of expertise. The SVC was able to decode expertise when trained and tested on OP (p=0.027). However, it was not able to generalize to the other states. Only the average of the three states was able to decode expertise (p=0.021) and to generalize to the OP (p=0.02) and RS (p=0.046) states. These results suggest that the average of the three states was the best way to capture an effect of expertise that would be present across all states. Additionally, the results confirm that the compassion state was very similar between experts and novices, as it was not possible to decode expertise in this state or when trained on it. There was no group difference during LKC (rv=0.30, p=0.21) confirming the finding with the multivariate approach. We confirmed this negative finding on the individual dispersion measures, which showed minimal variation between experts and novices, converging with the previous analysis. By contrast, there was an overall group difference on the dispersion measures (rv=0.079, p=0.024) during OP/OM states. This global effect was driven again by reduced dispersion, yet with a somewhat distinct pattern from RS.

To further functionally specify the group difference in eccentricity, we explored the multidimensional differences within the cognitive hierarchy, and between and within standard brain networks, as recently proposed (21,37) (Figure 2A). Using a multivariate non-parametric approach (38), we found an overall trend difference between groups between these various dispersion metrics (rv=0.055, p=0.066). We then investigated the group effect on each of these dispersion metrics as described in Figure 2A, as an exploratory analysis. These findings indicate that experts' dispersion metrics during the RS were lower compared to novices in specific networks. However, as experts are expected to exhibit a trait and possibly a trait by state effect during RS, these differences may not reflect the sole signature of expertise.

To address this concern, we repeated these analyses for LKC and OP/OM meditative states, in order to identify common characteristics among these states that could be considered as an effect of expertise. First, and following our hypothesis based on ego dissolution during psychedelics, we tested whether there was a global reduction of eccentricity. We computed a mixed ANOVA between groups and states to test for differences of mean eccentricity. There was a state effect ($F(2,136)=4.78$, $p=0.009$), but contrary to our hypothesis, no interaction effect ($F(2,136)=0.83$, $p=0.43$) and only a trend effect for the groups ($F(1,68)=2.54$, $p=0.1$). Regarding the group effect, we still performed an exploratory analysis as one of our hypotheses was that experts' connectome could show decreased eccentricity, mainly during OP, as in psychedelics study (22,23), albeit much weaker. This prediction was confirmed only during OP state which showed lower mean eccentricity for experts compared to novices ($t(68)=0.5$; $p=0.048$; (95% CI: 0.01,1.04)), while the LKC state showed the lowest difference between groups ($t(68)=0.15$; $p=0.55$; (95% CI: -0.37,0.66)).

Similarly to RS, we computed the large-scale networks' dispersion metrics for these meditative states, revealing different contrasts (Figure 2). These findings indicate an expertise-effect during OM/OP meditation but not LKC meditation, which only partially overlapped with the pattern found during RS. This suggests that the effect of expertise on large-scale networks may vary depending on the cognitive state or task and that segregation of networks, as captured by gradient dispersion, may reflect global modes of function.

## SVC EXPERTISE ANALYSIS

Our analyses revealed that each state showed a somewhat different signature of expertise, making it difficult to characterize its enduring dynamical characteristics. Although the so-called RS, where the participant is asked not to engage in any specific cognitive activity, is a gold standard approach to measure it, its instruction turns out to be ambiguous for many expert meditators. This instruction can typically be understood, on the one hand, as an invitation to spontaneously engage in OP meditation, a style of non-dual meditation, or on the other hand, as an invitation to actively try not to meditate by spontaneously following thoughts, such as during mind wandering.

To tackle this issue, we used a machine learning approach. We trained a support vector classifier (SVC) on a subset of a given state to distinguish experts from novices, and then tested its ability to both decode the same state and to generalize to the other states. Our rationale was that if the expertise effect was an enduring dynamical characteristic present in every state, the SVC should be able to generalize to the other states as well. Specifically, the state with the least amount of noise around the effect of expertise should be the most susceptible to generalize when tested on another state (Figure 2E). We were able to decode expertise only for the OP state, but we were not able to generalize its classification on the RS nor on the LKC state. Next, to reduce the contribution of the trait-by-state effect, and to make the effect of expertise more salient, we averaged all three states together, and again trained the classifier using the average pattern. As expected, the model demonstrated significant expertise decoding ability when trained on the average state and subsequently tested on the remaining test set. Interestingly, unlike its performance when trained on OP, the model was also able to generalize its classification ability when tested on different states, specifically RS and OP. In the line with the previous results (Figure 2C), the decoder was not able to distinguish experts from novices during LKC.

## AVERAGED STATE DISPERSION ANALYSIS

To further characterize the averaged state that best captured the fingerprint of meditation expertise, we examined the dispersion metrics of this averaged state using both a 3D space exploratory analysis (Figure 3A) and a surface-based analysis (Figure 3B). We also computed a surface-based analysis of the averaged state's eccentricity. All clusters showed reduced eccentricity for experts when compared to novices (Table 3). To investigate the behavioral relevance of these group differences, we then studied the individual contribution of various features, including sex, age, group, hours of meditation practice in life, and trait psychometric measures (DDS, BDI, FFMQ) to decode the dispersion metrics. To do so, we fitted a B2B model to control for the co-variance between features while optimizing the linear combination of dispersion metrics to detect the encoding information (Figure 3C). The output of this model is a set of beta coefficients, one for each feature. Here, only the DDS, a scale reflecting a person's capacity to cognitively defuse thoughts, and emotions, yielded a significant contribution to the decoding ($\beta$=0.31; p=0.043). We then applied the same B2B model to each dispersion metric individually, meaning that we used all previous features to predict dispersion metrics. We only present the results for DDS, as it was the only scale to demonstrate a significant relationship (Figure 3D). Here, our goal was to identify which dispersion metrics predicted by the set of features exhibited a significant contribution from the DDS. Importantly, the only associations were negative correlations, where a higher DDS score was associated with a lower dispersion metric, consistent with the fingerprint found in Figure 3A. However, contrary to Figure 3A, a higher DDS score was not associated with a change of dispersion within networks. To summarize, our analysis suggested that the capacity to put psychological distance between thoughts and emotions was associated with reduced network dispersion between and across specific networks, largely overlapping with the expert-related trait signature (Figure 3A), indicating a potential link between trait-like measures and neural activity during meditation. These findings shed light on the neural mechanisms underlying expertise effects in meditation and highlight the importance of considering state-averaging approaches in future studies.

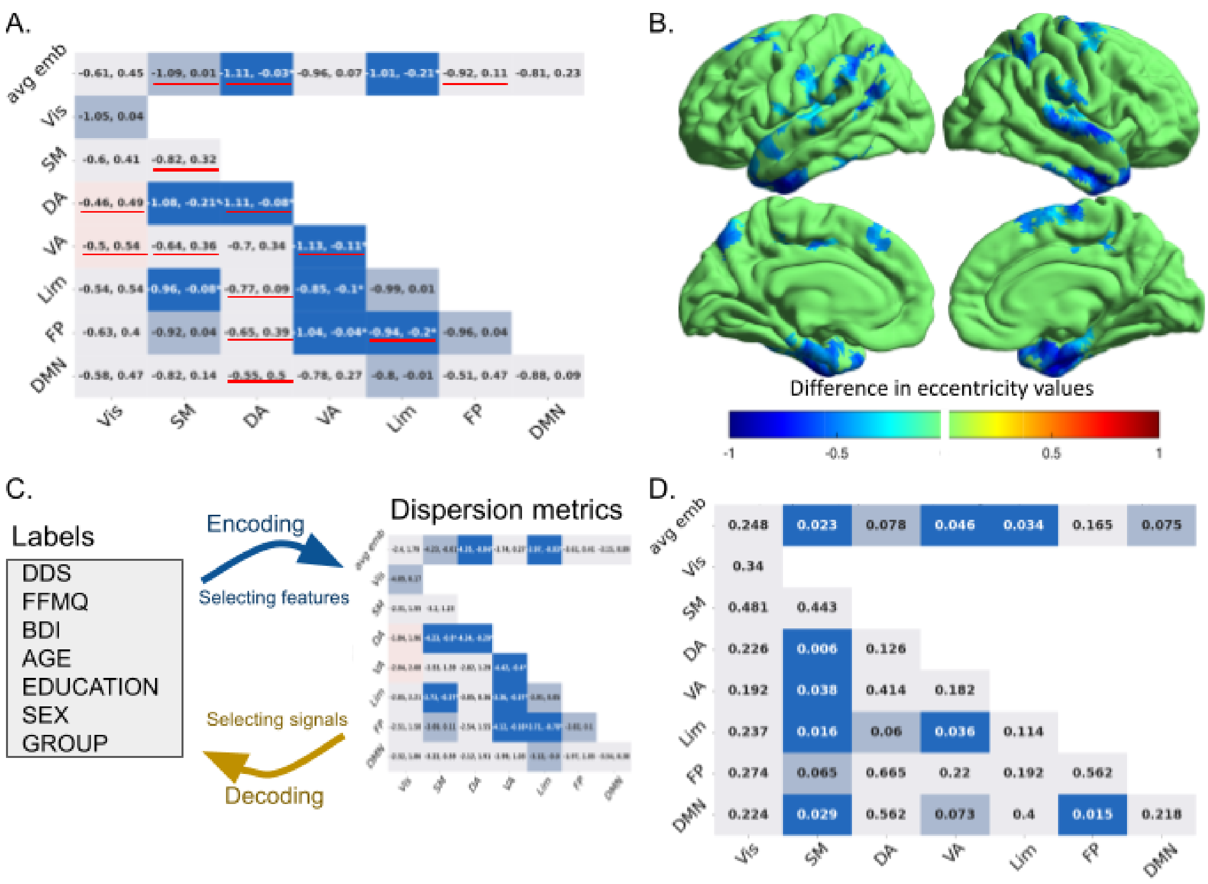

**Figure 3 Expertise and traits effect on networks' dispersion:** (A) Dispersion metrics after averaging RS, OP and LKC gradients. Colors are explained in Figure 2A. The red underlines indicate which dispersion metrics contribute significantly to the decoding of expertise (Figure 2E). Again, the results suggest that the dispersion of many networks decrease in experts compared to novices. This effect should reflect the expertise effect on the brain's dispersion metrics and be less influenced by state modulation. We found a significant decrease average eccentricity for the experts of the dorsal attention using a Studentized bootstrap ($t=0.54$; $p=0.046$), limbic ($t=0.61$; $p=0.017$) networks, close to significance for the somatomotor ($t=0.51$; $p=0.058$) network and within the dorsal ($t=0.57$; $p=0.029$) and ventral attention ($t=0.6$; $p=0.02$) networks. Additionally, we observed a decrease in dispersion between the limbic network and the somatomotor ($t=0.52$; $p=0.038$), ventral attention ($t=0.48$; $p=0.039$), and fronto-parietal ($t=0.56$; $p=0.021$) networks, as well as between the ventral attention and the fronto-parietal ($t=0.51$; $p=0.037$) networks, and between the dorsal attention network and the somatomotor cortex ($t=0.62$; $p=0.012$). (B) Surface-wide statistical comparisons between novices and experts after averaging RS, OP and LKC eccentricity maps. Many clusters mainly belonging to the sensorimotor (SM), dorsal attention (DA), ventral attention (VA), and limbic (Lim) networks show less dispersed vertices for experts than for novices. (C) The B2B regression method was computed, using the indicated labels as features and dispersion metrics as signals. An encoder was used on top of a decoder to determine the importance of features despite their shared covariance. The DDS was found to be the only significant feature ($\beta=0.31$; $p=0.043$). (D) The dispersion metrics were used to predict DDS scores using B2B on each metric. The color code is the same as in Figure 2A with the difference that blue corresponds to an anti-correlation and red to a correlation between the DDS and the corresponding dispersion metric. The results show that the DDS can predict mainly the same metrics that are significant in Figure 3A. A higher DDS trait score is associated with less dispersion in these metrics. More specifically, a higher DDS score was associated with lower average eccentricity in SM ($\beta=0.13$; $p=0.023$), VA ($\beta=0.09$; $p=0.046$) and limbic ($\beta=0.11$; $p=0.034$) networks. A higher DDS score was also associated with less dispersion between the SM network on one side and the DA ($\beta=0.21$; $p=0.006$), VA ($\beta=0.1$; $p=0.038$), limbic networks ($\beta=0.15$; $p=0.016$), and DMN ($\beta=0.11$; $p=0.029$) on the other side, and between the limbic and the VA networks ($\beta=0.11$; $p=0.036$), and between the FP network and the DMN ($\beta=0.15$; $p=0.015$).

| Cluster region | *p*FDR | *kE* | *Tmax* | *x* | *y* | *z* |
|---|---|---|---|---|---|---|
| L superior temporal | <0.001 | 854 | -4.00 | -50 | 0 | -6 |
| R parahippocampal | <0.001 | 774 | -3.93 | 32 | -11 | -39 |
| R visuomotor | <0.001 | 379 | -3.9 | 37 | -47 | 50 |
| R premotor + SMA | <0.001 | 219 | -3.5 | 5 | 13 | 71 |
| L supramarginal | <0.001 | 148 | -3.3 | -48 | -40 | 46 |
| L premotor + SMA | 0.001 | 105 | -2.8 | -32 | -1 | 52 |
| L angular | 0.002 | 128 | -3.4 | -31 | -50 | 38 |
| L visuomotor | 0.002 | 105 | -3.3 | -9 | -46 | 41 |
| R dorsal PCC | 0.004 | 95 | -3.9 | -10 | -12 | 43 |
| L ventral ACC | 0.015 | 106 | -3.2 | -10 | -12 | 43 |
| L medial temporal | 0.027 | 85 | -3.2 | -45 | -52 | 12 |

**Table 3:** Surface-based analysis of the averaged states' eccentricity. All eccentricity cluster were lower for experts than for novices. SMA: Supplementary motor area. PCC: Posterior Cingulate Cortex. ACC: Anterior Cingulate Cortex.

# DISCUSSION:

In this study, we first measured the vertex-wise eccentricity of the diffusion map embedding gradient, which reflects the functional integration (low eccentricity) and segregation (high eccentricity) along a scalar value (21,37,42). We found a higher integration for experts during OP (Figure 2D) and as a trend during RS (Figure 2D) in the overall mean eccentricity values. In line with these findings, previous studies have observed increased integration in long-term meditation practitioners' brains using different methodologies, such as graph analysis (43) and diffusion-weighted imaging (44–46). Additionally, using diffusion map embedding, similar but much more pronounced patterns of increased integration have been identified in studies investigating the acute effects of psychedelics (22,23). This consistency across different research approaches provides some support to our initial hypothesis of a compression of the cognitive hierarchy associated with the lessening of self-related/discursive processes during non-dual meditation akin to OP meditation. This observed increase in brain integration has been proposed to be a consequence of a heightened state of brain entropy, which has been observed both in psychedelic experiences (47) and during meditation (48). For instance, the REBUS model theorizes that this heightened brain entropy state is associated with "a relaxation of the precision weighting of priors that coincide with liberation of bottom-up signaling" (49). This means that the brain becomes less reliant on its pre-existing expectations or beliefs and pays more attention to the incoming sensory information. This mechanism also aligns with the description provided within the free energy principle framework (50) of the deconstructive meditation family (51) as cultivated in OM meditation and non-dual meditative states such as OP. Hence, our study provides some support with these current theories, which should guide the future empirical studies of these non-dual meditations.

Next, we showed specific group-related dispersion metrics effects (Figure 2B) which were not identical across the three states, suggesting both a group-by-state effect and a state independent trait effect (Figure 2A,C). To be able to identify a state independent trait measure, we used a SVC to decode expertise and to test whether its training generalized to other states. First, we managed to decode the group only when it was trained and tested on the OP state, suggesting that this state was functionally the most different between groups. Yet, this pattern did not generalize as a trait, meaning that the SVC weights likely captured also a trait-by-state effect. Next, we repeated the same procedure on the average of the three states, as the trait effect has been characterized by low-variability functional connectivity (52). If OP-related group differences was reflecting only a state effect, the predictability should decrease, because noise was added during the averaging. If, instead, the average of the three states was reducing noise by repeating a trait-like feature, then the predictability and its ability to generalize to other states should increase. We found some evidence for the latter (Figure 2E), suggesting that the average eccentricity was the best characterization of a trait-like effect in our sample.

Subsequent analyses specified the specific fingerprint of meditation expertise. Experts exhibited reduced average eccentricity in dorsal attention (DA) and limbic networks, and, at tendency, in the SM cortex suggesting that these networks were more integrated within the cognitive hierarchy for experts, allowing for enhanced information exchange with other networks (21,37). In line with these findings, our surface-based analysis revealed clusters exhibiting solely decreased eccentricity among the expert group (Figure 3B) in the parahippocampal gyrus, premotor gyrus, and supplementary motor area, thereby confirming the results obtained from the average eccentricity analysis described above. Experts also demonstrated a more reduced within-network dispersion in the DA and VA networks compared to novices, as previously reported in the literature on meditation traits (14,26,27), suggesting an enhanced spread of information within these networks, as their voxels exhibit stronger connectivity. Finally, for experts only, the limbic network displayed increased connectivity with the SM, VA, and FP networks, the SM cortex exhibited stronger connectivity with the DA network, while the VA network demonstrated enhanced connectivity with the FP network. Numerous brain imaging studies on meditation have similarly highlighted the role of these attention and affective brain networks during meditation practices (for reviews see Tang et al. (24), Lutz et al. (53), and Sezer et al. (54)). The functional coupling of these networks with the SM network in meditation is more rarely reported (55), even if it is consistent with the embodied nature of this practice (56,57). This finding pointed toward an important functional modulation of the SM cortex in meditation practice, which have often not been utilized as seeds or networks of interest in previous ICA studies.

We reported that several metrics capturing the meditation expertise fingerprint were correlated with the ability to create psychological distance between thoughts and emotions, as measured by the DDS. These correlations were assessed while considering the co-variation of all metrics included in the demographic table (Table 1), including expertise. Specifically, and in line with the trait fingerprint, a higher DDS score was associated with reduced averaged eccentricity in the SM cortex and limbic network. Additionally, the DDS negatively correlated with dispersion between the DA network and SM cortex, as well as between the limbic network and the SM and VA networks. These correlations between a higher DDS score and more integrated limbic, SM and VA networks, may explain how experts manage to modify their emotional processing. For example, Zorn and colleagues showed on the same sample of participants that these experts were more able to reduce and to decouple the unpleasantness of a painful stimulus from its intensity than novices (58). Moreover, they also showed that the DDS was a core mechanism to explain the stronger sensory-affective uncoupling of pain found in experts (59).

Finally, consistent findings in the study of meditation traits involve connectivity modifications between the FP and DMN, with both increased and decreased connectivity reported in the literature (13,14,26,27,29,30,60–62). However, our expertise fingerprint was not associated with such differences, despite a negative correlation between the DDS and dispersion between the FP network and the DMN. Several factors may explain these discrepancies (see Sezer et al. (54) for discussion).

Our study had several limitations. The cross-sectional nature of the study limits our ability to establish causal relationships between variables. Although efforts were made to control for potential confounding variables by matching experts and novices for age, sex, and education, there may still be unaccounted factors that could explain the observed differences. In addition, our study was mainly exploratory as we used diffusion embedding to study the effect of long-term meditation practice on the brain, thus our findings will require replication by future studies.

In conclusion, our study investigated the effects of long-term meditation practice on brain functional architecture and connectivity patterns, focusing on the development of shared characteristics associated with expertise in meditation. We identified large-scale networks associated with meditation expertise, which were not limited to specific meditative states, and which shed new light on the neural mechanisms of cognitive defusion as measured by DDS.

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All authors of the present article declare no conflicting interests.

# CREDIT AUTHORSHIP CONTRIBUTION STATEMENT:

**Gaël Chetelat**: Supervision. **Sébastien Czajko**: Methodology, Validation, Formal analysis, Data curation, Writing, Visualization. **Loïc Daumail**: Formal analysis. **Antoine Lutz**: Methodology, Conceptualization, Supervision, Writing – editing, Funding acquisition. **Daniel Margulies**: Methodology, Supervision. **Ronan Perry**: Methodology, Validation. Formal analysis. **Joshua Volgestein:** Methodology, Supervision **Jelle Zorn**: Conceptualization, Methodology, Investigation. Data Collection

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# Supplemental Material (SM)

# METHOD

## PARTICIPANT:

Initial studies on expertise suggested that 10.000 hours of deliberate practice were required to reach such a stage (1). However, meta-analyses have shown that experience itself is not a sufficient criteria to quantify such a construct (2,3), as it is not always a stronger predictor than a measure related to the domain of expertise (4). Following these recommendations, we operationalized here meditation expertise by combining objective and intersubjective criteria. Firstly, practitioners had to have learned and practiced the same styles of meditations, as it is proposed in traditional 3-year meditation retreat in the Kagyu or Nyingma school of Tibetan Buddhism, where the practitioners meditate 8-12 hours a day. They also had to have the minimum requirement of 10,000 hours of formal practice as well as a regular daily practice, ensuring a certain degree of expertise of the kind of meditation practice we wished to investigate. Secondly, a research assistant (CB) highly experienced with these practices checked that the practitioners' Buddhist communities perceived them as meditators sufficiently skilled and experienced in these practices. As such, we required experts to have an extensive and regular practice in the Karma Kagyu (Mahamudra) or Nyingma (Dzogchen) schools or both. More precisely, practitioners should have accumulated ≥ 10000 hours of experience, followed at least one formal 3-year meditation retreat, and have a regular daily practice of ≥ 45 minutes in the year preceding the study. Novices should have no experience in meditation or comparable mind-body practices (e.g. yoga, tai chi, qi-gong etc.). Notable exclusion criteria regardless of group were pre-existing neurological or psychiatric conditions (e.g. epilepsy, depression) and/or any condition involving sensitization to pain (e.g. chronic pain, fibromyalgia), had no family history of epilepsy, their score to the Beck Depression Inventory (BDI) had to be below 20, had no severe hearing loss, use of medication that could interfere with relevant cognitive functions such as the central nervous system (e.g. antidepressants, opioids) or the pain system (e.g. nonsteroidal anti-inflammatory drugs). Women had not to be pregnant, breastfeeding or having given birth in the last 6 months. Subjects had to be compatible for MRI sessions, such as not being claustrophobic, not having metal implants and dental prostheses.

## MEDITATION PRACTICE:

Mindfulness meditation consists of cultivating a vigilant awareness of one's own thoughts, actions, emotions and motivations (5). The participant learns to intentionally pay attention to his or her internal or external experiences in the present moment, without making any value judgment. The aim is that the present moment is lived in a more open and flexible way and is less dominated by mental conditioning that is a source of suffering. Two standard styles of mindfulness meditation are FA meditation and OM meditation (6). FA involves sustaining one's attentional focus on a particular object, either internal (e.g., breathing) or external (e.g., a candle flame). The practitioner is instructed to monitor their attention, notice episodes of distraction (mind-wandering), and bring their attention back to the object. OM practices aims to cultivate and sustain an effortless, open and accepting awareness of present moment experience, without changing, being reactive or absorbed in its contents. Such a dereifying perspective purportedly allows one to recognize that all components of conscious experience are simply mental events, and thus do not necessarily need to be acted upon. Thus, the aim of this training is not to explicitly change, alter or suppress experiential content, but rather to change one's relation to it. As such, an unpleasant experience might be perceived with equal or even increased vividness during an OM state, without the fear and emotional reactivity that usually accompanies such experience. "Open Presence" (OP) in Tibetan Buddhist traditions is a paradigmatic case of a so-called non-dual mindfulness meditation. Styles of meditation that cultivate OP are described as inducing a phenomenal experience where the intentional structure involving the duality between object and subject is attenuated, as captured by the notion of non-duality. OP style of practices can be found in both the Dzogchen (Tibetan, Rdzogschen) and Mahamudra or Chagchen (Tibetan, phyagchen) traditions of Tibetan Buddhist meditation (Schaik), that are highly overlapping and have as central tenet the cultivation of the OP state (Tibetan, "rig pa cog gzhag", pronounced "rigpachokshak", literally "freely resting in what consciousness manifests"). An exemplary instruction goes as follows: "Within a state free of hopes and fears, devoid of evaluation or judgment, be carefree and open. And within that state, do not linger on the past; do not invite the future; place awareness within the present, without alteration, without hopes or fears" (as quoted in (7)). Based on its traditional presentation (Namgyal, Third Dzochen Rinpoche), OP practice is considered here an advanced form of OM practice, where practitioners might have reached various stages of accomplishment. Theoretically, OP meditation consists of a state where the phenomenological qualities of effortlessness, openness, and acceptance are vividly experienced, and control-oriented elaborative processes are reduced to a minimum. Getting familiar and stabilizing the suspension of these elaborative processes requires substantial training. For this reason, the term OP will be used here only for expert meditators, even if both experts and novices received the same meditation instructions during the task. Below we use the term OM for both novices and experts, and we will assume that for experts only it will also qualify as a practice of OP. In addition, both groups practice a state of compassion meditation. Building on the non-judgmental monitoring capacity developed in OM/OP, the participants learn to cultivate kindness toward oneself for instance in relation to one's negative thoughts, distractions, difficult emotions, unpleasant physical sensations to foster appreciation toward positive qualities of one's mind (joy, contentment, …). The participants then learn to extend a similar attitude of care and loving-kindness toward their loved ones, toward neutral persons or toward difficult persons, ultimately recognizing that the need for comfort, security, and happiness is shared by all living beings. Experts referred to this practice as 'nonreferential compassion'' (dmigs med snying rje in Tibetan) or unconditional loving-kindness and compassion, which is described as an ''unrestricted readiness and availability to help living beings.' (8)

## PARADIGM (STATES INSTRUCTIONS)

At rest, participants were instructed to keep their eyes closed and let their thoughts run freely without falling asleep. For the open-monitoring condition, participants would start the meditation by anchoring their attention in their body, while keeping both body and mind relaxed. Subsequently, they were instructed to imagine their mind as a vast and clear space and to allow any arising experiences to occur naturally without resistance, while simultaneously not engaging in any distractions. As for the compassion state, participants were asked to relax and bring to their mind the image of a close relative, and to fill their mind with love directed to this person and the wish of good for oneself, and that any suffering may be dispelled. After each fMRI scan, we asked participants to respond to self-reports of subjective experience during the different states using a 1-10 item Likert scale as follows:

clarity of mind during a block, serenity -peace of mind during a block- and valence -how positive their experience was during a block.

## PSYCHOMETRIC SCALES

The Drexel defusion scale (DDS) measures a person's ability to distance from a variety of psychological experiences using a 10-item questionnaire. The questionnaire begins with an introduction to the concept of cognitive defusion, a concept closed to dereification (9), which is intended to help respondents understand the construct. Participants were asked to indicate the extent to which they would be able to cognitively defuse themselves from hypothetical situations with negative thoughts or feelings on a 6-point Likert scale ranging from "not at all" (0) to "very much" (5). Higher scores indicate better cognitive defusion. The DDS showed good preliminary internal consistency ($\alpha = .83$), and high convergent and divergent validity (10).

The Five facets mindfulness questionnaire (FFMQ) is a 39-item questionnaire that measures five purported mindfulness dimensions including: observing (noticing or attending to internal/external experiences), describing (labeling internal experiences with words), acting with awareness (attending to present moment experience), non-judging (adopting a non-evaluative stance toward thoughts and feelings), and non-reacting (allowing thoughts and feelings to pass). Participants indicate to what degree they experience these dimensions in their daily life on a 5-point Likert-type scale ranging from 1 (never or very rarely true) to 5 (very often or always true). Scores are calculated separately for subscales, with higher scores reflecting higher mindfulness. The FFMQ facets have been found to demonstrate adequate to good internal consistency, with α coefficients ranging from .75 to .91 (11).

The Beck depression inventory (BDI) is a 21-item questionnaire that measures characteristic attitudes and symptoms of depression. Each question has four scores ranging from 0 (symptom not present) to 3 (symptom very intense). A total sum score is calculated to reflect depression severity. The BDI-I has shown good internal consistency, with α coefficients of .86 and .81 for psychiatric and non-psychiatric populations, respectively, and good concurrent and discriminant validity (12).

## Data acquisition

Neuroimaging data was collected on a 3T Siemens Prisma scanner (Erlangen, Germany) with a 64-channel head/neck coil. Functional imaging data were acquired using an EPI sequence with the following parameters: TR = 2100 ms, TE = 30 ms, number of slices = 39, slice thickness = 3.1 mm, gap between slices = 3.1 mm, voxel size = 2.8 x 2.8 x 3.1 $mm^3$, flip angle = 80˚. In addition to that, the following structural images were recorded for anatomical reference: T1-weighted (TR = 2500 ms, TE = 2.83 ms, flip angle = 6˚, voxel size = 1 x 1 x 1 $mm^3$), T2-weighted (TR = 2500 ms, TE = 261 ms, flip angle = 120˚, voxel size 1 x 1 x 1 $mm^3$) and non-EPI T2*-weighted (TR = 3680 ms, TE = 30 ms, flip angle = 90˚, voxel size = 1 x 1 x 1 $mm^3$).

## Pre-processing

Results included in this manuscript come from preprocessing performed using fMRIprep version 1.2.6-1 (13,14), a Nipype (15,16) based tool. Each T1w (T1-weighted) volume was corrected for INU (intensity non-uniformity) using N4BiasFieldCorrection v2.1.0 (17) and skull-stripped using antsBrainExtraction.sh v2.1.0 (using the OASIS template).

Brain surfaces were reconstructed using recon-all from FreeSurfer v6.0.1 (18), and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical gray-matter of Mindboggle (19). Spatial normalization to the ICBM 152 Nonlinear Asymmetrical template version 2009c (20) was performed through nonlinear registration with the antsRegistration tool of ANTs v2.1.0 (21), using brain-extracted versions of both T1w volume and template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1w using fast (FSL v5.0.9) (22).

Functional data was slice time corrected using 3dTshift from AFNI v16.2.07 (23) and motion corrected using mcflirt (FSL v5.0.9) (24). This analysis was followed by co-registration to the corresponding T1w using boundary-based registration (25) with nine degrees of freedom, using bbregister (FreeSurfer v6.0.1). Motion correcting transformations, BOLD-to-T1w transformation and T1w-to-template (MNI) warp were concatenated and applied in a single step using antsApplyTransforms (ANTs v2.1.0) using Lanczos interpolation. Physiological noise regressors were extracted applying CompCor (26). Principal components were estimated for the two CompCor variants: temporal (tCompCor) and anatomical (aCompCor). A mask to exclude signal with cortical origin was obtained by eroding the brain mask, ensuring it only contained subcortical structures. Six tCompCor components were then calculated, including only the top 5% variable voxels within that subcortical mask. For aCompCor, six components were calculated within the intersection of the subcortical mask and the union of CSF and WM masks calculated in T1w space, after their projection to the native space of each functional run. Frame-wise displacement (27) was calculated for each functional run using the implementation of Nipype. ICA-based Automatic Removal Of Motion Artifacts (AROMA) was used to generate aggressive noise regressors as well as to create a variant of data that is non-aggressively denoised (28).

Many internal operations of FMRIPREP use Nilearn (29), principally within the BOLD-processing workflow. For more details of the pipeline see https://fmriprep.readthedocs.io/en/stable/workflows.html.

## Connectome gradient construction

Following Hong and colleagues (30), we projected grey-matter voxels on the brain surface to 10242 vertices per hemisphere and downsampled the time-series data. Then, using the fMRI time-series matrix in each subject, we calculated functional connectomes based on Pearson correlations. As in Margulies et al. (2016) (31) and other studies (30,32,33), we z-transformed and thresholded this matrix, leaving only the top 10% of weighted connections per row, and calculated a cosine similarity matrix that captures similarity in connectivity profiles between vertices. We applied diffusion map embedding (31,34), a nonlinear reduction technique, to identify principal gradient components explaining connectome variance in descending order. In this study, we followed the previous recommendation (30,31) and set $\alpha = 0.5$, a choice that retains the global relations between data points in the embedded space. We performed Procrustes rotation (https://github.com/satra/mapalign) to align components of each individual to the group-level embedding based on the Human Connectome Project S1200 sample (41).

## Identifying co-varying factors

This method first fits a decoding model **G** on half of the dataset to find the combination of dispersion measures that maximally decodes a given trait factor. Then, it fits an encoding model **Hf** on the other half of the dataset to estimate whether the decoded **f-hat** predictions are specific to **f** and/or attributable to other covariant factors.

Interestingly, it can be shown that the coefficients of **Hf** tend to a positive value if and only if **f** is linearly and specifically coded in the measures and not reducible to its covariant factors. In other words, this means that if a trait factor is not encoded in the neural responses, there will be no reliable match between the true factor and the model prediction. Thus, it is possible to use these coefficients as a measure of specific decoding performance and test whether they are statically greater than zero or not.

In order to implement the B2B regression, we first fitted a ridge regression model **G** on a random half of the dataset, across the 35 measures of dispersion, using the RidgeCV function of scikit-learn (35). Then we again used a RidgeCV function on the remaining half of the dataset to implement the **Hf** encoding model for each of the encoded trait factor. We repeated this process 10,000 times, shuffling half the dataset at each repetition. We used the same procedure to compute a B2B regression model with permuted features, with replacement, to draw the null hypothesis distribution. Significance p-values were obtained by comparing the proportion of times the averaged coefficient of a given factor exceeds the null model coefficient.

In our case, only the DDS score had a positive value in **Hf**. To better characterize the specific dispersion measures underlying this effect, we ran 35 exploratory B2B regression analyses to predict each dispersion measure. We reported on Figure 3D, the p-value testing the hypothesis that the coefficients of **Hf** for the DDS was positive for every dispersion measure.

## Statistical analysis

### Voxel-wise analysis

We statistically compared gradient component scores between experts and controls using surface-based linear models implemented in SurfStat (http://www.math.mcgill.ca/keith/surfstat/) for Matlab. Surface-based findings were corrected for family-wise errors (FWE) due to multiple comparisons using a random field theory of pFWE < 0.05.

### Dispersion metrics analysis

In order to show group differences through all dispersion metrics, we used the package hyppo (36) to perform a multivariate non-parametric two-sample test. To compare differences of dispersion, we computed Studentized bootstrap tests with 10,000 repetitions also called bootstrap-t test (37) rather than classical parametric Student tests. Although these tests are more computationally demanding, they are also less stringent regarding their assumptions, as they do not require assessing the normality of the sample, for example.

### Generalization

We used the stochastic gradient descent classifier in scikit-learn (29) with a modified-huber loss to predict expertise based on the within-network dispersion, the between-networks dispersion and average dispersion of the different states or the average of the three states. The goal was to assess which state predicts the best expertise trait, as it is very likely that expertise leads to state differences, even for the RS. We used a 5-fold cross-validation to separate training and test data. We repeated this procedure 5000 times with different sets of training and test data to avoid bias for separating subjects. Because markers of expertise should be present in every state, we expected that the trained classifier trained on a given state should also be able to predict expertise if tested on the other states. Thus, we also included test batches from the other states to see how well the classifier generalized. Given the fact that we had an unbalanced number of samples, we did not use accuracy as a metric of prediction performances. We preferred the area under the curve (AUC) of the receiver operating curve to compare the prediction performances, which does not suffer from unbalanced samples. The score for each test was defined as the mean AUC of the 5000 repetitions. We tested these scores for significance by computing the null-distribution using permuted labels (38).

### Identifying co-varying factors

In order to disentangle the respective contribution of collinear trait factors described in the Demographic (Table 1) on the measures of dispersion (Figure 2A), we computed a back-to-back (B2B) regression (39,40). This approach takes advantage of both encoding and decoding techniques: encoding models can disentangle the specific contribution of co-variable trait factors on a given dispersion measure, while decoding can combine these multiple factors into a linear model, in order to better capture the signal of interest despite a low signal-to-noise ratio.  In our case, only the DDS score had a significant contribution. To better characterize the specific dispersion measures underlying this effect, we ran 35 exploratory B2B regression analyses to predict each dispersion measure. We reported on Figure 3D, the p-value testing the hypothesis that the coefficients for the DDS was positive for every dispersion measure.


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Study 2: Insights into the Brain's Connectivity: Compassion and Loving-kindness Meditation Effects on the Connectome in Novice and Expert Meditators

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# ABSTRACT:

Mindfulness-based meditation practices have been linked to widespread beneficial effects on cognition, mental health, physiology, and behavior. Neuroimaging research has provided insights into the brain mechanisms underlying these effects, showing that mindfulness impacts activity and connectivity in fronto-parietal, default mode, and salience networks involved in attention, emotion regulation, and self-processing. In contrast, few studies have examined the neural impact of loving-kindness (LKC) meditation focused on cultivating care and compassion. This fMRI study investigated differences in intrinsic functional connectivity between LKC, open presence meditation, and resting states in long-term practitioners versus novices. We applied gradient connectivity methods sensitive to alterations in large-scale network dynamics. Results showed distinct state-dependent modulations - LKC reduced integration metrics compared to rest, specifically within sensorimotor, limbic, default mode, and attention regions. Critically, novices exhibited greater state-related changes in

eccentricity and between-network dispersion compared to experts. These findings demonstrate that meditation training induces neuroplastic adaptations, characterized by focused consolidation of affective, interoceptive, and attentional networks. The modulatory impact of LKC was more pronounced early in training, while extensive practice led to stable connectivity profiles across meditative states. Characterizing transient and enduring neural effects of compassion training has important implications for understanding contemplative neuroplasticity.

# INTRODUCTION:

In our prior work, we highlighted the association between meditation expertise as a trait measure and distinct, enduring patterns of functional connectivity (Czajko et al., 2023). These patterns were observed during both meditative states (involving open monitoring/open presence and loving-kindness compassion meditations) and non-meditative resting states. This association was quantified using connectivity gradients, as detailed in our previous study (Czajko et al., 2023). In the current investigation, our objective is to delve deeper into the characterization of the correlates associated with these meditative states. This exploration will be accomplished again through an analysis of large-scale functional magnetic resonance imaging (fMRI) connectomics.

In general, mindfulness meditation is often described as the cultivation of present-moment awareness accompanied by open and accepting attitudes (Kabat-Zinn et al.). However, a more nuanced understanding may be achieved through a family resemblance approach, acknowledging a diverse family of practices that align themselves with mindfulness-related principles (Lutz et al., 2015). A growing body of studies have documented beneficial effects of mindfulness across domains including cognitive performance, mental health, physiology, and behavior (Keng et al., 2011; Kral et al., 2018; Kral, Lapate, et al., 2022). Neuroimaging research has been instrumental in unraveling the brain mechanisms underlying these effects.

Functional MRI (fMRI) studies show mindfulness meditation impacts activity and connectivity between the fronto-parietal (FP) network, default-mode-network (DMN) and salience network involved in attention control, emotion regulation, and self-processing (Bremer et al., 2022; Tang et al., 2015). Structural MRI studies reveal alterations in gray matter morphology and white matter connectivity in meditators versus controls, indicative of experience-dependent neuroplasticity (Fox et al., 2014), even though short 8-week interventions are not enough to induce them (Kral, Davis, et al., 2022). Overall, findings from neuroscience research contextualize mindfulness within a brain-based framework and provide biological plausibility for behavioral and clinical effects. Looking ahead, additional research is needed to further characterize the time course, dose-effects, and individual differences modulating mindfulness-based neural plasticity.

Loving-kindness (metta) and compassion (karuna) are foundational Buddhist meditative practices traditionally cultivated to generate benevolence towards oneself and others. Loving-kindness meditation (LKM) involves directing feelings of warmth, care, and goodwill towards a sequence of imagined beings (Fredrickson et al., 2008; Salzberg, 1995). Practitioners repeat phrases like "may you be happy" to evoke a heartfelt intention for wellbeing. In contrast, compassion meditation (CM) trains one to meet suffering with empathy, acceptance, and the sincere desire to alleviate pain. Phrases wishing freedom from suffering are repeated (e.g. "may you be free from pain"). While related in their focus on kindness, LKM and CM differ regarding their object of concentration. LKM emphasizes cultivating positivity and joy, whereas CM requires staying present with suffering (Gilbert, 2009). However, compassion training programs often integrate both techniques given their conceptual links within Buddhist teachings on the "four immeasurables" - compassion, loving-kindness, empathetic joy, and equanimity. Compared to mindfulness practices, LKM and CM, below referred as LKCM meditation, have a stronger interpersonal orientation which may more directly enhance prosocial emotions and behaviors (Hutcherson et al., 2008; Kang et al., 2014).

A growing body of research has begun elucidating how LKCM practices impact functional brain activity. Neuroimaging studies indicate LKCM activate fronto-limbic and default-mode networks involved in emotional sharing, embodied simulation, and socio-affective processing (Desbordes et al., 2012; Fox et al., 2016; Klimecki et al., 2013; Mascaro et al., 2013). These areas are involved in self-referential processing and often show reduced activation during LKC, reflecting decreased self-focused rumination (Garrison et al., 2014). LKC practice has also been found to alter functioning in limbic areas including the insula and amygdala. Increased insula activation is thought to reflect enhanced interoceptive awareness and embodied simulation of others' affective states (Desbordes et al., 2012; Lutz, Brefczynski-Lewis, et al., 2008). Changes in somatosensory and motor regions implicate bodily and movement related networks in embodying compassion during LKC (Fox et al., 2016). This aligns with perspectives highlighting visceral feedback and facial expressions as important drivers of empathic responses. Lastly, Valk and colleagues (Valk et al., 2023) found that targeted socio-affective skills training led to distinct changes in cortical function and microstructure compared to other training modules. Unlike attention-mindfulness and socio-cognitive training, affect training, including LKM, did not alter overall integration or segregation of intrinsic brain networks. However, it resulted in widespread decreases in cortical myelin content in middle and deep cortical layers, especially in frontal and parietal regions implicated in emotional processing. In this way, LKM and CM can be distinguished from mindfulness and open monitoring techniques while similarly inducing beneficial functional neural changes.

An important consideration in studying meditation is the role of practice expertise. Several studies have compared functional and structural brain changes in novice versus expert meditators. These works reveal crucial insights into how repeated practice sculpts enduring trait-level shifts in brain function (Brewer et al., 2011; Hasenkamp & Barsalou, 2012; Hölzel et al., 2008). In novices, LKC induces transient state-dependent increases in activity and connectivity across limbic, default mode, and sensorimotor networks. The observed modulation reflects an explicit reorganizing of functional neural dynamics during active meditation (Desbordes et al., 2012; Garrison et al., 2014). However, experts exhibit fewer state-related changes. Their brain activation patterns remain consistent across rest and meditation conditions. This implies extensive LKC training produces stable trait-level transformations. Longitudinal studies support this notion, demonstrating enhanced structural plasticity and sustained functional connectivity patterns in experts following repeated practice over years (Fox et al., 2016; Hölzel et al., 2008; Lutz, Brefczynski-Lewis, et al., 2008). The cultivation of enduring neural dispositions is thought to support spontaneous experiences of compassion, even outside formal meditation.

Overall, these findings demonstrate that loving-kindness compassion meditation induces robust yet complex changes in brain function and connectivity patterns. Further research on the complex interplay between transient and enduring effects of compassion training will be important for comprehensively mapping functional changes underlying socio-affective processing. Advancing such models has implications for understanding contemplative neuroplasticity and for optimizing real-world applications of LKC-based interventions.

The purpose of the present study is to investigate the cognitive architecture associated with LKC meditation in long-term practitioners and novices. We aim to characterize differences in intrinsic functional connectivity between LKC and resting states using a data-driven gradient connectivity approach. As, we saw, only a few studies have examined the neural impact of compassion-focused practices. Furthermore, the role of expertise in modulating meditation states remains unclear. To address these gaps, we compare intrinsic connectivity networks between LKC and rest in practitioners with over 10,000 hours of experience versus novices. We apply gradient connectivity methods, which capture cortical organization along continuous dimensions (Coifman et al., 2005; Margulies et al., 2016; Valk et al., 2023). This technique is sensitive to alterations in large-scale brain dynamics and can reveal meditation-related changes in network integration (Czajko et al., 2023; Valk et al., 2023). Based on prior findings, we predict LKC will modulate connectivity gradients compared to rest, with the strongest effects in novices. Specifically, we hypothesize LKC will compress the cognitive hierarchy gradient and enhance integration of fronto-parietal and salience networks. Testing these predictions will advance understanding of how compassion-based contemplative states shape functional connectivity architectures. Characterizing network dynamics underlying socio-affective processes has important implications for future translational applications.

# METHOD:

## Participants

Participants were recruited for the Brain and Mindfulness ERC-funded project, which includes a cross-sectional observational neuroscientific study on the effect of mindfulness meditation on experiential, cognitive and affective processes conducted in Lyon, France, from 2015 to 2018. Participants included novice and long-term meditation practitioners (hereafter referred to as experts), who were recruited through multiple screening stages (see the Brain & Mindfulness Project Manual, Abdoun et al., 2018).

The participants included for this study were 75 cognitively normal adults aged 35-66 years old (SD 7.7) including 28 expert meditators and 47 control participants (hereafter referred to as novices) matched on age and gender ($p>0.5$, see Table 1 in Czajko et al., 2023). Novices attended a one-weekend meditation training program prior to any measurements to get familiarized with the meditation techniques. Initial studies on expertise suggested that 10,000 hours of deliberate practice were required to reach such a stage (Anders Ericsson, 2008). However, meta-analyses have shown that experience itself is not a sufficient criterion to quantify expertise, as it is not always a stronger predictor than a measure related to the domain of expertise (Hambrick et al., 2018).

Following these recommendations, we operationalized meditation expertise here by combining objective and intersubjective criteria. Firstly, practitioners had to have learned and practiced the same styles of meditations, as proposed in traditional 3-year meditation retreats in the Kagyu or Nyingma schools of Tibetan Buddhism, where practitioners meditate 8-12 hours a day. They also had to meet the minimum requirement of 10,000 hours of formal practice as well as maintain a regular daily practice, ensuring a certain degree of expertise in the specific meditation practices we aimed to investigate. Secondly, a research assistant (C.B.) highly experienced with these practices verified that the practitioners' Buddhist communities perceived them as sufficiently skilled and experienced in these practices.

As such, we required experts to have an extensive and regular practice in the Karma Kagyu (Mahamudra) or Nyingma (Dzogchen) schools or both. More precisely, practitioners should have accumulated ≥10,000 hours of experience, completed at least one formal 3-year meditation retreat, and maintained a regular daily practice of ≥45 minutes in the year preceding the study. Novices should lack significant experience in meditation or comparable mind-body practices (e.g. yoga, tai chi, qi-gong).

Notable exclusion criteria regardless of group were: pre-existing neurological or psychiatric conditions (e.g. epilepsy, depression); any condition involving sensitization to pain (e.g. chronic pain, fibromyalgia); family history of epilepsy; Beck Depression Inventory (BDI) score above 20; severe hearing loss; use of medication that could interfere with relevant cognitive functions such as the central nervous system (e.g. antidepressants, opioids) or the pain system (e.g. nonsteroidal anti-inflammatory drugs); pregnancy, breastfeeding or having given birth in the last 6 months.

Subjects also had to be compatible for MRI sessions, such as not being claustrophobic, not having metal implants and dental prostheses. Finally, subjects had to be affiliated with a social security system. All participants received information on the experimental procedures during a screening session, and signed an informed written consent. The study was approved by the regional ethics committee on Human Research (CPP Sud-Est IV, 2015-A01472-47).

In order to control for potential motion effects, all participants with more than 0.3mm/degree movement were considered outliers (2 experts and 3 novices) and thus removed from the analysis (Power et al., 2014).

## Meditation practices

Mindfulness meditation consists of cultivating a vigilant awareness of one's own thoughts, actions, emotions and motivations (Kabat-Zinn, 1982). The participant learns to intentionally pay attention to his or her internal or external experiences in the present moment, without making any value judgment. The aim is that the present moment is lived in a more open and flexible way and is less dominated by mental conditioning that is a source of suffering.

Two standard styles of mindfulness meditation are focused attention (FA) meditation and open monitoring (OM) meditation (Lutz, Brefczynski-Lewis, et al., 2008). FA involves sustaining one's attentional focus on a particular object, either internal (e.g., breathing) or external (e.g., a candle flame). The practitioner is instructed to monitor their attention, notice episodes of distraction (mind-wandering), and bring their attention back to the object.

OM practices aim to cultivate and sustain an effortless, open and accepting awareness of present moment experience, without changing, being reactive or absorbed in its contents. Such a dereifying perspective purportedly allows one to recognize that all components of conscious experience are simply mental events, and thus do not necessarily need to be acted upon. Thus, the aim of this training is not to explicitly change, alter or suppress experiential content, but rather to change one’s relation to it. As such, an unpleasant experience might be perceived with equal or even increased vividness during an OM state, without the fear and emotional reactivity that usually accompanies such experience.

“Open Presence” (OP) in Tibetan Buddhist traditions is a paradigmatic case of a so-called non-dual mindfulness meditation. Styles of meditation that cultivate OP are described as inducing a phenomenal experience where the intentional structure involving the duality between object and subject is attenuated, as captured by the notion of non-duality ((Dahl et al., 2015).

Open Presence (OP) style of practices can be found in both the Dzogchen (Tibetan, Rdzogschen) and Mahamudra or Chagchen (Tibetan, phyagchen) traditions of Tibetan Buddhist meditation (Schaik 2016), that are highly overlapping and have as central tenet the cultivation of the OP state (Tibetan, “rig pa cog gzhag”, pronounced “rigpachokshak”, literally “freely resting in what consciousness manifests”). An exemplary instruction goes as follows: “Within a state free of hopes and fears, devoid of evaluation or judgment, be carefree and open. And within that state, do not linger on the past; do not invite the future; place awareness within the present, without alteration, without hopes or fears” (as quoted in Dunne, 2011).

Based on its traditional presentation (Namgyal and Rinpoche 2004), OP practice is considered here an advanced form of OM practice, where practitioners might have reached various stages of accomplishment. Theoretically, OP meditation consists of a state where the phenomenological qualities of effortlessness, openness, and acceptance are vividly experienced, and control-oriented elaborative processes are reduced to a minimum. Getting familiar and stabilizing the suspension of these elaborative processes requires substantial training. For this reason, the term OP will be used here only for expert meditators, even if both experts and novices received the same meditation instructions during the task.

Below we use the term OM for both novices and experts, and we will assume that for experts only it will also qualify as a practice of OP. In addition, both groups practice a state of compassion meditation. Building on the non-judgmental monitoring capacity developed in OM/OP, the participants learn to cultivate kindness toward oneself for instance in relation to one's negative thoughts, distractions, difficult emotions, unpleasant physical sensations to foster appreciation toward positive qualities of one's mind (joy, contentment, ...).

The participants then learn to extend a similar attitude of care and loving-kindness toward their loved ones, toward neutral persons or toward difficult persons, ultimately recognizing that the need for comfort, security, and happiness is shared by all living beings. Experts referred to this practice as 'nonreferential compassion' (dmigs med snying rje in Tibetan) or unconditional loving-kindness and compassion (LKC), which is described as an ''unrestricted readiness and availability to help living beings' (Dahl et al., 2015).

## Paradigm

All participants attended a single fMRI session in which we first acquired their structural image (T1 weighted MRI scan). We then acquired functional scans (BOLD fMRI), starting with a 10-minute resting state scan. We also acquired 10-minute scans during meditative states of loving-kindness compassion (LKC) meditation and open presence (OP) meditation. The order of acquisition of the two meditative states was randomized across participants. At rest, participants were instructed to keep their eyes closed and let their thoughts run freely without falling asleep. For LKC, participants were instructed to relax and bring to mind the image of a close friend or relative, filling their mind with feelings of warmth, care, and goodwill directed toward that person. For OP, participants were asked to relax, open the scope of their awareness and rest in a state of open, non-judging awareness, not focusing on anything specific but allowing anything arising in the field of awareness without distraction or resistance.

After each fMRI scan, participants completed self-report ratings of their subjective experience during the different states on a 10-point Likert scale, including: clarity of mind, serenity/peace of mind, and valence (how positive the experience was).

We used three self-report questionnaires: the Drexel Defusion Scale (DDS) which measures cognitive defusion or the ability to distance from one's thoughts and feelings (Forman, 2012); the Five Facet Mindfulness Questionnaire (FFMQ) which assesses five aspects of mindfulness in daily life including observing, describing, acting with awareness, non-judging, and non-reactivity (Baer et al., 2008); and the Beck Depression Inventory (BDI) which assesses characteristic attitudes and symptoms of depression (Beck et al., 1988).

The DDS is a 10-item scale where participants rate their perceived ability to defuse from hypothetical negative situations on a 6-point Likert scale. Higher scores indicate greater cognitive defusion. The DDS has demonstrated good internal consistency and validity (Forman, 2012).

The FFMQ is a 39-item scale with subscales corresponding to five facets of mindfulness. Participants rate statements on a 5-point Likert scale with higher scores reflecting higher mindfulness. The FFMQ facets have shown good internal consistency and validity (Baer et al., 2008).

The BDI is a 21-item questionnaire measuring severity of depressive symptoms. Each item is scored from 0 to 3 reflecting symptom intensity. The BDI has exhibited strong psychometric properties including internal consistency and validity (Beck et al., 1988).

In summary, validated self-report questionnaires were used to quantify mindfulness, cognitive defusion, and depressive symptoms in participants undergoing fMRI during rest and two meditation states. Subjective experiences were also rated after each scan. This allows relating neural changes to psychological traits and state experiences.

## Data acquisition

Neuroimaging data acquisition was performed using a 3T Siemens Prisma MRI scanner with a 64-channel phased-array head coil. Functional MRI data were acquired using a T2*-weighted echo planar imaging (EPI) sequence with the following parameters: repetition time (TR) = 2100 ms, echo time (TE) = 30 ms, 39 interleaved axial slices, slice thickness = 3.1 mm with a 3.1 mm interslice gap, in-plane resolution = 2.8 x 2.8 mm, flip angle = 80 degrees. The EPI sequence provided whole-brain coverage and was used to acquire data during resting state and meditation conditions.

In addition, high-resolution anatomical images were acquired for registration and localization of functional data, including:

- T1-weighted image with TR = 2500 ms, TE = 2.83 ms, flip angle = 6 degrees, and 1 mm isotropic resolution.
- T2-weighted image with TR = 2500 ms, TE = 261 ms, flip angle = 120 degrees, and 1 mm isotropic resolution.
- Non-EPI T2*-weighted gradient echo with TR = 3680 ms, TE = 30 ms, flip angle = 90 degrees, and 1 mm isotropic resolution.

The T1-weighted provided high-contrast gray/white matter anatomical reference. The T2-weighted provided complementary anatomical contrast sensitive to fluid. The non-EPI T2* enabled correction of EPI spatial distortion artifacts.

## Pre-processing

Results included in this manuscript come from preprocessing performed using FMRIPREP version stable 1.2.6-1 (Esteban et al., 2019, 2023), a Nipype (K. Gorgolewski et al., 2011; K. J. Gorgolewski et al., 2017) based tool. Each T1w (T1-weighted) volume was corrected for INU (intensity non-uniformity) using N4BiasFieldCorrection v2.1.0 (Tustison et al., 2010) (Tustison et al., 2010) and skull-stripped using antsBrainExtraction.sh v2.1.0 (using the OASIS template). Brain surfaces were reconstructed using recon-all from FreeSurfer v6.0.1 (Dale et al., 1999), and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical gray-matter of Mindboggle (Klein et al., 2017). Spatial normalization to the

ICBM 152 Nonlinear Asymmetrical template version 2009c (Fonov et al., 2009) was performed through nonlinear registration with the antsRegistration tool of ANTs v2.1.0 (Avants et al., 2008), using brain-extracted versions of both T1w volume and template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1w using fast (FSL v5.0.9) (Zhang et al., 2001).

Functional data was slice time corrected using 3dTshift from AFNI v16.2.07 (Cox, 1996) and motion corrected using mcflirt (FSL v5.0.9) (Jenkinson et al., 2002). This analysis was followed by co-registration to the corresponding T1w using boundary-based registration (Greve & Fischl, 2009) with nine degrees of freedom, using bbregister (FreeSurfer v6.0.1). Motion correcting transformations, BOLD-to-T1w transformation and T1w-to-template (MNI) warp were concatenated and applied in a single step using antsApplyTransforms (ANTs v2.1.0) using Lanczos interpolation. Physiological noise regressors were extracted applying CompCor (Behzadi et al., 2007). Principal components were estimated for the two CompCor variants: temporal (tCompCor) and anatomical (aCompCor). A mask to exclude signal with cortical origin was obtained by eroding the brain mask, ensuring it only contained subcortical structures. Six tCompCor components were then calculated, including only the top 5% variable voxels within that subcortical mask. For aCompCor, six components were calculated within the intersection of the subcortical mask and the union of CSF and WM masks calculated in T1w space, after their projection to the native space of each functional run. Frame-wise displacement (Power et al., 2014) was calculated for each functional run using the implementation of Nipype. ICA-based Automatic Removal Of Motion Artifacts (AROMA) was used to generate aggressive noise regressors as well as to create a variant of data that is non-aggressively denoised (Pruim et al., 2015).
Many internal operations of FMRIPREP use Nilearn (Abraham et al., 2014), principally within the BOLD-processing workflow. For more details of the pipeline see https://fmriprep.readthedocs.io/en/stable/workflows.html.

## Connectome gradient construction

Functional connectivity gradients have emerged as a powerful tool for characterizing the large-scale organization of the human cerebral cortex (Huntenburg et al., 2018; Margulies et al., 2016; Valk et al., 2023). As described by Hong and colleagues (Hong et al., 2019), we can derive these gradients by first calculating a functional connectivity matrix between all grey matter voxels or surface vertices based on resting-state fMRI data. Pearson correlations are computed between the fMRI time series of each voxel/vertex pair to generate this matrix (Friston et al., 1993; Greicius et al., 2003).

To reduce the dimensionality of this highly dense matrix, we follow several standard processing steps. First, the connectivity matrix is Fisher z-transformed to normalize its distribution. Next, it is thresholded by retaining only the top 10% of positive connections for each voxel/vertex (Bethlehem et al., 2020; Hong et al., 2019; Margulies et al., 2016; Valk et al., 2023). This focuses the analysis on the strongest functional connections. The thresholded z-matrix is then used to calculate a cosine similarity matrix, which quantifies the similarity of whole-brain functional connectivity profiles between each voxel/vertex pair.

This cosine similarity matrix is provided as input to a nonlinear dimension reduction technique known as diffusion map embedding (Coifman et al., 2005; Margulies et al., 2016). This identifies the primary axes or gradients along which functional connectivity patterns vary across the cortex. The first principal gradient delineates the transition from unimodal sensory and motor regions to transmodal association cortex (Margulies et al., 2016). Higher-order gradients capture additional dimensions of connectivity variation. In this study, we followed the previous recommendation (Hong et al., 2019; Margulies et al., 2016) and set $\alpha = 0.5$, a choice that retains the global relations between data points in the embedded space.

To align gradients across subjects, we applied Procrustes rotation methods (https://github.com/satra/mapalign, Langs et al., 2015). Individual subject gradients were rotated to maximally align with group-level gradient components derived from the Human Connectome Project S1200 release (Demirtaş et al., 2019). This Procrustes transformation aligns the coordinate systems while preserving the topology and spatial relationships of the gradients.

Functional connectivity gradients provide a powerful data-driven approach for distilling the complex network architecture of the human brain into continuous mapping functional hierarchies. Our analytic pipeline follows established best practices for gradient derivation, dimensionality reduction, and cross-subject alignment.

# 3D gradient metrics

As described by Margulies et al. (2016), the leading connectivity gradients (Figure 1B) explain a substantial portion of the total variance in resting-state functional connectivity patterns. For instance, the first three gradient components typically account for over 50% of the variance (Hong et al., 2019; Margulies et al., 2016).

These dominant gradient dimensions can be combined to construct a multidimensional gradient space (Bethlehem et al., 2020; Valk et al., 2023). Each brain vertex is situated within this space based on its functional connectivity profile across the gradients (Figure 1 A). The geometric relationships between brain regions in this space reflect their functional affiliations and interactions.

# FIGURE 1

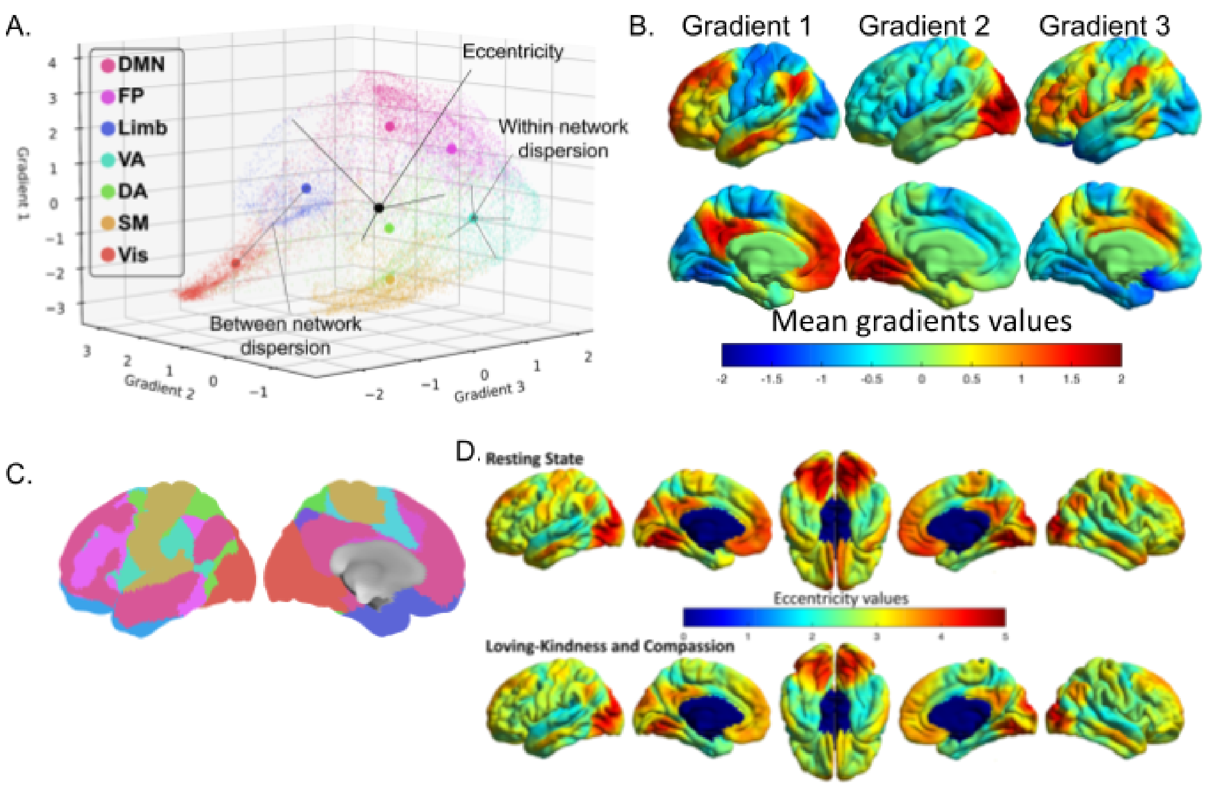


**Figure 1 Effect of loving-kindness and compassion on eccentricity maps:** (A) Visualization of dispersion metrics within the three-gradient space. The eccentricity value of a vertex corresponds to the Euclidean distance from the barycenter (depicted by a black dot) of the 3D space. For each individual 3D map, we computed an eccentricity map defined by the Euclidean distance from each vertex to the individual barycenter of the 3D space (Bethlehem et al., 2020; Valk et al., 2023). These maps reflect the integration (low eccentricity) and segregation (high eccentricity) within the connectome for each voxel of each participant. We then quantified the dispersion metrics, which represents the segregation of large-scale networks (Valk et al., 2023). The within-network dispersion is calculated as the sum squared Euclidean distance of network vertices to the network barycenter. Between-networks dispersion is quantified as the Euclidean distance between network barycenters (Valk et al., 2023). (B) The first gradient (left) denotes the cognitive hierarchy of the brain, ranging from the unimodal cortex to the DMN. The second gradient (middle) differentiates the visual cortex from other networks, while the third gradient (right) segregates the limbic network. (C) Visualization of the cortical parcellation (Thomas Yeo et al., 2011) used in (A). The color code corresponds to the one in Figure 1A (D) The eccentricity map describes a continuous coordinate system, where a lower value signifies that the vertex is closer to the barycenter of the 3D space. Sensory regions, including the visual and somatomotor cortex, and the DMN tend to be the least integrated regions. Although the average maps of LKC and RS are similar overall, it is evident that eccentricity values are visually lower during LKC than during RS, except within the visual cortex. Figures 1.A-C adapted from Czajko et al., 2023

To quantify integration and segregation within this gradient space, we can compute an eccentricity map for each individual (Bethlehem et al., 2020; Valk et al., 2023). The eccentricity value of a vertex indicates its Euclidean distance to the barycenter, or centroid, of the multidimensional gradient space. Regions with low eccentricity reside close to the space centroid, reflecting high integration with diverse functional systems. In contrast, regions with high eccentricity are peripherally situated, indicating segregation into specialized functional modules.

The dispersion of resting-state networks can also be assessed within this multidimensional gradient space (Bethlehem et al., 2020; Valk et al., 2023). The within-network dispersion quantifies the compactness of a network based on the summed squared Euclidean distances between its vertices and the network barycenter. Between-network dispersion is measured as the Euclidean distance separating the barycenters of different networks. Together, these dispersion metrics indicate the degree of integration versus segregation between distributed functional modules.

In summary, the multidimensional functional gradient framework provides a powerful means to quantify principles of connectome organization, including the integration and segregation of cortical regions and systems. Examining differences in gradient space topography and dispersion may reveal key neural markers associated with meditative states.

## Statistical analysis

### Voxel-wise analysis

We performed voxel-wise statistical comparisons of gradient component scores between meditative states and resting state using surface-based linear models implemented in SurfStat (http://www.math.mcgill.ca/keith/surfstat/) for Matlab. SurfStat allows flexible specification of linear models using model formulas. Multiple testing correction was performed using random field theory to control the family-wise error rate at $p_{FWE} < 0.05$ across the cortical surface.

### Dispersion metrics analysis

Rather than classical parametric t-tests, we computed Studentized bootstrap tests, also known as bootstrap-t tests (Efron & Tibshirani, 1994), to assess differences in dispersion. Bootstrap resampling methods make fewer assumptions about underlying data distributions and are more robust for smaller sample sizes. For each dispersion metric, we resampled the data with replacement 10,000 times to derive a bootstrap distribution of mean differences between the meditative states and RS. P-values were computed as the proportion of bootstrap samples where the mean difference was different from zero (two-tailed test).

### Decoding

We used a support vector classifier (SVC) implemented in scikit-learn (Abraham et al., 2014) to predict meditative states from resting state (RS) based on gradient dispersion metrics. The SVC was trained using stochastic gradient descent with a modified Huber loss function. Our goal was to determine if the meditative states' dispersion metrics were predictable when compared to RS, under the hypothesis that meditative states induced enduring alterations observable in dispersion metrics. We focused on three dispersion metrics - within-network, between-network, and average dispersion - derived from the gradient matrices. A 5-fold cross-validation routine was used to partition data into separate training and test sets. To reduce variability, we repeated this cross-validation procedure 5000 times with different data folds. At each iteration, we trained the SVC on the dispersion metrics from a subsample of participants, then evaluated its performance in classifying the two states on the remaining participants in the batch. As in our previous analysis, we quantified predictive performance using the area under the receiver operating characteristic curve (AUC). The mean AUC over the 5000 repetitions provided an unbiased estimate. To determine significance, we derived a null distribution of AUC values by repeating the analysis with permuted class labels (Ojala & Garriga, 2009).

# RESULTS:

## MEDITATIVE STATE ANALYSIS

Previous research by our group (Czajko et al., 2023) demonstrated that the density of eccentricity map values was modulated by LKC meditation compared to RS across participant groups. Based on these findings, we hypothesized that large-scale fMRI connectomics measures (Figure 1A) would be differentially modulated by the state-like effects of LKC meditation versus RS, but not open presence (OP) meditation versus RS. To test this hypothesis, we trained a support vector classifier (SVC) on the dispersion metrics from a subset of participants to distinguish meditative states from RS. The classifier successfully decoded LKC from RS (AUC=0.62; p=0.018) but not OP from RS (AUC=0.59; p=0.678), confirming our prior reported results (Czajko et al., 2023).

We next conducted surface-wide statistical comparisons of eccentricity values between meditative states and RS using surface-based linear models implemented in SurfStat (Worsley et al., 2009). As shown in Figure 2A and Table 1, we found statistically significant differences in eccentricity between LKC and RS, particularly decreased eccentricity during LKC compared to RS in bilateral primary motor cortices and superior temporal gyri. More specifically, participants also exhibited decreased eccentricity during LKC versus RS in the right parahippocampal and orbitofrontal gyri, and left fusiform gyrus, temporal pole, and dorsal posterior cingulate cortex. In contrast, when comparing OP and RS states (Figure 2B, Table 2), only four clusters showed differential eccentricity, including increased values in the right angular and left medial temporal gyri and decreased values in the right visual associative cortex and left parahippocampal gyrus during OP versus RS.

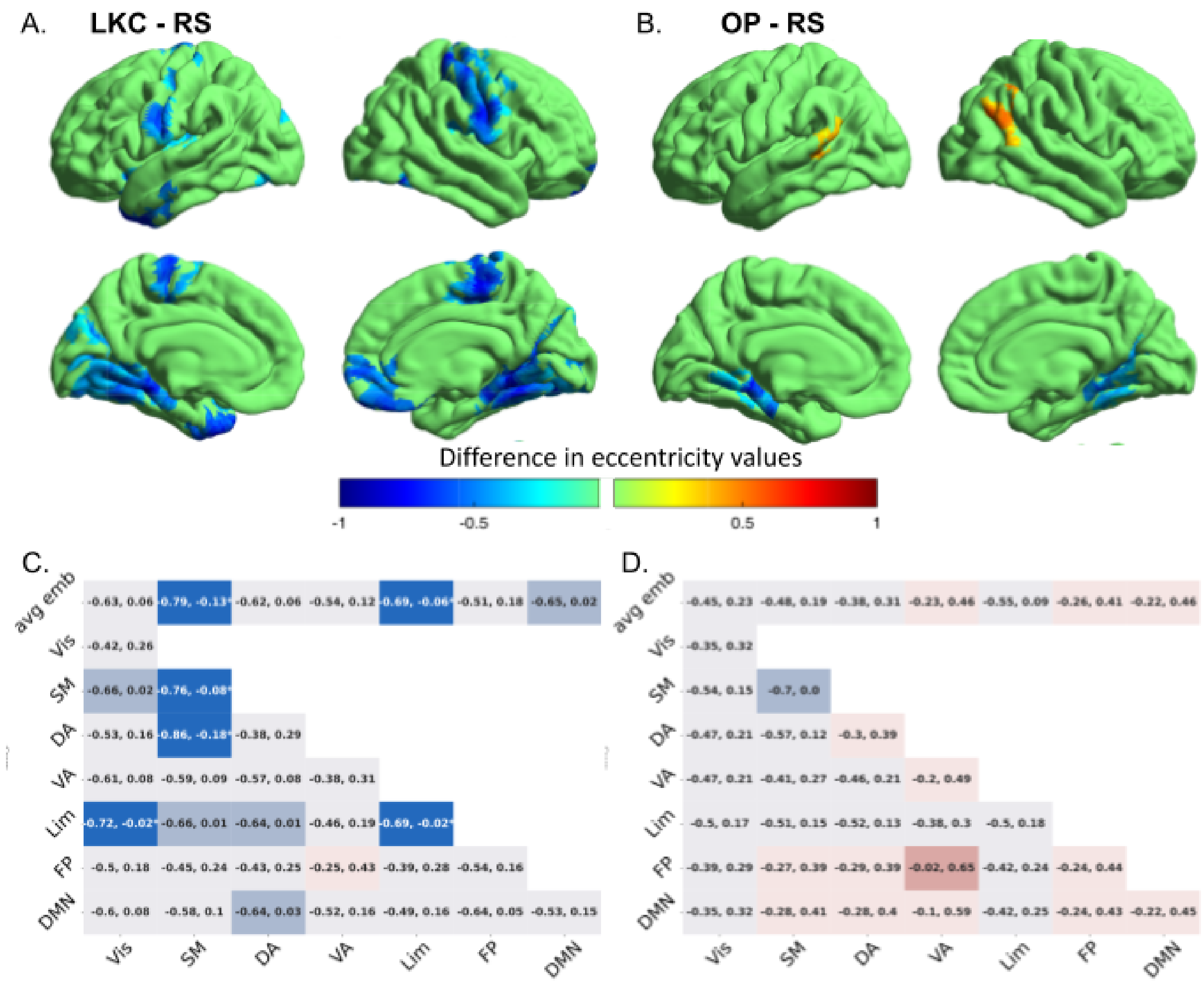


**Figure 2 Effect of meditative states on eccentricity maps:** (A) Surface-wide statistical comparisons between LKC and RS across groups are presented, with increases/decreases during LKC shown in red/blue. Findings were obtained using surface-based linear models implemented in SurfStat (Table 1). (B) Same analysis but between OP and RS this time. (C) Dispersion metrics analysis between LKC and RS. The first row of each matrix shows the average vertex-wise eccentricity of each network, referred to as the average embedding (Valk et al., 2023). The remaining dispersion metrics display the within-network dispersion (diagonal in green) and the between-network dispersion (the remaining squares). Significance was tested using bootstrap-t tests (Efron & Tibshirani, 1994), and 95% confidence intervals are plotted in each box. The color blue (respectively red) indicates that the average eccentricity value was lower (respectively higher) during LKC than RS. Bright-colored boxes indicate significant tests (p<0.05), medium-colored boxes indicate a trend (0.1>p>0.05), while light-colored boxes indicate non-significant tests (p>0.1). (D) Dispersion metrics analysis between OP and RS. No results were significant.

| Cluster region | *p*FDR | *kE* | *Tmax* | *x* | *y* | *z* |
|---|---|---|---|---|---|---|
| R primary motor | <0.001 | 1104 | -5.2 | 41 | -14 | 34 |
| R parahippocampal | <0.001 | 702 | -4.4 | 18 | -41 | -12 |
| L fusiform | <0.001 | 746 | -4.1 | -21 | -42 | -14 |
| L primary motor | <0.001 | 503 | -4.2 | -55 | -8 | 20 |
| L temporal pole | 0.005 | 175 | -3 | -21 | 8 | -42 |
| L dorsal PCC | 0.011 | 135 | -3.2 | -8 | -25 | 48 |
| L superior temporal | 0.13 | 105 | -3.5 | -41 | -2 | -19 |
| R obitofrontal | 0.33 | 166 | -3 | 7 | 57 | -25 |
| R superior temporal | 0.35 | 84 | -4.1 | 43 | -8 | -13 |

**Table 1: Surface-based analysis of the LKC's eccentricity compared to RS**. All eccentricity cluster were lower during LKC than during RS. PCC: Posterior Cingulate Cortex. L: Left; R: Right

| Cluster region | *p*FDR | *kE* | *Tmax* | *x* | *y* | *z* |
|---|---|---|---|---|---|---|
| R angular | 0.017 | 148 | 3.2 | 45 | -70 | 26 |
| L medial temporal | 0.033 | 105 | 3.2 | -59 | -40 | 4 |
| R visual associative | <0.001 | 261 | -3.3 | 13 | -53 | 0 |
| L parahippocampal | 0.001 | 201 | -3.5 | -16 | -32 | -14 |

**Table 2: Surface-based analysis of the OP's eccentricity compared to RS.** L: Left; R: Right

To further characterize state differences in eccentricity from a functional perspective, we explored the multidimensional differences using Studentized bootstrap (Efron & Tibshirani, 1994) within the cognitive hierarchy and between and within standard resting state networks (Figure 1A), as proposed by recent studies (Bethlehem et al., 2020; Valk et al., 2023). These analyses indicated that participants' dispersion metrics were lower during LKC meditation compared to RS specifically within somatomotor, limbic, and default mode networks (Figure 2C). In particular, the average eccentricity decreased within the somatomotor ($t=0.457$; $p=0.008$) and limbic ($t=0.383$; $p=0.035$) networks, and marginally within the default mode network (DMN; $t=0.307$; $p=0.076$), during LKC versus RS. Eccentricity also decreased within the somatomotor ($t=0.41$; $p=0.02$) and limbic ($t=0.354$; $p=0.043$) networks. Furthermore, we observed decreased dispersion between the dorsal attention network and the somatomotor network ($t=0.52$; $p=0.003$) as well as between the visual cortex and limbic network ($t=0.354$; $p=0.043$) during LKC compared to RS. In comparison, the same analysis comparing OP and RS states (Figure 2D) revealed no significant differences.

In summary, our findings demonstrate distinct modulations of large-scale brain functional connectivity by loving-kindness compassion versus open presence meditative states. The state-dependent decreases in eccentricity and between-network dispersion specifically during loving-kindness compassion meditation suggest this practice induces a particular dynamical integration of sensorimotor, affective, and attentional processes.

## WITHIN GROUP ANALYSIS AND INTERACTION

To further understand if expertise may play a role in the contrast between LKC and RS we computed the same analyses within novice and expert groups separately. Even after splitting our sample, the SVC trained on the dispersion metrics of a subset of each group still managed to significantly decode LKC from RS states in novices (AUC=0.67; p=0.01) but not in experts (AUC=0.53; p=0.36). The surface-wide statistical comparisons between LKC and RS also revealed different contrasts between groups (Figure 3A). In novices, numerous clusters exhibited decreased eccentricity during LKC compared to RS, with only one cluster in the left visual associative cortex showing increased eccentricity. Specifically, eccentricity decreased to a greater extent during LKC versus RS in novices within the bilateral anterior prefrontal cortex, right dorsal posterior cingulate cortex, left premotor and supplementary motor area, right angular, fusiform and supramarginal gyri, right temporal pole and left medial temporal gyrus (Table 3). In contrast, for experts, LKC compared to RS only increased eccentricity in the right fusiform gyrus while decreasing eccentricity in the left secondary visual area and right visual associative cortex (Figure 3B, Table 4).

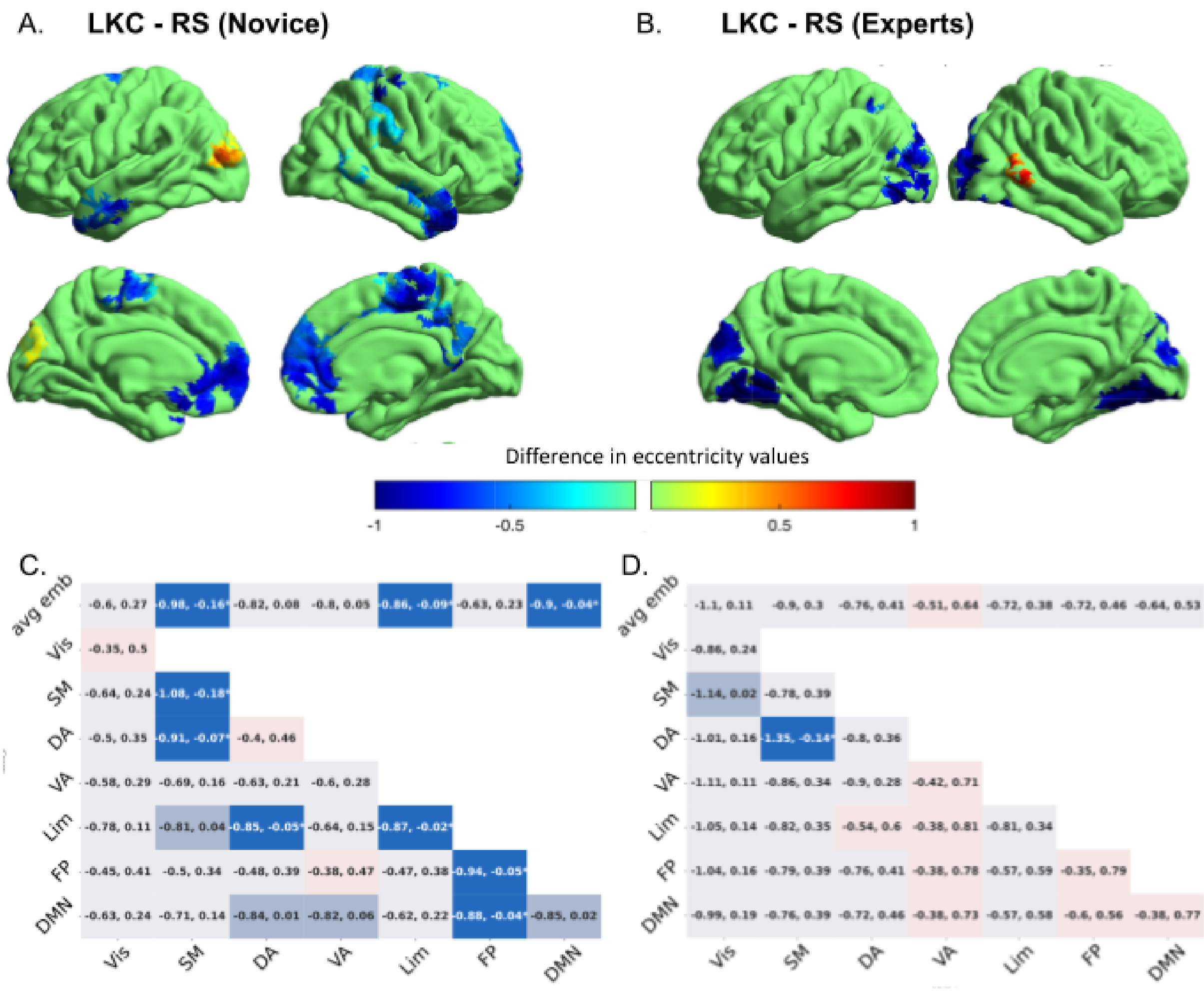


**Figure 3 Within-group LKC vs RS analysis:** (A) Surface-wide statistical comparisons between LKC and RS within novices are presented, with increases/decreases during LKC shown in red/blue. (B) Surface-wide statistical comparisons between LKC and RS within experts. (C) Dispersion metrics analysis between LKC and RS within novices. Color code and explanations are presented in Figure 2C. (D) Dispersion metrics analysis between LKC and RS within experts.

| Cluster region | *p*FDR | *kE* | *Tmax* | *x* | *y* | *z* |
|---|---|---|---|---|---|---|
| **L visual associative** | **<0.001** | **277** | **3.3** | **-40** | **-85** | **12** |
| **R dorsal PCC** | **<0.001** | **440** | **-4** | **13** | **-29** | **44** |
| **R angular** | **<0.001** | **195** | **-3.8** | **36** | **-45** | **38** |
| **L premotor + SMA** | **<0.001** | **162** | **-3.3** | **-3** | **-14** | **64** |
| **R temporal pole** | **<0.001** | **192** | **-4.2** | **48** | **13** | **-37** |
| **R anterior PFC** | **<0.001** | **307** | **-3.4** | **14** | **46** | **7** |
| **R dorsal PCC** | **<0.001** | **226** | **-3.2** | **4** | **-49** | **46** |
| **R supramarginal** | **<0.001** | **134** | **-3.1** | **64** | **-35** | **33** |
| **R fusiform** | **0.006** | **83** | **-3.3** | **55** | **-60** | **7** |
| **L anterior PFC** | **0.017** | **206** | **-3.1** | **-6** | **59** | **11** |
| **L medial temporal** | **0.036** | **93** | **-4** | **-57** | **-3** | **-24** |

**Table 3: Surface-based analysis of the LKC's eccentricity compared to RS in novices**. All eccentricity cluster were lower during LKC than during RS excepted for the left visual associative cortex. SMA: Supplementary motor area. PCC: Posterior Cingulate Cortex. PFC: Pre Frontal Cortex. L: Left; R: Right

| Cluster region | *p*FDR | *kE* | *Tmax* | *x* | *y* | *z* |
|---|---|---|---|---|---|---|
| **R fusiform** | **0.037** | **68** | **3.5** | **55** | **-59** | **10** |
| **L secondary visual** | **<0.001** | **1010** | **-3.9** | **-9** | **-77** | **23** |
| **R visual associative** | **<0.001** | **610** | **-4.0** | **39** | **-85** | **13** |

**Table 4: Surface-based analysis of the LKC's eccentricity compared to RS in experts**. PCC: Posterior Cingulate Cortex. L: Left; R: Right

To further characterize state differences in eccentricity functionally, we explored multidimensional differences within the cognitive hierarchy and between/within resting state networks in novices and experts separately (Figure 1A), as proposed by recent work (Margulies et al. 2016; Haak et al. 2018). In novices, dispersion metrics were lower during LKC versus RS specifically within somatomotor, limbic, and default mode networks (Figure 3C). The average eccentricity decreased within somatomotor (t=0.57; p=0.009), limbic (t=0.49; p=0.03) and default mode (t=0.46; p=0.04) networks in 3D space during LKC compared to RS. Eccentricity also decreased within somatomotor (t=0.6; p=0.006), limbic (t=0.433; p=0.049) and fronto-parietal (t=0.47; p=0.04) networks. We additionally observed decreased dispersion between dorsal attention and somatomotor (t=0.48; p=0.03) as well as limbic (t=0.453; p=0.04) networks, and between fronto-parietal and default mode networks (t=0.458; p=0.04) during LKC versus RS. Although the SVC did not decode LKC from RS in experts, for comparison we show these results (Figure 3D). The only significant finding was reduced dispersion between dorsal attention and somatomotor cortex (t=0.696; p=0.018) during LKC versus RS.

Given the differential LKC vs RS contrasts in experts and novices, these results provide evidence for a state-by-trait interaction. We directly tested this using a surface-based analysis (Figure 4) which revealed clusters present in the previous contrasts. Eccentricity decreased more during LKC than RS in novices compared to experts within the right dorsal posterior cingulate cortex, supramarginal gyrus, temporal pole, and fusiform gyrus. Conversely, eccentricity in the left secondary visual area and right visual associative area decreased more in LKC versus RS for experts than novices (Table 5).

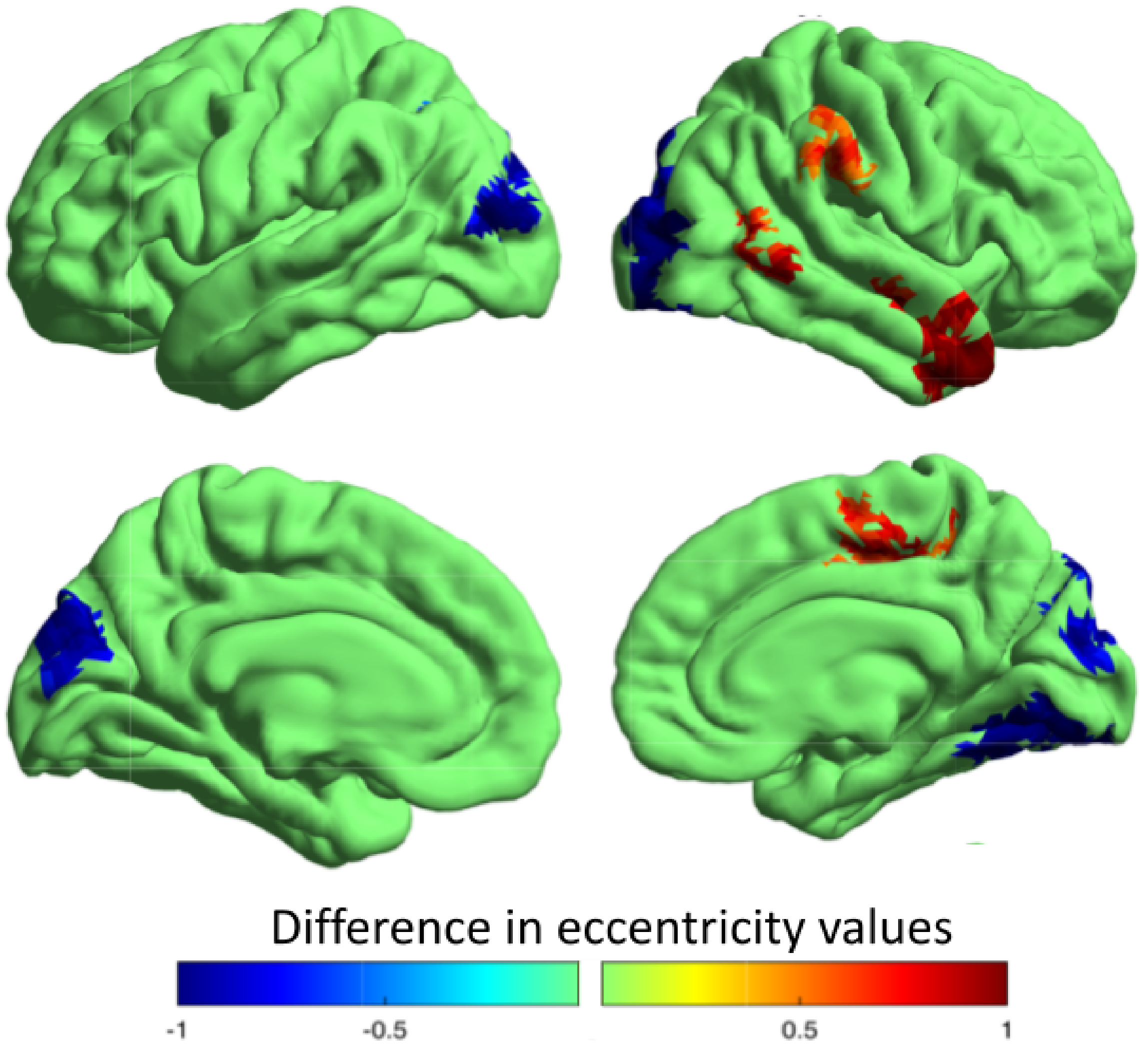


**Figure 4 Interaction between states (LKC and RS) and traits (experts and novices):**
For both participant groups (novices and experts), the eccentricity values from the RS were subtracted from those obtained during the LKC meditation state. Subsequently, these contrasts were subjected to a statistical comparison between the two groups. Any increases or decreases in eccentricity values for

experts were denoted in red and blue, respectively. According to the findings depicted in Figures 3A and 3B, during LKC meditation, experts exhibited a more pronounced decrease in eccentricity within the secondary visual area compared to novices. Conversely, novices demonstrated a more substantial reduction in the eccentricity of the right dorsal posterior cingulate cortex,  during LKC relative to the RS, as opposed to experts.

| Cluster region | *p*FDR | *kE* | *Tmax* | *x* | *y* | *z* |
|---|---|---|---|---|---|---|
| **R dorsal PCC** | **0.002** | **132** | **3.2** | **13** | **-29** | **44** |
| **R supramarginal** | **0.005** | **101** | **3** | **63** | **-29** | **35** |
| **R temporal pole** | **0.005** | **143** | **3.51** | **47** | **13** | **-37** |
| **R fusiform** | **0.008** | **81** | **3.6** | **55** | **-60** | **7** |
| **L secondary visual** | **<0.001** | **801** | **-3.4** | **-9** | **-77** | **23** |
| **R visual associative** | **<0.001** | **470** | **-3.9** | **39** | **-85** | **13** |

**Table 5: Surface-based interaction of the LKC's eccentricity contrasted with the RS compared between novices and experts** . All eccentricity clusters that were lower during this interaction are also present in the novices alone contrast while all the eccentricity clusters that are higher, are also present in the experts alone contrast. PCC: Posterior Cingulate Cortex. L: Left; R: Right

In summary, these analyses demonstrate distinct state-by-trait interactions wherein LKC meditation induces decreased eccentricity and between-network dispersion compared to resting state specifically in novices. The modulatory effects were localized to sensorimotor, limbic, attention and default mode regions. In contrast, experts exhibited fewer differences between LKC and rest overall, suggesting they maintain a more stable functional connectivity profile across states.

# DISCUSSION

The current study investigated how LKC and OP meditation states modulate large-scale brain functional connectivity compared to RS in both novice and expert meditators. Using a multivariate decoding approach, we found that LKC could be distinguished from RS based on dispersion metrics, while OP could not. In surface-based statistical analyses, LKC induced decreased eccentricity relative to RS, primarily in sensorimotor and limbic regions. Comparatively, few differences were observed between OP and RS states. When examining state changes within and between intrinsic connectivity networks, LKC specifically reduced dispersion metrics in the somatomotor, limbic, and default mode networks. Furthermore, important state-by-trait interactions were uncovered. Novice meditators exhibited greater decreases in eccentricity and between-network dispersion during LKC compared to RS. The effects were localized to sensorimotor, limbic, attentional, and default mode regions, which showed significantly reduced dispersion metrics during LKC in novices. In contrast, experts displayed fewer differences between LKC and RS overall, indicative of a more stable trait-like functional connectivity signature. In summary, these findings demonstrate that LKC meditation induces robust changes in large-scale brain dynamics, characterized by network-specific effects on dispersion measures. The modulatory impact of LKC was state-dependent and more pronounced in novice meditators, while experts maintained a consistent connectivity profile across meditative states.

The current findings add to a growing body of research demonstrating that loving-kindness and compassion meditation can induce changes in functional brain connectivity (Desbordes et al., 2012; Klimecki et al., 2013; Lutz, Brefczynski-Lewis, et al., 2008; Mascaro et al., 2013). Our results specifically show that the loving-kindness compassion state reduces network dispersion and eccentricity compared to rest, primarily in sensorimotor and limbic areas. This aligns with prior work linking loving-kindness practices to increased activation and connectivity of brain regions involved in empathy, embodied simulation, and emotion regulation, such as the insula and cingulate cortex (Engen & Singer, 2015; Klimecki et al., 2013). The decreased dispersion and localized effects observed here suggest that loving-kindness compassion meditation consolidates and focuses connectivity within key networks involved in interoceptive, affective, and social processing. As suggested by Desbordes et al., (2012), practices cultivating positivity, care, and concern for others may decrease self-referential processing in cortical midline regions of the default mode network. This effect may underlie the decreased dispersion and eccentricity observed within the default mode network here. Reduced dispersion in sensorimotor and limbic systems also corroborates studies highlighting the key role of these regions in embodying compassion (Engen & Singer, 2015; Fox et al., 2016). Our results indicate LKC strengthens connectivity within affective and interoceptive networks involved in empathic responses. The state-dependent modulation of network dispersion implies a dynamic reorganization of functional brain architectures underlying compassion.

In contrast, the open presence state did not produce robust differences compared to rest, indicating it may rely less on alterations in functional dynamics. This highlights how different forms of meditation can modulate brain function through distinct mechanisms (Lutz, Slagter, et al., 2008). Furthermore, our finding that loving-kindness compassion effects were more pronounced in novices accords with the principle that meditation-related neuroplasticity is often greater earlier in training (Davidson & McEwen, 2012). The reduced dispersion and eccentricity exhibited by novices likely reflects early reorganization processes as brain networks become more selective and efficient at generating loving-kindness compassion states. Meanwhile, experts showed stability across states, consistent with long-term, trait-like changes associated with extensive meditation practice (Brewer et al., 2011).

Our study had several limitations worth noting. First, the cross-sectional design limits inferences about causality between meditation expertise and brain changes. While we matched groups on demographic variables, uncontrolled confounds could still influence results. Second, our exploratory connectomics analysis requires replication in future work to validate the findings. Third, expert meditators are rare and thus our sample size was modest, reducing statistical power to detect smaller effects. Relatedly, uneven group sizes could affect the stability of the results. Overall, while results suggest trait-like brain differences in experts, longitudinal training studies are needed to establish a causal link between meditation practice and functional brain changes. Further research with larger, more diverse samples would strengthen the reliability and generalizability of the findings.

Overall, these discoveries shed light on how iterative activation of temporary states, as occurs during focused meditation training, can induce neuroplastic adaptations underlying enduring traits. Elucidating state-related connectivity signatures and their transformation with expertise provides crucial insights into the brain changes shaped by contemplative practices. Further research should explore if the altered connectivity patterns observed here mediate enhancements in socio-affective skills following loving-kindness compassion training.

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All authors of the present article declare no conflicting interests.

# CREDIT AUTHORSHIP CONTRIBUTION STATEMENT:

**Sébastien Czajko**: Methodology, Validation, Formal analysis, Data curation, Writing, Visualization. **Antoine Lutz**: Methodology, Conceptualization, Supervision, Writing – editing, Funding acquisition. **Daniel Margulies**: Methodology, Supervision. **Jelle Zorn**: Conceptualization, Methodology, Investigation. Data Collection

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# Study 3: Characterizing Acute State and Long-Lasting Trait Neural Signatures of Intensive Mindfulness Meditation

## INTRODUCTION:

Mindfulness meditation has surged in popularity as an intervention for reducing stress (Creswell, 2014), improving wellbeing (Keng et al., 2011), and treating mental health conditions like anxiety and depression (Hofmann et al., 2010). Standard mindfulness-based interventions (MBIs) like Mindfulness-Based Stress Reduction (Kabat-Zinn, 1982) and Mindfulness-Based Cognitive Therapy (Segal et al., 2002) involve 8 weekly 2-3 hour sessions spanning 8 weeks in total. Participants are guided in a variety of mindfulness exercises during sessions and instructed to practice 20-45 minutes daily at home (MacCoon et al., 2012). This amounts to ~12-26 hours of total practice time over the 8 weeks (Parsons et al., 2017). However, research indicates the total dose of mindfulness training strongly mediates outcomes. Parsons et al. (2017) showed total minutes of mindfulness practice over 8 weeks significantly predicted improvements in wellbeing. Meanwhile, abbreviated 4-week versions of MBIs with ~half the total practice time show reduced benefits compared to the standard 8-week format (Demarzo et al., 2015). This reliance on practice time suggests condensed MBIs may limit training effects. Intensive meditation retreats promise accelerated and amplified benefits compared to standard MBIs by having practitioners meditate 8-15 hours daily over consecutive days or weeks (Lutz et al., 2008). For instance, a typical 10-day retreat involves 80-100 hours of intensive mindfulness practice, greatly exceeding the total practice time in an 8-week MBI format (Dor-Ziderman et al., 2021). However, scientific investigation of such retreats using rigorous experimental designs remains scarce. Retreats therefore represent a unique and under-investigated intensive training model for rapidly reduce anxiety, depression, and stress while increasing self-compassion and satisfaction (Dor-Ziderman et al., 2021). These subjective changes suggest retreats may confer rapid improvements in mental health and wellbeing. However, the biological mechanisms underlying such benefits are unknown.

Prior neuroimaging research has demonstrated that standard 8-week mindfulness-based interventions (MBIs) can alter resting-state functional connectivity between large-scale brain networks involved in executive control, self-referential processing, and attentional regulation (Fox et al., 2014). Several studies have shown experienced mindfulness meditators exhibit increased functional connectivity within fronto-parietal regions and the salience network compared to meditation-naïve controls (Brewer et al., 2011; Chen et al., 2013; Doll et al., 2015). The fronto-parietal network supports executive control and directed attention (Marek et al., 2018), while the salience network is involved in redirecting attention to salient stimuli (Uddin, 2015). Enhanced connectivity within these networks aligns with the attentional and self-regulatory skills cultivated during mindfulness training (Lutz et al., 2015).

Conversely, meditators show decreased connectivity between the default mode network (DMN) and fronto-parietal regions (Brewer et al., 2011; Chen et al., 2013; Berkovich-Ohana et al., 2016). The DMN is active during self-referential processing and mind wandering (Andrews-Hanna et al., 2014). Reduced DMN-frontoparietal connectivity may reflect improved ability to monitor and regulate self-related thoughts during meditation (Brewer et al., 2011). Despite these trends, findings remain inconsistent, as some studies report increased

DMN-frontoparietal connectivity following mindfulness training (Kilpatrick et al., 2011; Xue et al., 2011).

Critically, the long-term impact of intensive meditation retreats on functional connectivity patterns remains unclear (Dor-Ziderman et al., 2021). Only a few studies have examined connectivity changes immediately before and after retreats. For instance, one study found increases in DMN integration and fronto-parietal segregation during rest following 8 days of focused attention meditation (Mooneyham et al., 2020). But whether retreat-related connectivity alterations persist over weeks or months is unknown. Rigorously investigating the lasting neural effects of intensive training could elucidate its mechanisms and inform translation into clinical applications.

To address open questions regarding the lasting impact of intensive mindfulness training, here we leveraged an innovative experimental framework combining a 10-day intensive retreat with longitudinal neuroimaging. We recruited participants with existing meditation experience and randomly assigned them to either an active retreat training group or a waitlist control group. The active group underwent fMRI scans at three timepoints: before, immediately after, and crucially, several weeks following the retreat. This enabled examination of both acute state effects during intensive practice as well as assessment of longer-term trait changes persisting after retreat completion. At each scan session, participants completed resting state, open monitoring meditation, and focused attention meditation scans. Importantly, resting state instructions differed from typical - participants were requested to actively engage in mind wandering while refraining from meditation. Specifically, they focused on objects of the past and future, like what they were doing earlier or needed to do after the scan. This aimed to avoid inadvertent meditation and improve consistency with mind wandering in meditation-naïve participants. The three scan conditions allowed isolation of the neural signatures underlying these distinct meditative states (Lutz et al., 2008; Xue et al., 2011). During open monitoring, practitioners relax attentional engagement and remain receptive to any experiences without fixation on a single object (Lutz et al., 2008). In contrast, focused attention meditation involves sustained attentional focus on a chosen object like the breath (Manna et al., 2010). Acquiring both states longitudinally enabled precise characterization of how each one is impacted by intensive training.The inclusion of mind wandering resting state scans provided a task-free baseline for comparing intrinsic functional connectivity patterns unbiased by active meditation (Brewer et al., 2011).

To characterize whole-cortex functional organization, we applied gradient connectivity analysis, a novel data-driven technique that reveals continuous dimensions of intrinsic functional connectivity (Margulies et al., 2016). This analysis extracts principal gradients that span cortical regions from unimodal sensory to transmodal association cortex (Haak et al., 2018). The first principal gradient captures the cognitive hierarchy as it progresses from primary sensorimotor regions to transmodal default mode and frontoparietal regions (Huntenburg et al., 2018). The second and third gradients represent additional topological dimensions (Haak et al., 2018).

Applying this technique to each scan session revealed the expression of connectivity gradients during resting state, open monitoring meditation, and focused attention meditation. Comparing gradient expression before versus after the retreat enabled broad evaluation of network integration and segregation induced by intensive training. We showed in a previous study that the long-term practice of mindfulness meditation reduces globally the eccentricity within the brain, and more specifically increases the somatomotor cortex and the attentional and fronto-parietal networks integration (Czajko et al., 2023). Thus, we hypothesized the intense mindfulness practice during the retreat would have similar longitudinal effects on gradient expression and network integration. We also expected a stronger state effect than during our previous study, when comparing meditative states with RS, as we explicitly asked the participants to not meditate during the RS and to actively engage in mind-wandering. Prior work shows that mind-wandering engages the default mode network (DMN), including medial prefrontal cortex and posterior cingulate regions (Mason et al., 2007). Areas of the DMN also deactivate during goal-directed tasks (Greicius et al., 2003). Prior research on functional connectivity in long-term meditation practitioners and MBI provide some clues, but remain inconsistent. Some studies have shown meditators exhibit reduced connectivity between the default mode network (DMN) and fronto-parietal regions (Brewer et al., 2011; Berkovich-Ohana et al., 2016). This aligns with the decoupling of self-referential processing from executive control networks cultivated during mindfulness (Farb et al., 2007). Conversely, other studies report experienced meditators display increased connectivity between the DMN and fronto-parietal networks compared to novices (Kilpatrick et al., 2011; Xue et al., 2011). Similarly, the links between salience network and fronto-parietal regions have been shown to both increase (Doll et al., 2015) and decrease (Brewer et al., 2011) with meditation expertise. By promoting mind-wandering, we expected to see divergent gradient expression compared to meditative conditions, but the exact direction of this divergence is difficult to predict and requires open empirical investigation.

This innovative design combining intensive retreat training, longitudinal fMRI, and whole-cortex gradient analysis enabled novel insights into systems-level neural plasticity underlying an extreme mindfulness regimen.

# METHOD:

## Participants

Briefly, the full study enrolled 54 healthy adults aged 18-67 years with established meditation experience. Participants practiced mindfulness-related techniques (e.g., Mindfulness-Based Stress Reduction, Vipassana, Tibetan Buddhism) for ≥20 minutes/day, ≥3 days/week over the past year. They also completed a silent meditation retreat involving ≥6 hours of practice daily for ≥2 days. Major exclusion criteria included neurological/psychiatric history, chronic pain, sensory/motor deficits, and central nervous system-acting medications.

Participants provided informed consent and were compensated. This investigation, entitled LONGIMED, was approved by an independent ethics committee (CPP Est IV 2020-A00669-30).

In order to control for potential motion effects, all participants with more than 0.3mm/degree movement were considered outliers (2 participants) and thus removed from the analysis (Power et al., 2014). More over, two participants could not attend one or more MRI sessions and were not included in the analysis. In the end, the analysis concerned 50 participants.

## Paradigm

Between fall 2020 and spring 2021, we coordinated three distinct 10-day mindfulness meditation retreats at the IZARIAT retreat center in Le Poizat-Lalleyriat, France, as detailed on the center's website (https://izariat.com/). These sessions were designed to accommodate a substantial number of attendees within a brief timeframe, a challenge compounded by the concurrent Covid-19 pandemic. Each retreat mirrored the others in format, yet participant numbers varied owing to initial recruitment hurdles and health protocols. The first retreat comprised 11 individuals (8 women, 3 men), the second 23 (15 women, 8 men), and the third

19 (10 women, 9 men), with one individual withdrawing from the final retreat before it began. The first session concluded a day prematurely due to a Covid-related national lockdown, limiting attendees to 9 days of intensive practice. The retreats were led by Stephane Offort, a meditation instructor trained in the Tibetan Buddhist Karma-Kagyu lineage under Guendune Rinpoché. Despite his traditional training, Offort provided secular teachings aligned with typical mindfulness-based interventions. These retreats were held in silence, minimizing communication except for essential discussions with the instructor and research team. Electronic device usage, particularly smartphones, was discouraged. Daily, participants engaged in about 7 hours of meditation, distributed over eight sessions ranging from 45 to 60 minutes each. They had the option to sit on a cushion or chair, with a focus on maintaining a comfortable yet attentive posture. Initial sessions were fully guided, later transitioning to more independent practice with preliminary instructions. Evenings included a one-hour Q&A session with Offort. The retreat's first three days concentrated on attentional focus, utilizing breathing and bodily sensations. As days progressed, auditory and visual senses were incorporated. From the fifth day, participants were encouraged to embrace an open awareness, letting thoughts and emotions flow freely. They were taught to view these experiences as transient and to maintain equanimity. Towards the retreat's end, instructions became sparse, fostering independence, with occasional mentions of compassion for oneself and others. In addition to meditation, daily sessions of movement meditation, inspired by yoga and qi gong, were also conducted, heightening awareness of movement and sensation.

Each retreatant underwent three fMRI scans: immediately before, right after (J+0 to J+2), and two to three weeks post-retreat. We first acquired their functional scans (BOLD fMRI), including a 10-minute resting state scan, 10-minute open monitoring meditation and 10-minute focused attention meditations, sequenced randomly for participants. During rest, participants kept their eyes open without succumbing to sleep, while actively engaging in mind-wandering. For FA, they focused on feelings and sensations within one of their arm (sides were randomized across participants, but kept consistent longitudinally), while OP involved an expansive, unbiased state of awareness. After each fMRI scan, participants provided self-assessments on stability, object orientation, meta-awareness, cognitive fusion, reification, sleepiness, effort, nervousness, and mind wandering on 1-10 Likert scales. We also checked participants focused on their instructed arm side during FA. This allowed verification of distinct phenomenology between conditions and longitudinal effects.

## Data acquisition

Neuroimaging data acquisition was performed using a 3T Siemens Prisma MRI scanner with a 64-channel phased-array head coil. Functional MRI data were acquired using a T2*-weighted echo planar imaging (EPI) sequence with the following parameters: TR=1.72s, TE=30ms, flip angle=90 degrees, FOV=220x220mm, matrix=92x88, slice thickness=2.4mm. 45 slices were acquired with a multiband acceleration factor of 2. The scan was 450 volumes in duration, corresponding to a scan length of 13.5 minutes. The EPI sequence provided whole-brain coverage and was used to acquire data during resting state and meditation conditions. A dual-echo fieldmap scan was also acquired with matched geometry to the functional scan to allow for correction of geometric distortions. Additionally, a high-resolution T1-weighted anatomical scan was acquired using an MPRAGE sequence with the following parameters: TR=3s, TE=3.7ms, TI=1100ms, flip angle=8 degrees, FOV=224x280mm, matrix=320x280, slice thickness=0.8mm.

## Pre-processing

Results included in this manuscript come from preprocessing performed using fMRIPrep 22.0.2 (Esteban, Markiewicz, et al. (2018); Esteban, Blair, et al. (2018); RRID:SCR_016216), which is based on Nipype 1.8.5 (K. Gorgolewski et al. (2011); K. J. Gorgolewski et al. (2018); RRID:SCR_002502).

A total of 3 fieldmaps were found available within the input BIDS structure for each subject. A B0 nonuniformity map (or fieldmap) was estimated from the phase-drift map(s) measure with two consecutive GRE (gradient-recalled echo) acquisitions. The corresponding phase-map(s) were phase-unwrapped with prelude (FSL 6.0.5.1:57b01774).

A total of 3 T1-weighted (T1w) images were found for each subject. All of them were corrected for intensity non-uniformity (INU) with N4BiasFieldCorrection (Tustison et al. 2010), distributed with ANTs 2.3.3 (Avants et al. 2008, RRID:SCR_004757). The T1w-reference was then skull-stripped with a Nipype implementation of the antsBrainExtraction.sh workflow (from ANTs), using OASIS30ANTs as target template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1w using fast (FSL 6.0.5.1:57b01774, RRID:SCR_002823, Zhang, Brady, and Smith 2001). A T1w-reference map was computed after registration of 3 T1w images (after INU-correction) using mri_robust_template (FreeSurfer 7.2.0, Reuter, Rosas, and Fischl 2010). Brain surfaces were reconstructed using recon-all (FreeSurfer 7.2.0, RRID:SCR_001847, Dale, Fischl, and Sereno 1999), and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical gray-matter of Mindboggle (RRID:SCR_002438, Klein et al. 2017). Volume-based spatial normalization to two standard spaces (MNI152NLin2009cAsym, MNI152NLin6Asym) was performed through nonlinear registration with antsRegistration (ANTs 2.3.3), using brain-extracted versions of both T1w reference and the T1w template. The following templates were selected for spatial normalization: ICBM 152 Nonlinear Asymmetrical template version 2009c (Fonov et al. (2009), RRID:SCR_008796; TemplateFlow ID: MNI152NLin2009cAsym), FSL's MNI ICBM 152 non-linear 6th Generation Asymmetric Average Brain Stereotaxic Registration Model (Evans et al. (2012), RRID:SCR_002823; TemplateFlow ID: MNI152NLin6Asym).

Functional data preprocessing

For each of the 9 BOLD runs found per subject (across all tasks and sessions), the following preprocessing was performed. First, a reference volume and its skull-stripped version were generated by aligning and averaging 1 single-band references (SBRefs). Head-motion parameters with respect to the BOLD reference (transformation matrices, and six corresponding rotation and translation parameters) are estimated before any spatiotemporal filtering using mcflirt (FSL 6.0.5.1:57b01774, Jenkinson et al. 2002). The estimated fieldmap was then aligned with rigid-registration to the target EPI (echo-planar imaging) reference run. The field coefficients were mapped on to the reference EPI using the transform. BOLD runs were slice-time corrected to 0.816s (0.5 of slice acquisition range 0s-1.63s) using 3dTshift from AFNI (Cox and Hyde 1997, RRID:SCR_005927). The BOLD reference was then co-registered to the T1w reference using bbregister (FreeSurfer) which implements boundary-based registration (Greve and Fischl 2009). Co-registration was configured with six degrees of freedom. First, a reference volume and its skull-stripped version were generated using a custom methodology of fMRIPrep. Several confounding time-series were calculated based on the preprocessed BOLD: framewise displacement (FD), DVARS and three region-wise global signals. FD was computed using two formulations following Power (absolute sum of relative motions, Power et al. (2014)) and Jenkinson (relative root mean square displacement between affines, Jenkinson et al. (2002)). FD and DVARS are calculated for each functional run, both using their implementations in Nipype (following the definitions by Power et al. 2014). The three global signals are extracted within the CSF, the WM, and the whole-brain masks. Additionally, a set of physiological regressors were extracted to allow for component-based noise correction (CompCor, Behzadi et al. 2007). Principal components are estimated after high-pass filtering the preprocessed BOLD time-series (using a discrete cosine filter with 128s cut-off) for the two CompCor variants: temporal (tCompCor) and anatomical (aCompCor). tCompCor components are then calculated from the top 2% variable voxels within the brain mask. For aCompCor, three probabilistic masks (CSF, WM and combined CSF+WM) are generated in anatomical space. The implementation differs from that of Behzadi et al. in that instead of eroding the masks by 2 pixels on BOLD space, a mask of pixels that likely contain a volume fraction of GM is subtracted from the aCompCor masks. This mask is obtained by dilating a GM mask extracted from the FreeSurfer's aseg segmentation, and it ensures components are not extracted from voxels containing a minimal fraction of GM. Finally, these masks are resampled into BOLD space and binarized by thresholding at 0.99 (as in the original implementation). Components are also calculated separately within the WM and CSF masks. For each CompCor decomposition, the k components with the largest singular values are retained, such that the retained components' time series are sufficient to explain 50 percent of variance across the nuisance mask (CSF, WM, combined, or temporal). The remaining components are dropped from consideration. The head-motion estimates calculated in the correction step were also placed within the corresponding confounds file. The confound time series derived from head motion estimates and global signals were expanded with the inclusion of temporal derivatives and quadratic terms for each (Satterthwaite et al. 2013). Frames that exceeded a threshold of 0.5 mm FD or

1.5 standardized DVARS were annotated as motion outliers. Additional nuisance timeseries are calculated by means of principal components analysis of the signal found within a thin band (crown) of voxels around the edge of the brain, as proposed by (Patriat, Reynolds, and Birn 2017). The BOLD time-series were resampled onto the following surfaces (FreeSurfer reconstruction nomenclature): fsaverage5, fsaverage. Automatic removal of motion artifacts using independent component analysis (ICA-AROMA, Pruim et al. 2015) was performed on the preprocessed BOLD on MNI space time-series after removal of non-steady state volumes and spatial smoothing with an isotropic, Gaussian kernel of 6mm FWHM (full-width half-maximum). Corresponding "non-aggresively" denoised runs were produced after such smoothing. Additionally, the "aggressive" noise-regressors were collected and placed in the corresponding confounds file. Grayordinates files (Glasser et al. 2013) containing 91k samples were also generated using the highest-resolution fsaverage as intermediate standardized surface space. All resamplings can be performed with a single interpolation step by composing all the pertinent transformations (i.e. head-motion transform matrices, susceptibility distortion correction when available, and co-registrations to anatomical and output spaces). Gridded (volumetric) resamplings were performed using antsApplyTransforms (ANTs), configured with Lanczos interpolation to minimize the smoothing effects of other kernels (Lanczos 1964). Non-gridded (surface) resamplings were performed using mri_vol2surf (FreeSurfer).

Many internal operations of fMRIPrep use Nilearn 0.9.1 (Abraham et al. 2014, RRID:SCR_001362), mostly within the functional processing workflow. For more details of the pipeline, see the section corresponding to workflows in fMRIPrep's documentation.

## Connectome gradient construction

In a recent cross-sectional study, we employed the same methodology with the aim of replicating certain findings. The study leveraged functional connectivity gradients as a potent method for delineating the extensive organization of the human cerebral cortex, building off the work of Huntenburg et al., 2018; Margulies et al., 2016; and Valk et al., 2023. Following the protocol outlined by Hong et al. in 2019, these gradients were obtained by initially computing a functional connectivity matrix that spans all grey matter voxels or surface vertices, drawing on resting-state fMRI data. This matrix is created by calculating Pearson correlations between the time series of fMRI data for each pair of voxels or vertices (Friston et al., 1993; Greicius et al., 2003).

To streamline the data from the dense connectivity matrix, we adhered to a series of established steps. The matrix first undergoes a Fisher z-transformation to standardize its distribution. Then, we apply a threshold, preserving only the strongest 10% of positive connections for each voxel or vertex, thereby concentrating on the most significant functional connections (Bethlehem et al., 2020; Hong et al., 2019; Margulies et al., 2016; Valk et al., 2023). Following this, we compute a cosine similarity matrix to measure the similarity of functional connectivity profiles across the brain for each voxel or vertex pair.

As a next step, we enter this cosine similarity matrix into a non-linear dimensionality reduction process called diffusion map embedding (Coifman et al., 2005; Margulies et al., 2016). This process discerns the main gradients or axes that showcase the variation in functional connectivity across the cortex. The primary gradient identifies the continuum from sensory and motor areas to transmodal association areas (Margulies et al., 2016). Subsequent gradients reveal further dimensions of variation in connectivity. In line with earlier research (Hong et al., 2019; Margulies et al., 2016), we set the parameter α to 0.5 to maintain the overarching relationships within the embedded data representation.

For gradient alignment across different individuals, Procrustes rotation techniques were utilized (as available at https://github.com/satra/mapalign, Langs et al., 2015). This approach involved rotating individual subject gradients to achieve maximum congruence with the group-level gradient components extracted from the Human Connectome Project S1200 release (Demirtaş et al., 2019). Such Procrustes transformations ensure that the topology and spatial relationships of the gradients are preserved while aligning the coordinate systems.

In summary, functional connectivity gradients serve as a sophisticated, data-driven strategy for simplifying the intricate network structure of the human brain into a continuum that maps out functional hierarchies. Our analytical approach is consistent with the established best practices for generating gradients, reducing dimensionality, and aligning subjects in a cross-sectional framework.

## 3D gradient metrics

Margulies et al. (2016) illustrate that the primary connectivity gradients capture a significant amount of the variance in the connectivity patterns observed during the resting state. Specifically, the initial three gradients are often responsible for explaining more than half of this variance (Hong et al., 2019; Margulies et al., 2016).

These principal gradient dimensions, when synthesized, form a complex gradient space (Bethlehem et al., 2020; Valk et al., 2023). Within this framework, each vertex of the brain is positioned according to its connectivity profile in relation to the gradients. The spatial arrangement of brain regions within this configuration mirrors their functional connections and interactions.

An individual's integration and segregation in this gradient framework can be evaluated by calculating an eccentricity map (Bethlehem et al., 2020; Valk et al., 2023). A vertex's eccentricity is defined by its Euclidean distance from the central point, or barycenter, of the gradient space. Vertices with lesser eccentricity are closer to this central point, signifying their integrated role across various functional networks. Conversely, vertices with greater eccentricity lie at the periphery, suggesting their involvement in more specialized, distinct functional units.

Furthermore, the spread of resting-state networks is measurable within this gradient space (Bethlehem et al., 2020; Valk et al., 2023). The compactness of a network is determined by the cumulative squared Euclidean distances of its vertices from the network's own barycenter, known as within-network dispersion. The separation of different networks is reflected in the between-network dispersion, calculated by the Euclidean distances between the networks' barycenters. These dispersion measures help to understand the extent of integration or segregation among different functional groups.

In essence, the framework of multidimensional functional gradients serves as an effective tool for delineating connectome organization principles, such as the integration and segregation across cortical areas and systems. By analyzing variations in the topography and dispersion within gradient space, one might uncover crucial brain signatures linked to states of meditation.

## Statistical analysis

Instead of traditional parametric t-tests, our analysis utilized Studentized bootstrap tests, often referred to as bootstrap-t tests (Efron & Tibshirani, 1994), to evaluate the variance in dispersion. This bootstrap resampling technique is less dependent on the assumptions regarding the data's distribution and offers increased reliability when dealing with smaller datasets. For each measure of dispersion, we performed resampling of the dataset with replacement 10,000 times, which allowed us to establish a bootstrap distribution for the average differences observed between meditative states and the resting state (RS). We then calculated p-values based on the frequency of bootstrap samples that exhibited a mean difference significantly distinct from zero, applying a two-tailed testing approach

# DISCUSSION:

This PhD thesis delved into the impact of mindfulness meditation on brain function by employing diffusion embedding techniques with resting-state fMRI data. The research retrospectively charted the neural correlates of meditation, capturing both the ephemeral nature of meditative states and the more lasting traits shaped by long-term practice. The methodological innovations presented in this work sought to shed light on the cognitive processes influenced by mindfulness, with an eye toward enhancing mental health and well-being.

The initial phase of the investigation centered on the neural underpinnings that accompany meditation expertise (Study 1). The study contrasted long-term Buddhist practitioners with meditation novices, revealing enduring neural modifications attributable to substantial meditation experience. The investigation drew on prior research to hypothesize that such expertise would manifest specific, detectable changes in the brain's network activity, observable across various states of consciousness. Particularly, it was anticipated that the effects observed in non-dual meditation might mirror, albeit less intensively, those elicited by psychedelic substances in terms of cognitive hierarchy alterations.

Building upon the foundational insights from the first study, the subsequent research phase probed the brain connectivity patterns evoked by the loving-kindness compassion (LKC) meditation (Study 2). This segment of the thesis aimed to identify the distinct neural signatures of the LKC state, emphasizing the altered connectivity in socio-affective processing regions. The focus was on discerning how LKC meditation differentially affected the brain's functional connectivity, with a special interest in comparing the experiences of meditative novices and seasoned experts.

The concluding study of the thesis shifted focus to the persistent influences of intensive mindfulness practice on intrinsic brain function (Study 3). Analyzing the aftermath of a 10-day meditation retreat, the research sought to determine whether such concentrated practice could forge long-standing changes in functional connectivity.

In sum, the thesis utilized state-of-the-art neuroimaging and analytical methods to chart the brain's functional networks as they are sculpted by the practice of mindfulness meditation. The discoveries made throughout this research journey elucidated the dynamic interplay between meditative states and traits, with potential therapeutic implications for harnessing mindfulness in the pursuit of mental health optimization and cognitive enhancement.

## What are the enduring impacts of sustained mindfulness meditation practice on the cognitive hierarchy?

In the first study of the thesis, a deeper understanding of the neural dynamics associated with meditation expertise was uncovered. The results demonstrated that experts in open presence meditation exhibited a heightened level of functional integration, not only during meditation but also at resting state, albeit to a lesser extent. This suggested an increased capacity for information exchange across various brain networks, analogous with the concept of amplified brain entropy and a more relaxed set of cognitive priors typically held during meditation. Furthermore, these experts displayed notable reductions in eccentricity and dispersion within the dorsal attention, limbic, and sensorimotor networks. This finding implied a significant boost in the connectivity both within these networks and between them, as corroborated by surface analyses pinpointing the same reduced eccentricity in specific brain regions. A particularly striking insight was the increased connectivity observed between limbic, sensorimotor, ventral attention, and frontoparietal networks among the meditation experts. This inter-network connection underscores the integration of attentional processes, emotional regulation, and embodied cognition, which are all pivotal components of meditation expertise. Crucially, the effects observed in meditation expertise did not seem confined to the meditative states alone but pointed to broader trait-related changes. An average across various states yielded the most accurate depiction of these enduring traits. Interestingly, a higher ability to decouple thoughts and emotions, as indicated by the DDS score, was correlated with more integrated limbic, sensorimotor, and ventral attention networks in these expert meditators. This suggests that the practice of meditation may lead to an enhanced capacity for sensory-affective uncoupling—a key feature of mindfulness. The insights gained from this study illuminate the intricate neural mechanisms that underlie cognitive defusion, a concept measured by the DDS and commonly experienced by long-term meditators. However, the pioneering use of the diffusion embedding technique underscores the need for further research to replicate these findings and solidify the understanding of these complex neural patterns.

## How Does Mindfulness Meditation Acutely Influence the Cognitive Hierarchy?

The second study of the thesis provided a detailed exploration of the state effects of Loving-Kindness Compassion (LKC) meditation, distinguishing it from the resting state (RS) by evaluating changes in brain connectivity patterns. During LKC meditation, participants experienced a reduction in connectivity dispersion, particularly in sensorimotor and limbic regions, when compared to their resting state. This suggests that LKC meditation prompts a more integrated network within regions associated with emotional processing and physical sensations. Additionally, there was a specific decrease in dispersion within the somatomotor, limbic, and default mode networks during LKC meditation. Intriguingly, novice meditators demonstrated more significant connectivity changes when engaging in LKC, pointing to early stages of neuroplasticity. In contrast, expert meditators showed a consistent connectivity pattern across states, hinting at the ingrained, trait-like nature of their meditation practice. When examining Open Presence (OP) meditation, the study found that this practice did not exhibit substantial changes in connectivity compared to the resting state. This suggests that OP may operate through mechanisms different from those of LKC meditation. The findings of this study are in line with previous research that has identified shifts in functional connectivity resulting from practices of loving-kindness and compassion meditation. These practices are understood to potentially diminish self-referential processing while simultaneously enhancing brain functions related to empathy. The implications of this research are multifaceted. LKC meditation appears to concentrate connectivity within neural networks that are crucial for empathy and the regulation of emotions. The pronounced neuroplastic alterations in novice practitioners indicate that meditation can swiftly affect brain function, while the stable brain patterns in experts suggest a long-term, trait-like reconfiguration of brain connectivity. Furthermore, the contrast in effects between LKC and OP meditations underscores the possibility that different meditation practices may shape brain function through unique neural pathway.

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